

GENERAL PHYSICS I

PHYS 1985

Course Notes

CONCEPTS

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CONTENTS

Chapter 1	Mathematical Review for Physics	1
1.1	Basic Algebra Review	1
1.2	Linear and quadratic equations	2
1.3	Differential Calculus	3
1.4	Geometry	7
1.5	Trigonometry	10
1.6	Vector Calculus	12
Chapter 2	Units and Dimensional Analysis	17
2.1	Physical Quantities and Base Units	17
2.2	Dimensional analysis	20
2.3	Fundamental Constants	21
Chapter 3	Kinematics	23
3.1	One-Dimensional Kinematics	23
3.2	Three-Dimensional Kinematics	28
3.3	Changing Coordinate Systems	31
3.4	Special Types of Motion	32
Chapter 4	Dynamics	38
4.1	Forces	38
4.2	Inertial frames	41
4.3	Newton's 3 Laws	42
4.4	Mass and Weight	44
4.5	Friction forces	46
Chapter 5	Power, Work, Kinetic and Potential Energy	49
5.1	Review of Forces	49
5.2	Power, work and kinetic energy	50
5.3	Work of a force along a path	51
5.4	Conservative forces and potential energy	54
5.5	Conservation of mechanical energy	56
Chapter 6	Momentum, Centre of Mass and Collisions	59
6.1	Momentum	59
6.2	Movement of the centre of mass	60
6.3	Collisions	61
Chapter 7	Torque, Angular Momentum and Rotation	66
7.1	Torque of a force	66
7.2	Angular momentum	67
7.3	Dynamic and Static Equilibrium	70
7.4	Rotation around a fixed axis	71
7.5	Analogy of rotation and translation	75
7.6	Overview	76
Chapter 8	Mechanical Waves	77
8.1	One-Dimensional Mechanical Waves	78
8.2	Doppler effect	81

CHAPTER 1

MATHEMATICAL REVIEW FOR PHYSICS

Remark. This review considers only differentiable functions of a real variable x .

1.1 Basic Algebra Review

For every integer n, m , every $a, b, c, d \neq 0$ and every $x, y \neq 0$, we have

FRACTIONS, EXPONENTS AND LOGARITHMS

$$\circ \frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm cb}{bd}$$

$$\circ \frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

$$\circ \frac{a}{b} \div \frac{c}{d} = \frac{ad}{bc}$$

$$\circ \left(\frac{a}{b}\right)^{-1} = \frac{b}{a}$$

$$\circ \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$\circ x^0 = 1$$

$$\circ x^1 = x$$

$$\circ x^n x^m = x^{n+m}$$

$$\circ x^n / x^m = x^{n-m}$$

$$\circ x^{1/n} = \sqrt[n]{x}$$

$$\circ (x^n)^m = x^{nm}$$

$$\boxed{x = a^z \Leftrightarrow z = \log_a x}$$

$$\circ \log(xy) = \log x + \log y$$

$$\circ \log(x/y) = \log x - \log y$$

$$\circ \log(x^n) = n \log x$$

$$\circ \ln e^x = x, \quad \ln(1/x) = -\ln x$$

$$\circ \log x = \frac{\ln x}{\ln 10}$$

In the right-hand column, $a > 0$ and $a \neq 1$ define the base of the logarithm $\log_a$; $\log \equiv \log_{10}$ is the common logarithm and $\ln \equiv \log_e$ is the natural logarithm (with base $e \approx 2.71828$). The arguments of logarithms must be strictly positive: $x > 0$ and $y > 0$ in the identities above.

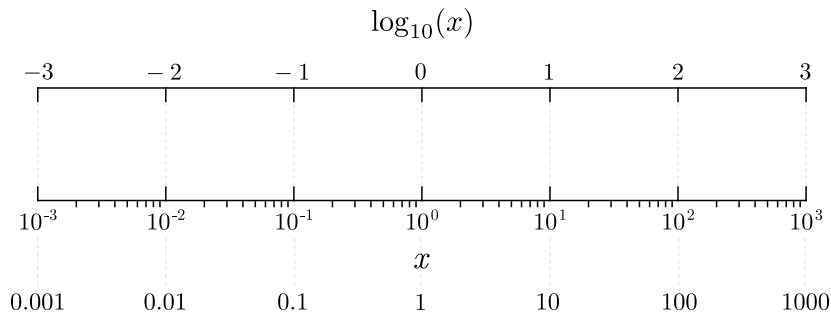


Figure 1: Logarithmic scale.

1.2 Linear and quadratic equations

Affine and linear functions An affine function of a real variable x , $f(x) = ax + b$ with $a, b \in \mathbb{R}$, has a straight-line graph with slope a . It intersects the vertical axis at $y = b$ and the horizontal axis at $x = -b/a$ when $a \neq 0$. When $b = 0$, the function is called linear.

Linear equation The linear equation $ax + b = 0$ has a unique solution when $a \neq 0$,

$$f(x) = ax + b = 0 \quad \Leftrightarrow \quad \boxed{x = -b/a} \quad a \neq 0$$

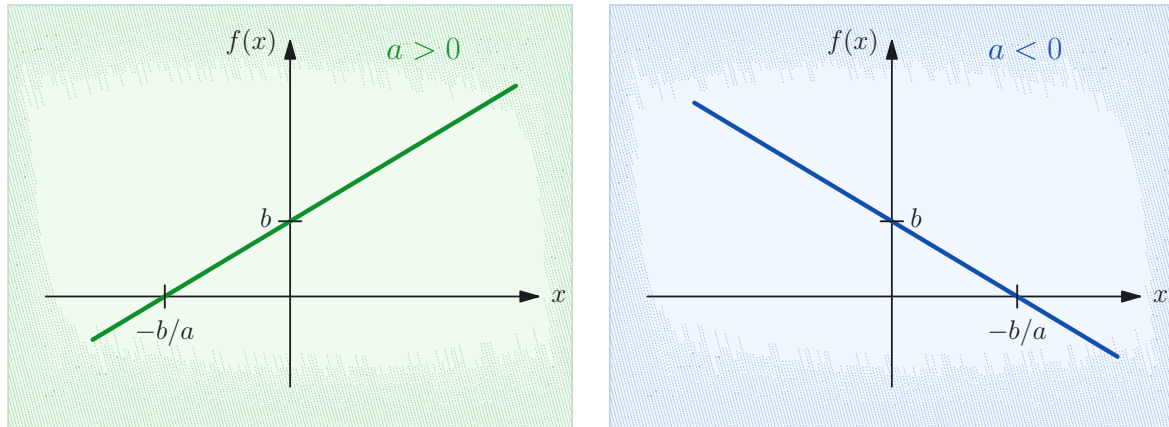


Figure 2: Graph of an affine function $f(x) = ax + b$: a straight line with slope a .

Quadratic function A quadratic function of a real variable x , $f(x) = ax^2 + bx + c$ with $a, b, c \in \mathbb{R}$ and $a \neq 0$, has a parabolic graph. The parabola is convex (concave upwards) if $a > 0$ and concave (concave downwards) if $a < 0$. It intersects the vertical axis at $y = c$ and intersects the horizontal axis when the discriminant

$$\boxed{\Delta \stackrel{\text{def}}{=} b^2 - 4ac}$$

is positive or zero.

Quadratic equation The equation $ax^2 + bx + c = 0$ has no real solution when the discriminant is negative, one real solution $x = -b/(2a)$ when it is zero, and two real solutions

$$\boxed{x = \frac{-b \pm \sqrt{\Delta}}{2a}} \quad (1)$$

when the discriminant is (strictly) positive.

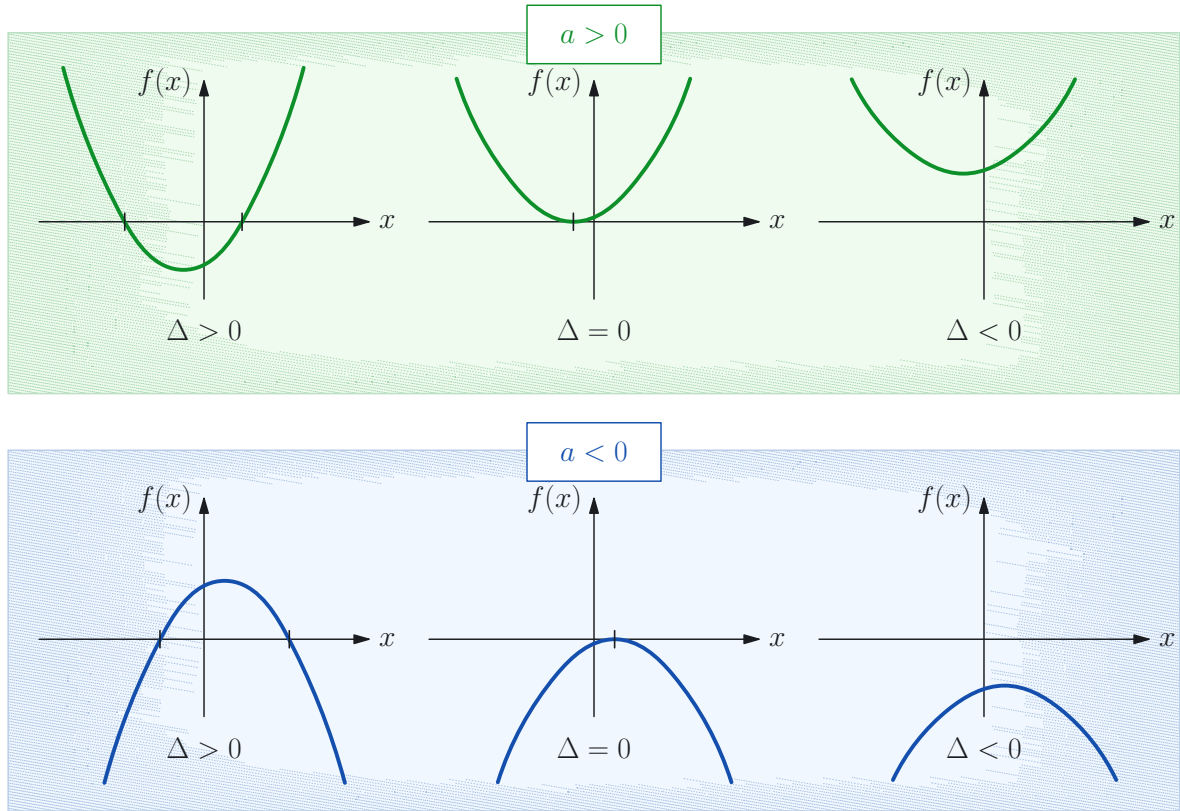


Figure 3: Typical graphs of the quadratic function $f(x) = ax^2 + bx + c$: a parabola, convex if $a > 0$ and concave if $a < 0$, with discriminant $\Delta = b^2 - 4ac$ positive (left), zero (middle) or negative (right).

Remark. The equation $f(x) = ax^2 + bx + c = 0$ *always* has two solutions over the complex numbers, counted with multiplicity (these solutions need not be real):

$$f(x) = ax^2 + bx + c = 0 \Leftrightarrow x = \frac{-b \pm \sqrt{\Delta}}{2a} \in \mathbb{C} \quad (2)$$

1.3 Differential Calculus

Derivative The derivative of a function $f(x)$ of a real variable x is defined by

$$\frac{df(x)}{dx} \stackrel{\text{def}}{=} \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \quad (3)$$

When the value of $f(x)$ is plotted on the vertical axis (y) and the value of x on the horizontal axis, the derivative at a point equals the slope¹ of the tangent to the graph at that point (see Figure 4). The second derivative of a function $f(x)$ is the derivative of its derivative, namely

$$\frac{d^2 f(x)}{dx^2} \stackrel{\text{def}}{=} \frac{d\left(\frac{df(x)}{dx}\right)}{dx},$$

and similarly for higher-order derivatives.

¹The slope of a line is defined as the constant ratio $\frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} \quad \forall x_1, x_2$.

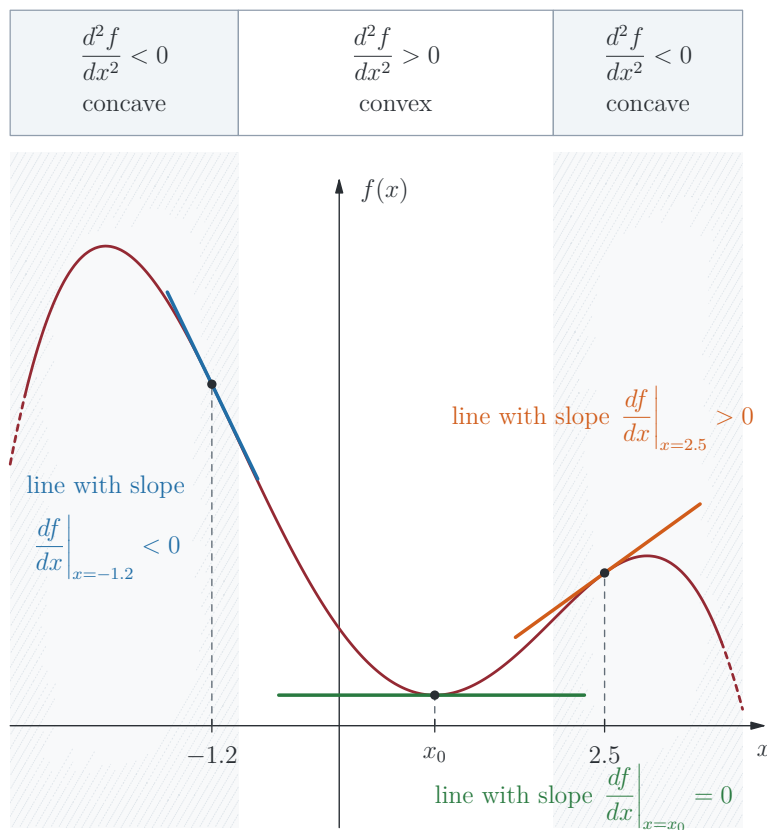


Figure 4: The first derivative of $f(x)$ at x_0 , denoted by $\frac{df}{dx}\big|_{x=x_0}$, gives the slope of the tangent to the graph of $f(x)$ at the point with horizontal coordinate $x = x_0$. The second derivative describes the curvature of the graph (concave when $\frac{d^2 f}{dx^2} < 0$ and convex when $\frac{d^2 f}{dx^2} > 0$).

Remark. Several notations are used for the derivative defined in equation (3); the most common are $\frac{df(x)}{dx}$, $f'(x)$ and $D_x(f(x))$.

COMMON DERIVATIVES

For any $a \neq 0$ and $n \in \mathbb{R}$,

- $\frac{d}{dx}(a) = 0$
- $\frac{d}{dx}(\sin(ax)) = a \cos(ax)$
- $\frac{d}{dx}(ax^n) = n a x^{n-1}$
- $\frac{d}{dx}(\cos(ax)) = -a \sin(ax)$
- $\frac{d}{dx}(e^{ax}) = a e^{ax}$
- $\frac{d}{dx}(\ln(ax)) = \frac{1}{x}$

DIFFERENTIATION RULES

For any differentiable functions f and g ,

$$\circ \frac{d}{dx}(f(x) + g(x)) = \frac{df(x)}{dx} + \frac{dg(x)}{dx} \quad (\text{sum})$$

$$\circ \frac{d}{dx}(f(x)g(x)) = \frac{df(x)}{dx}g(x) + f(x)\frac{dg(x)}{dx} \quad (\text{product})$$

$$\circ \frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x)\frac{df(x)}{dx} - f(x)\frac{dg(x)}{dx}}{[g(x)]^2} \quad (\text{quotient})$$

$$\circ \frac{d}{dx}(f \circ g)(x) = \frac{d}{dx}(f(g(x))) = \frac{df(g)}{dg} \frac{dg(x)}{dx} \quad (\text{composite function})$$

Example. Calculate the derivative of $f(x) = \frac{1}{\sin^2 x}$ with respect to x .

We can write $f(x) = g(h(x))$ with $g(h) = \frac{1}{h^2}$ and $h(x) = \sin x$, which leads to

$$\frac{df(x)}{dx} = \frac{d}{dx}g(h(x)) = \frac{dg(h)}{dh} \frac{dh(x)}{dx} = -\frac{2}{h^3} \cos x = -2 \frac{\cos x}{\sin^3 x}$$

Taylor Expansion The second-order Taylor expansion of a continuous, twice-differentiable function $f(x)$ in a neighbourhood of $x = x_0$ is

$$f(x) \approx f(x_0) + \underbrace{\frac{df(x)}{dx} \Big|_{x=x_0} (x - x_0)}_{\text{linear in } x} + \underbrace{\frac{1}{2} \frac{d^2 f(x)}{dx^2} \Big|_{x=x_0} (x - x_0)^2}_{\text{quadratic in } x} \quad (4)$$

Example. Find the second-order Taylor expansion of $f(x) = \sin x$ about $x_0 = 0$.

Since $\frac{df(x)}{dx} = \cos x$ and $\frac{d^2 f(x)}{dx^2} = -\sin x$, we have $f(x_0) = 0$, $\frac{df(x)}{dx} \Big|_{x=x_0} = 1$ and $\frac{d^2 f(x)}{dx^2} \Big|_{x=x_0} = 0$.

Taylor's formula (4) then gives

$$\sin x \approx x \quad \text{for } x \ll 1$$

Antiderivative An antiderivative of a function $f(x)$ is a function $F(x)$ such that $\frac{dF(x)}{dx} = f(x)$.

Remark. A given function $f(x)$ has infinitely many antiderivatives, differing only by an additive constant.

Example. Find the antiderivatives of $f(x) = \sin(2x)$.

It is readily verified that the derivative of $\cos(2x)$ is $-2\sin(2x)$. All antiderivatives of $f(x)$ therefore have the form

$$F(x) = -\frac{1}{2} \cos(2x) + c \quad \text{with } c \in \mathbb{R}$$

Integral The integral of a function $f(x)$ over an interval $[a, b]$ is the signed area bounded by the graph of $f(x)$, the horizontal axis and the lines $x = a$ and $x = b$ (see Figure 5). By definition, the area is positive where the graph of $f(x)$ lies above the horizontal axis and negative where it lies below. The integral is denoted by

$$\int_a^b f(x) dx$$

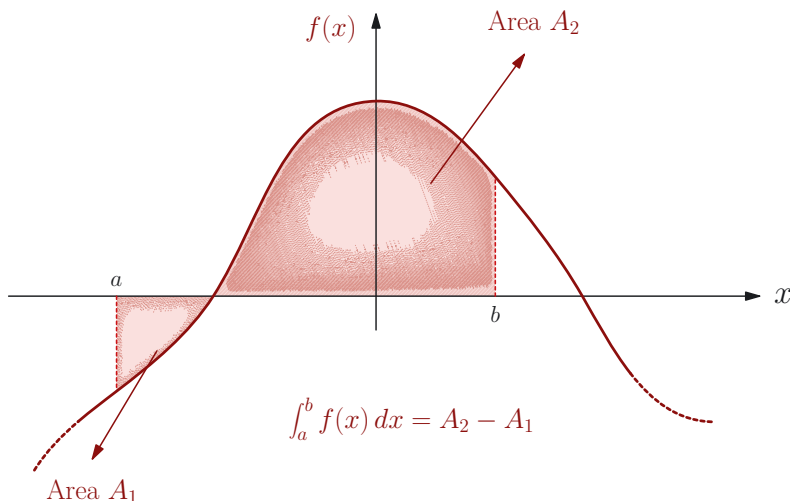


Figure 5: The integral of $f(x)$ from $x = a$ to $x = b$ also equals $F(b) - F(a)$, where $F(x)$ is an antiderivative of $f(x)$.

Fundamental Theorem of Calculus The integral of a function $f(x)$ over an interval $[a, b]$ equals the change in any antiderivative $F(x)$ between the endpoints of the interval:

$$\int_a^b f(x) dx = F(b) - F(a) \equiv F|_a^b$$

Example. Calculate the integral of $f(x) = \sin(2x) - x$ over the interval $x \in [0, \pi]$.

An antiderivative of $f(x)$ is $F(x) = -\frac{1}{2} (\cos(2x) + x^2)$. Hence,

$$\int_0^\pi [\sin(2x) - x] dx = -\frac{1}{2} [\cos(2\pi) + \pi^2 - \cos(0)] = -\frac{\pi^2}{2}$$

Relationship between integral and derivative

$$\frac{d}{du} \left(\int_a^u f(x) dx \right) = f(u)$$

Proof. For every $a \in \mathbb{R}$, we have $I_a(u) \equiv \int_a^u f(x) dx = F(u) - F(a)$ by the definition of an antiderivative F of f . Differentiating $I_a(u)$ with respect to u gives $\frac{dI_a(u)}{du} = \frac{dF(u)}{du} - \frac{dF(a)}{du} = \frac{dF(u)}{du} = f(u)$. $\square$

Integration by parts

$$\int_a^b f(x) \frac{dg(x)}{dx} dx = - \int_a^b \frac{df(x)}{dx} g(x) dx + f(x)g(x)|_a^b$$

Change of variable (substitution) We call $x = x(t)$ a regular change of variable between the intervals $[x_A, x_B]$ and $[t_A, t_B]$ if $x(t)$ is differentiable and defines a bijection from $[t_A, t_B]$ onto $[x_A, x_B]$. A change of variable transforms an integral according to

$$\int_{x_A}^{x_B} f(x) dx = \int_{t_A}^{t_B} \left(f(x(t)) \frac{dx(t)}{dt} \right) dt,$$

where $x_A = x(t_A)$ and $x_B = x(t_B)$.

COMMON ANTIDERIVATIVES

For any $a \neq 0$ and $n \in \mathbb{R} \setminus \{-1\}$,

$$\circ \int a dx = ax + c$$

$$\circ \int (\sin(ax)) dx = -\frac{\cos(ax)}{a} + c$$

$$\circ \int (ax^n) dx = \frac{a}{n+1} x^{n+1} + c$$

$$\circ \int (\cos(ax)) dx = \frac{\sin(ax)}{a} + c$$

$$\circ \int (e^{ax}) dx = \frac{e^{ax}}{a} + c$$

$$\circ \int \frac{1}{x} dx = \ln(|ax|) + c \quad (x \neq 0)$$

where c is a constant.

💡 What is differential calculus used for in mechanics?

Differential calculus is the mathematical tool used to study how quantities change.

In kinematics, it describes a body's motion through its position, velocity (rate of change of position) and acceleration (rate of change of velocity) as functions of time.

In dynamics, it relates force to changes in motion.

1.4 Geometry

Plane angle An angle *plane* expressed in radians (rad) is equal to the ratio of two lengths: the length of the circle arc s and the radius r (see Figure 6, on the left), i.e.

$$\theta \stackrel{\text{def}}{=} \frac{s}{r} \quad \theta \in [0, 2\pi]$$

The conversion of the angles to degrees and radians is as follows

$$\theta [\text{deg}] = \frac{180}{\pi} \times \theta [\text{rad}] \quad \Leftrightarrow \quad \theta [\text{rad}] = \frac{\pi}{180} \times \theta [\text{deg}]$$

Example. A right angle (90°) equals $\pi/2$ rad, and a straight angle (180°) equals π rad.

Fact. The angles of a plane triangle add up to 180° .

Solid angle An angle *solid* expressed in steradians (sr) is the ratio of two areas: the intercepted area A on a sphere of radius r , and the square of the radius, r^2 (see Figure 6, right):

$$\Omega \stackrel{\text{def}}{=} \frac{A}{r^2} \quad \Omega \in [0, 4\pi]$$

Example. The solid angle subtended by a hemisphere (half a sphere) is $\Omega = 2\pi$ sr.

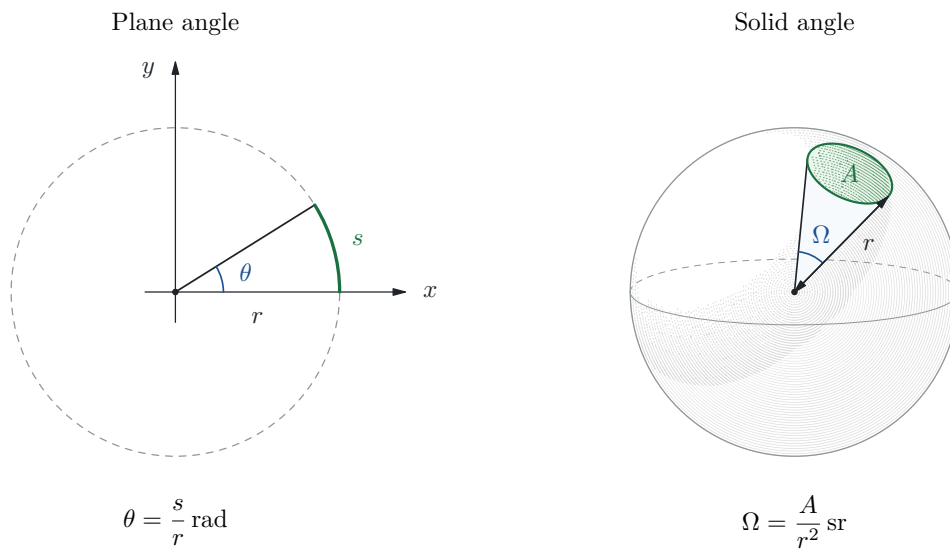


Figure 6: Left: angle *plane* in radians as the ratio of two lengths, $\theta = (s/r)$ rad. The same angle expressed in degrees is $\theta = \left(\frac{s}{r} \frac{180}{\pi}\right)^\circ$. Right: angle *solid* in steradians as the ratio of two areas, $\Omega = (A/r^2)$ sr.

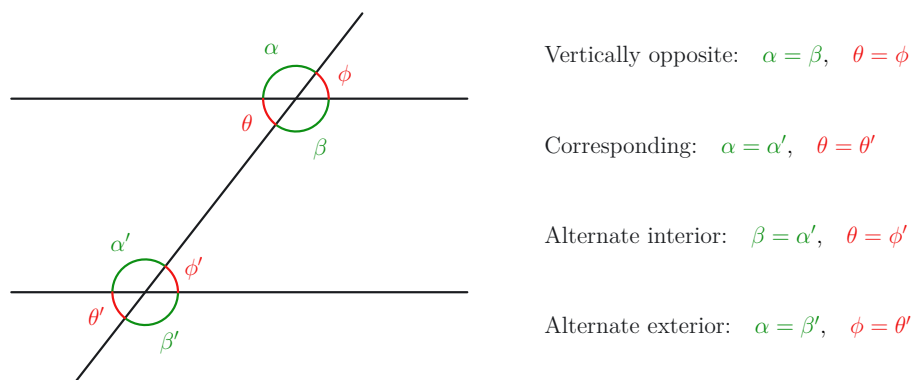


Figure 7: Relationships between angles.

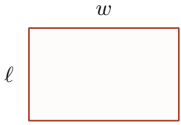
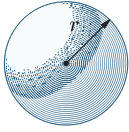
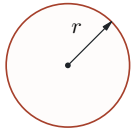
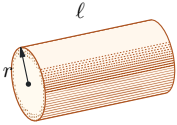
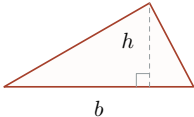
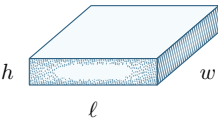
Shape	Area	Shape	Area or volume
 Rectangle	$\text{Area} = \ell w$	 Sphere	$\text{Surface} = 4\pi r^2$ $\text{Volume} = \frac{4}{3}\pi r^3$
 Circle	$\text{Area} = \pi r^2$ $\text{Circumference} = 2\pi r$	 Cylinder	$\text{Side surface} = 2\pi r\ell$ $\text{Volume} = \pi r^2\ell$
 Triangle	$\text{Area} = \frac{1}{2}bh$	 Rectangular box	$\text{Surface} = 2(\ell h + \ell w + hw)$ $\text{Volume} = \ell wh$

Figure 8: Areas and volumes of common shapes.

1.5 Trigonometry

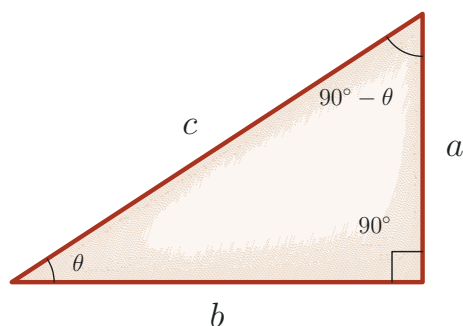
Trigonometric functions

In a *right-angled* triangle, ratios of side lengths define the elementary trigonometric functions $\sin$, $\cos$ and $\tan$ (see Figure 9).

a = opposite side

b = adjacent side

c = hypotenuse



$$\sin \theta \stackrel{\text{def}}{=} \frac{a}{c} = \frac{\text{opposite side}}{\text{hypotenuse}}$$

$$\cos \theta \stackrel{\text{def}}{=} \frac{b}{c} = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

$$\tan \theta \stackrel{\text{def}}{=} \frac{a}{b} = \frac{\text{opposite side}}{\text{adjacent side}}$$

$$\text{Pythagoras: } c^2 = a^2 + b^2$$

Figure 9: Definitions of $\sin$, $\cos$ and $\tan$ as ratios of side lengths in a right-angled triangle.

The functions $\sin$ and $\cos$ are periodic with period 2π (angles in radians; see Figure 10) and oscillate between ± 1 . The function $\tan$ has fundamental period π and is unbounded. We have $\tan \theta = \frac{\sin \theta}{\cos \theta}$.

α [deg]	0	30°	45°	60°	90°
α [rad]	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin \alpha$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \alpha$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\text{tg } \alpha$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	$\nexists$
$\text{cotg } \alpha$	$\nexists$	$\sqrt{3}$	1	$\frac{\sqrt{3}}{3}$	0

Table 1: Trigonometric function values for a few specific angles.

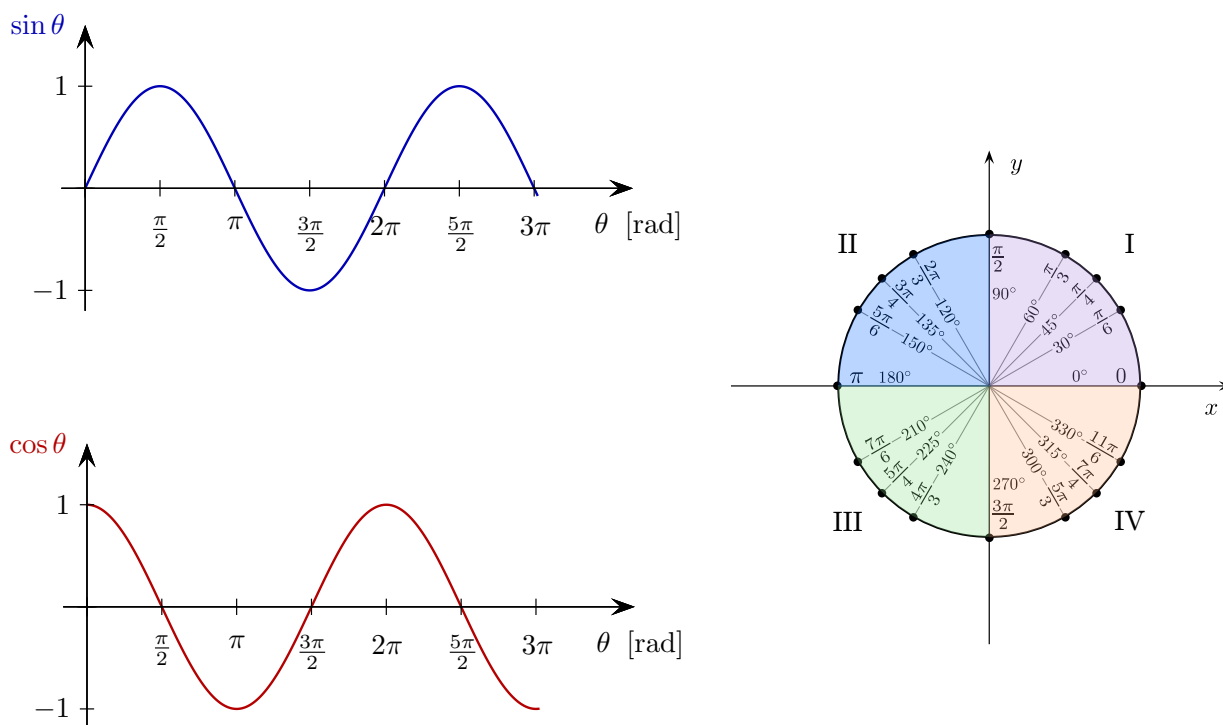


Figure 10: Graphs of the trigonometric functions $\sin$ and $\cos$ and the unit circle and its quadrants. In quadrant I, $\sin \theta > 0$ and $\cos \theta > 0$. In quadrant II, $\sin \theta > 0$ and $\cos \theta < 0$. In quadrant III, $\sin \theta < 0$ and $\cos \theta < 0$. In quadrant IV, $\sin \theta < 0$ and $\cos \theta > 0$.

TRIGONOMETRIC EQUATIONS

$$\sin \theta = \sin \alpha \Leftrightarrow \theta = \alpha + 2k\pi \quad \text{or} \quad \theta = \pi - \alpha + 2k\pi, \quad k \in \mathbb{Z}$$

$$\cos \theta = \cos \alpha \Leftrightarrow \theta = \alpha + 2k\pi \quad \text{or} \quad \theta = -\alpha + 2k\pi, \quad k \in \mathbb{Z}$$

$$\tan \theta = \tan \alpha \Leftrightarrow \theta = \alpha + k\pi, \quad k \in \mathbb{Z}$$

TRIGONOMETRIC IDENTITIES

$$\begin{aligned}
\circ \sin^2 \theta + \cos^2 \theta &= 1 & \circ \cos(\theta \pm \varphi) &= \cos \theta \cos \varphi \mp \sin \theta \sin \varphi & \circ \tan(2\theta) &= \frac{2 \tan \theta}{1 - \tan^2 \theta} \\
\circ \sin^2(\theta/2) &= (1 - \cos \theta)/2 & \circ \sin(\theta \pm \varphi) &= \sin \theta \cos \varphi \pm \cos \theta \sin \varphi & \circ \cos(2\theta) &= \cos^2 \theta - \sin^2 \theta \\
\circ \cos^2(\theta/2) &= (1 + \cos \theta)/2 & \circ \left| \tan\left(\frac{\theta}{2}\right) \right| &= \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} & \circ \sin(2\theta) &= 2 \sin \theta \cos \theta
\end{aligned}$$

In the identity containing a square root, $\cos \theta \neq -1$, and the right-hand side gives the absolute value of $\tan(\theta/2)$; its sign is determined by the quadrant containing $\theta/2$.

Supplementary $(-)$ and antisupplementary $(+)$ angles :

$$\circ \sin(\pi \pm \theta) = \mp \sin \theta \quad \circ \cos(\pi \pm \theta) = -\cos \theta \quad \circ \tan(\pi \pm \theta) = \pm \tan \theta \quad \circ \cot(\pi \pm \theta) = \pm \cot \theta$$

Complementary angles :

$$\circ \sin(\pi/2 - \theta) = \cos \theta \quad \circ \cos(\pi/2 - \theta) = \sin \theta \quad \circ \tan(\pi/2 - \theta) = \cot \theta \quad \circ \cot(\pi/2 - \theta) = \tan \theta$$

Opposite angles :

$$\circ \sin(-\theta) = -\sin \theta \quad \circ \cos(-\theta) = \cos \theta \quad \circ \tan(-\theta) = -\tan \theta \quad \circ \cot(-\theta) = -\cot \theta$$

Sum-to-product identities (Simpson's formulas) :

$$\begin{aligned}
\circ \cos \theta - \cos \varphi &= -2 \sin\left(\frac{\theta + \varphi}{2}\right) \sin\left(\frac{\theta - \varphi}{2}\right) \\
\circ \cos \theta + \cos \varphi &= 2 \cos\left(\frac{\theta + \varphi}{2}\right) \cos\left(\frac{\theta - \varphi}{2}\right) \\
\circ \sin \theta \pm \sin \varphi &= 2 \sin\left(\frac{\theta \pm \varphi}{2}\right) \cos\left(\frac{\theta \mp \varphi}{2}\right)
\end{aligned}$$

1.6 Vector Calculus

Vectors and vector additions

A free vector is a geometric object defined by a magnitude and a direction, including its sense. A bound vector is a free vector with a fixed initial point. Vector addition is defined by the parallelogram rule or by placing vectors head to tail. It has the following properties:

VECTOR ADDITION: PROPERTIES

- *Associativity* : $\vec{A} + (\vec{B} + \vec{C}) = (\vec{A} + \vec{B}) + \vec{C}$
- *Additive identity* : $\vec{A} + \vec{0} = \vec{0} + \vec{A} = \vec{A}$
- *Additive inverse* : $\vec{A} + (-\vec{A}) = \vec{A} - \vec{A} = \vec{0}$
- *Commutativity* : $\vec{A} + \vec{B} = \vec{B} + \vec{A}$

The length of a vector is also called its norm, denoted by $|\vec{A}|$. Multiplication by a scalar $\lambda \in \mathbb{R}$ produces a vector whose length is multiplied by $|\lambda|$ and whose direction is reversed if $\lambda < 0$.

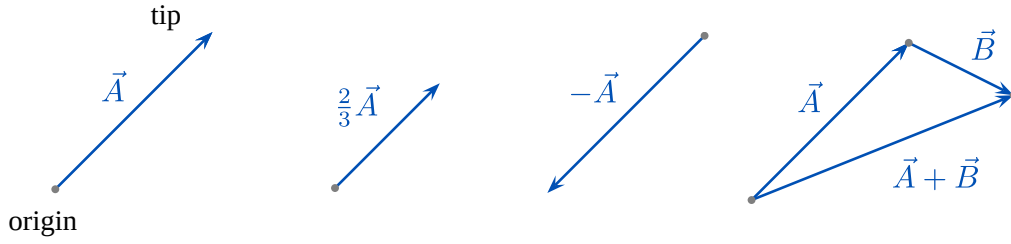


Figure 11: A bound vector $\vec{A}$, scalar multiplication of this vector, and vector addition with a vector $\vec{B}$.

Basis and components

Any vector in two-dimensional Euclidean space $\mathbb{R}^2$ can be decomposed as

$$\vec{A} = A_x \vec{e}_x + A_y \vec{e}_y,$$

where $\vec{e}_x$ and $\vec{e}_y$ are unit vectors (of length 1) pointing along the positive Ox and Oy axes (see Figure 12). Vectors $\vec{e}_x$ and $\vec{e}_y$ form a *basis* of $\mathbb{R}^2$. Similarly, any vector in three-dimensional Euclidean space $\mathbb{R}^3$ can be decomposed as

$$\vec{A} = A_x \vec{e}_x + A_y \vec{e}_y + A_z \vec{e}_z,$$

where $(\vec{e}_x, \vec{e}_y, \vec{e}_z)$ forms a basis of $\mathbb{R}^3$ (see Figure 13). The numbers A_x, A_y and A_z are called the *cartesian components* of $\vec{A}$, and we write $\vec{A} = (A_x, A_y, A_z)$. In component form, multiplication of a vector $\vec{A}$ by a scalar λ is simply $\lambda \vec{A} = (\lambda A_x, \lambda A_y, \lambda A_z)$.

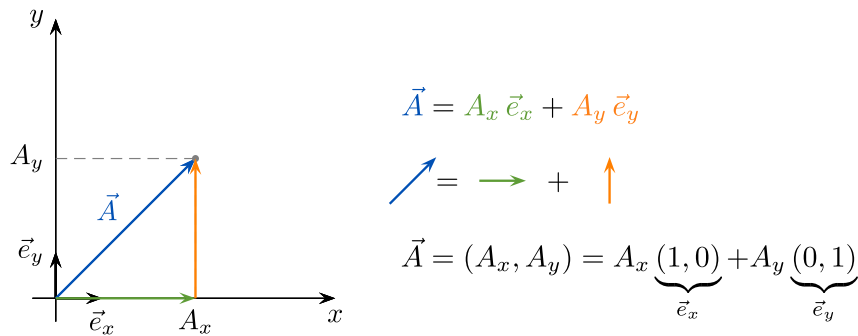


Figure 12: Decomposition of a vector $\vec{A}$ in $\mathbb{R}^2$ along basis vectors $(\vec{e}_x, \vec{e}_y)$, with Cartesian components (A_x, A_y) .

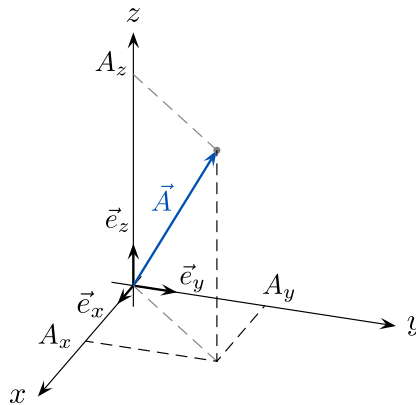


Figure 13: Decomposition of a vector $\vec{A}$ in $\mathbb{R}^3$ along basis vectors $(\vec{e}_x, \vec{e}_y, \vec{e}_z)$: $\vec{A} = A_x \vec{e}_x + A_y \vec{e}_y + A_z \vec{e}_z$.

Vector addition can be done directly on vector components since we have in the case of two vectors of $\mathbb{R}^2$ (see

Figure 14)

$$\begin{aligned}
\vec{A} + \vec{B} &= A_x \vec{e}_x + A_y \vec{e}_y + B_x \vec{e}_x + B_y \vec{e}_y \\
&= (A_x + B_x) \vec{e}_x + (A_y + B_y) \vec{e}_y \\
&= (A_x + B_x, A_y + B_y)
\end{aligned}$$

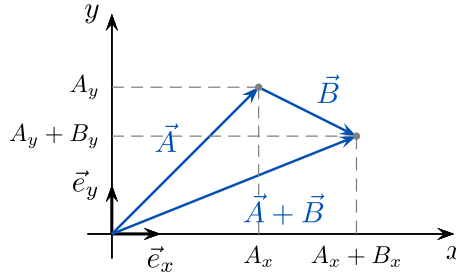


Figure 14: Vector addition in terms of Cartesian components.

Dot product

The dot product of two vectors $\vec{A}$ and $\vec{B}$, denoted by $\vec{A} \cdot \vec{B}$, is the scalar (positive, zero or negative)

$$\vec{A} \cdot \vec{B} \stackrel{\text{def}}{=} A_x B_x + A_y B_y + A_z B_z. \quad (\text{scalar})$$

An equivalent geometric definition is

$$\vec{A} \cdot \vec{B} \stackrel{\text{def}}{=} A B \cos \varphi,$$

where φ is the angle between vectors $\vec{A}$ and $\vec{B}$, with norms A and B (see Figure 15).

The dot product satisfies the following properties:

DOT PRODUCT: PROPERTIES

- $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$ (*commutativity*)
- $\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$ (*Distributivity*)
- $\vec{A} \cdot \vec{B} = 0 \Leftrightarrow \vec{A} \perp \vec{B} \quad (\vec{A}, \vec{B} \neq 0)$ (*criterion of orthogonality*)
- $\vec{A} \cdot \vec{A} = A^2 \Rightarrow |\vec{A}| = A = \sqrt{\vec{A} \cdot \vec{A}}$ (*norm of $\vec{A}$*)
- $A_x = \vec{A} \cdot \vec{e}_x, \quad A_y = \vec{A} \cdot \vec{e}_y, \quad A_z = \vec{A} \cdot \vec{e}_z$ (*components of $\vec{A}$ obtained by projection - see figure 15*)
- $|\vec{A} \cdot \vec{B}| \leq |\vec{A}| |\vec{B}|$ (*cauchy-Schwarz inequality*)

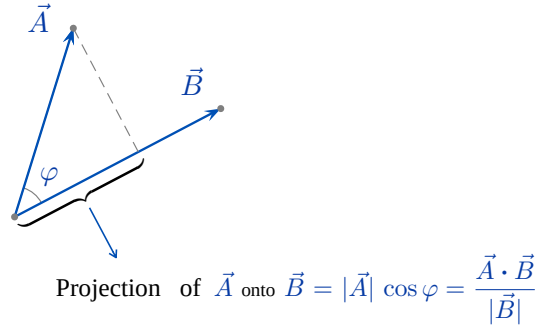


Figure 15: Using the dot product to calculate the projection of a vector onto an axis.

Cross product

The cross product of $\vec{A}$ by $\vec{B}$, noted $\vec{A} \times \vec{B}$, is the vector

$$\vec{A} \times \vec{B} \stackrel{\text{def}}{=} (A_y B_z - B_y A_z) \vec{e}_x + (A_z B_x - B_z A_x) \vec{e}_y + (A_x B_y - B_x A_y) \vec{e}_z \quad (\text{vector})$$

An equivalent geometric definition is as follows: the cross product of two vectors $\vec{A}$ and $\vec{B}$ is a vector whose norm equals the area of the parallelogram spanned by $\vec{A}$ and $\vec{B}$. It is perpendicular to both $\vec{A}$ and $\vec{B}$, with its direction given by the right-hand rule (see the caption of Figure 16).

The cross product satisfies the following properties:

CROSS PRODUCT: PROPERTIES

- $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$ (*anti-commutativity*)
- $\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$ (*Distributivity*)
- $(\lambda \vec{A}) \times \vec{B} = \lambda (\vec{A} \times \vec{B}) = \vec{A} \times (\lambda \vec{B})$ (*multiplication by a scalar*)
- $\vec{A} \times \vec{B} = 0 \Leftrightarrow \vec{A} \parallel \vec{B} \quad (\vec{A}, \vec{B} \neq 0)$ (*criterion of parallelism*)
- $|\vec{A} \times \vec{B}| = |\vec{A}| |\vec{B}| \sin \varphi$ (*norm*)

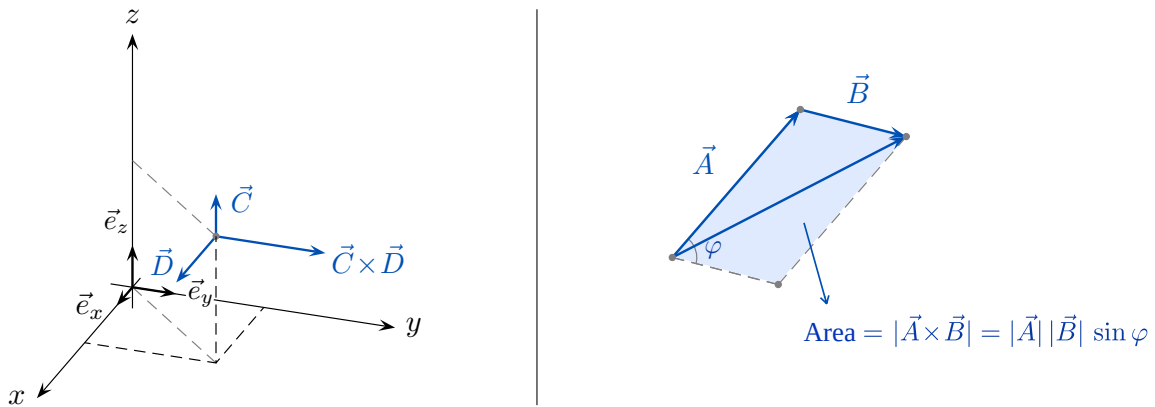


Figure 16: Two illustrations of the cross product. Left: $\vec{C} \parallel \vec{e}_z$ and $\vec{D} \parallel \vec{e}_x$ lie in a plane parallel to the Oxz plane. Their cross product $\vec{C} \times \vec{D}$ is perpendicular to this plane. Its direction follows the right-hand rule: curl the fingers of your right hand from the first vector towards the second through the smaller angle between $\vec{C}$ and $\vec{D}$ (between 0 and 180°); your thumb then points along $\vec{C} \times \vec{D}$. Right: the area of the parallelogram spanned by $\vec{A}$ and $\vec{B}$ is its height $h = |\vec{A}| \sin \varphi$ times its base $b = |\vec{B}|$, namely $|\vec{A}| |\vec{B}| \sin \varphi = |\vec{A} \times \vec{B}|$.

Example. Calculate the dot and cross products of vectors $\vec{A} = 2\vec{e}_x + 2\vec{e}_y - 3\vec{e}_z$ and $\vec{B} = -2\vec{e}_x + \vec{e}_y - \vec{e}_z$. Show that $\vec{A} \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\vec{A} \times \vec{B}) = 0$ and explain why.

We have $\vec{A} \cdot \vec{B} = 1$ and $\vec{A} \times \vec{B} = \vec{e}_x + 8\vec{e}_y + 6\vec{e}_z$. The scalar triple products $\vec{A} \cdot (\vec{A} \times \vec{B})$ and $\vec{B} \cdot (\vec{A} \times \vec{B})$ vanish because the cross product $\vec{A} \times \vec{B}$ is perpendicular to both $\vec{A}$ and $\vec{B}$, and the dot product of perpendicular vectors is zero.

Vector analysis

If a vector quantity $\vec{A}$ depends on time, it may be regarded as a vector-valued function $\vec{A}(t)$, where t is the time variable. Its derivative $\frac{d\vec{A}(t)}{dt}$ with respect to t is then

$$\frac{d\vec{A}(t)}{dt} = \frac{d}{dt}(A_x(t)\vec{e}_x + A_y(t)\vec{e}_y + A_z(t)\vec{e}_z) = \frac{dA_x(t)}{dt}\vec{e}_x + \frac{dA_y(t)}{dt}\vec{e}_y + \frac{dA_z(t)}{dt}\vec{e}_z,$$

since the basis vectors $\vec{e}_x, \vec{e}_y, \vec{e}_z$ do not depend on time. In Cartesian components, we have

$$\frac{d\vec{A}(t)}{dt} = \left(\frac{dA_x(t)}{dt}, \frac{dA_y(t)}{dt}, \frac{dA_z(t)}{dt} \right).$$

The usual differentiation rules readily give

$$\frac{d}{dt}(\vec{A}(t) \cdot \vec{B}(t)) = \frac{d\vec{A}(t)}{dt} \cdot \vec{B}(t) + \vec{A}(t) \cdot \frac{d\vec{B}(t)}{dt}$$

and

$$\frac{d}{dt}(\vec{A}(t) \times \vec{B}(t)) = \frac{d\vec{A}(t)}{dt} \times \vec{B}(t) + \vec{A}(t) \times \frac{d\vec{B}(t)}{dt}$$

Example. Show that a unit vector^a $\vec{e}(t)$ and its derivative relative to time are orthogonal.

^aA unit vector has norm 1 for every t .

Two vectors are perpendicular if and only if their dot product is zero. Since $\vec{e}(t)$ is a unit vector, $\vec{e}(t) \cdot \vec{e}(t) = 1$ holds for every t . Differentiating with respect to time gives

$$0 = \frac{d}{dt}(\underbrace{\vec{e}(t) \cdot \vec{e}(t)}_{=1}) = \frac{d\vec{e}(t)}{dt} \cdot \vec{e}(t) + \vec{e}(t) \cdot \frac{d\vec{e}(t)}{dt} = 2\vec{e}(t) \cdot \frac{d\vec{e}(t)}{dt} \Rightarrow \vec{e}(t) \perp \frac{d\vec{e}(t)}{dt}$$

 What are dot and cross products used for in mechanics?

The dot product: projections, angles, work, ...

The dot product measures the component of a vector along a given direction. It appears, for example, in the power and work of a force, describing an effect “along” a direction.

The cross product: describe orientation, rotation effects, ...

The cross product describes rotation and orientation in space. It appears in torque and angular momentum, describing an effect “about” an axis.

CHAPTER 2

UNITS AND DIMENSIONAL ANALYSIS

2.1 Physical Quantities and Base Units

A physical quantity is generally expressed as a number multiplied by a unit. The base quantities in the *International System of Units* (SI) are length, mass, time, electric current, thermodynamic temperature, amount of substance and luminous intensity (see the table below for their symbols). Their SI base units are the metre (m), kilogram (kg), second (s), ampere (A), kelvin (K), mole (mol) and candela (cd), respectively. Metres, kilograms and seconds appear mainly in mechanics; amperes in electromagnetism; kelvins in thermodynamics; moles in chemistry; and candelas in optics. These base units are defined below.

Base quantity		SI base unit	
Name	Symbol	Name	Symbol
length	$\ell, x, r, \dots$	metre	m
mass	m	kilogram	kg
time, duration	t	second	s
electric current	I, i	ampere	A
thermodynamic temperature	T	kelvin	K
amount of substance	n	mole	mol
luminous intensity	I_v	candela	cd

Table 1: SI base units and associated symbols.

The second

Définition. The second is the duration of 9,192,631,770 periods of radiation corresponding to the transition between the two hyperfine levels of the ground state of the undisturbed caesium 133 atom.

The metre

Définition. The metre is the length of the path travelled by light in vacuum during a time interval of $1/299\,792\,458$ s.

The kilogram

Définition. The kilogram, symbol kg, is the unit of mass of the SI. It is defined by taking the fixed numerical value of the Planck constant, h , equal to $6.626\,070\,15 \times 10^{-34}$ when expressed in Js, unit equal to $\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-1}$, the metre and the second being defined according to c and $\Delta\nu_{\text{Cs}}$.

The ampere

Définition. The ampere, symbol A, is the SI unit of electric current. It is defined by fixing the numerical value of the elementary charge e at $1.602\,176\,634 \times 10^{-19}$ when expressed in C, which equals A.s, with the second defined in terms of $\Delta\nu_{\text{Cs}}$.

The kelvin

Définition. The kelvin, symbol K, is the thermodynamic temperature unit of the SI. It is defined by taking the fixed numerical value of the Boltzmann constant, k , equal to $1.380\,649 \times 10^{-23}$ when expressed in $\text{J}\cdot\text{K}^{-1}$, unit equal to $\text{kg}\cdot\text{m}^2\cdot\text{s}^{-2}\text{K}^{-1}$, the kilogram, meter and second being defined according to h , c and $\Delta\nu_{\text{Cs}}$.

The Mole

Définition. The mole, symbol mol, is the SI unit of amount of substance. One mole contains exactly $6.022\,140\,76 \times 10^{23}$ elementary entities. This number, known as the Avogadro number, is the fixed numerical value of the Avogadro constant, N_A , when expressed in mol^{-1} . The amount of substance, symbol n , measures the number of specified elementary entities in a system. An elementary entity may be an atom, molecule, ion, electron or any other specified particle or group of particles.

The Candela

Définition. The candela, cd symbol, is the unit of the SI of luminous intensity in a given direction. It is defined by taking the fixed numerical value of the luminous efficacy of a monochromatic radiation of frequency 540×10^{12} Hz, K_{cd} , equal to 683 when expressed in $\text{lm}\,\text{W}^{-1}$, unit equal to $\text{cd}\cdot\text{sr}\cdot\text{W}^{-1}$, or $\text{cd}\cdot\text{sr}\cdot\text{kg}^{-1}\cdot\text{m}^{-2}\cdot\text{s}^3$, the kilogram, meter and second being defined according to h , c and $\Delta\nu_{\text{Cs}}$.

Derived units

SI derived units are formed from products of powers of the base units, following the algebraic relations that define derived quantities in terms of base quantities. Some examples are given in the table below.

Derived quantity		Coherent SI derived unit	
Name	Symbol	Name	Symbol
area	A	square metre	m^2
volume	V	cubic metre	m^3
velocity	v	meter per second	m/s
acceleration	a	meter per square second	m/s^2
wavenumber	$\sigma, \tilde{\nu}$	reciprocal metre	m^{-1}
density	ρ	kilogram per cubic metre	kg/m^3
surface mass density	ρ_A	kilogram per square metre	kg/m^2
specific volume	ν	cubic metre per kilogram	m^3/kg
current density	j	ampere per square metre	A/m^2
magnetic field	H	ampere per metre	A/m
amount concentration	c	mole per cubic metre	mol/m^3
mass concentration	ρ, γ	kilogram per cubic metre	kg/m^3
luminance	L_v	candela per square metre	cd/m^2
refractive index	n	one	1
relative permeability	μ_r	one	1

Table 2: A few SI-derived units.

Example. In one dimension, the instantaneous velocity of a particle is defined by $v = dx/dt$, where x is its position (in m) and t is time (in s). This relation gives the derived unit m/s for velocity. Other units can of course express the same physical quantity. For example, $v = 25$ m/s in SI units can also be written as $v = 90$ km/h $\approx 7.884 \times 10^{12}$ cm/century, where km/h and cm/century are non-SI units.

Common non-SI units

The SI is recommended for use in science, technology, engineering and commerce. However, some non-SI units remain widely used in scientific, technical and commercial publications. Certain common non-SI units are therefore accepted for use with the SI; the table below gives some examples.

Quantity	Unit name	Unit symbol	Value in SI units
time	minute	min	1 min = 60 s
	hour	h	1 h = 60 min = 3600 s
	day	d	1 d = 24 h = 86 400 s
plane angle	degree	°	1° = ($\pi/180$) rad
	minute	'	1' = (1/60)° = ($\pi/10\,800$) rad
	second	''	1'' = (1/60)' = ($\pi/648\,000$) rad
area	hectare	ha	1 ha = 1 hm ² = 10 ⁴ m ²
volume	liter	L, l	1 L = 1 l = 1 dm ³ = 10 ³ cm ³ = 10 ⁻³ m ³
mass	tonne	t	1 t = 10 ³ kg

Table 3: A few common units outside the SI.

Remark. A plane angle is a dimensionless quantity (its SI derived unit is one). The radian is a special name for the number one, used to express a plane angle as the ratio of two lengths (see Figure 1, left). In practice, the symbol for the unit one is usually omitted when stating values of dimensionless quantities.

Remark. A solid angle is a dimensionless quantity. The steradian is used to express a solid angle as the ratio of two areas (see figure 1, to the right).

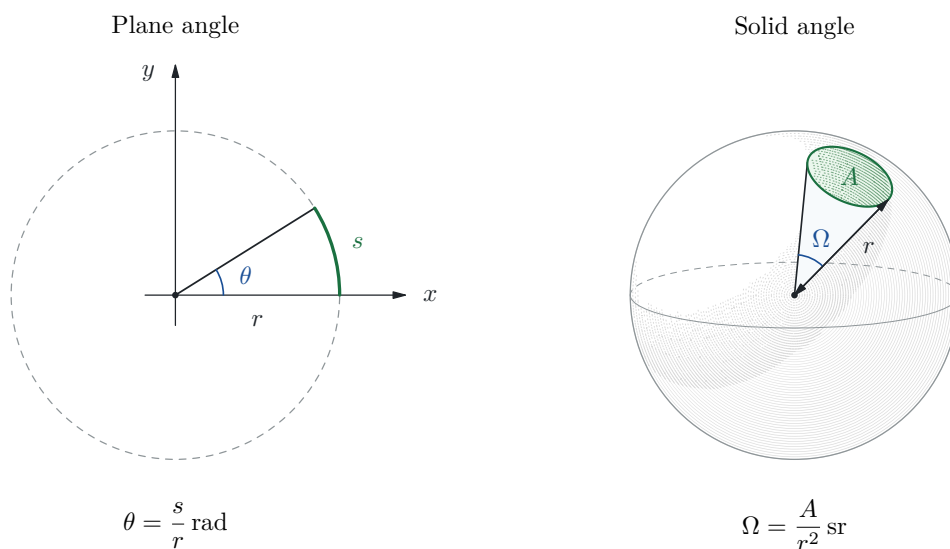


Figure 1: Left: definition of an angle *plane* in radians as the ratio of two lengths. The same angle expressed in degrees is $\theta = \left(\frac{s}{r} \frac{180}{\pi}\right)^\circ$. Right: angle *solid* in steradians as the ratio of two areas, $\Omega = (A/r^2) \text{ sr}$.

Prefixes

The group consisting of a prefix symbol (see table below) and a unit symbol constitutes a new inseparable unit symbol forming a multiple or submultiple of the unit in question.

Factor	Name	Symbol	Factor	Name	Symbol
10^1	deca	da	10^{-1}	deci	d
10^2	hecto	h	10^{-2}	centi	c
10^3	kilo	k	10^{-3}	milli	m
10^6	mega	M	10^{-6}	micro	μ
10^9	giga	G	10^{-9}	nano	n
10^{12}	tera	T	10^{-12}	pico	p
10^{15}	peta	P	10^{-15}	femto	f
10^{18}	exa	E	10^{-18}	atto	a
10^{21}	zetta	Z	10^{-21}	zepto	z
10^{24}	yotta	Y	10^{-24}	yocto	y
10^{27}	ronna	R	10^{-27}	ronto	r
10^{30}	quetta	Q	10^{-30}	quecto	q

Table 4: Prefixes used with SI.

Remark. The prefixes quetta, ronna, ronto and quecto were added in 2022 by the 27th CGPM. For the official text, see Resolution 3 of the 27th CGPM: <https://www.bipm.org/en/cgpm-2022/resolution-3>.

Example. $2.3 \text{ cm}^3 = 2.3 (\text{cm})^3 = 2.3 (10^{-2} \text{ m})^3 = 2.3 \times 10^{-6} \text{ m}^3$.

$1 \text{ cm}^{-1} = 1 (\text{cm})^{-1} = 1 (10^{-2} \text{ m})^{-1} = 10^2 \text{ m}^{-1} = 100 \text{ m}^{-1}$.

2.2 Dimensional analysis

The dimension of a quantity does not depend on the unit chosen. Quantities of the same kind necessarily have the same dimension, but quantities with the same dimension are not necessarily of the same kind. For example, the height and diameter of a cylinder both have the dimension of length.

PRINCIPLE OF DIMENSIONAL ANALYSIS

Only physical quantities of the same kind — hence of the same dimension — can be compared, equated, added or subtracted; their numerical values must first be converted to a common unit where necessary.

Dimensional analysis thus ensures that an equality between physical quantities can *a priori* have a physical meaning. Here are some examples that illustrate one of the purposes of dimensional analysis.

Example. Look for the dependence of the period of oscillation T of a pendulum on the relevant physical parameters.

The relevant physical parameters are the length of the pendulum ℓ , its mass m , the acceleration of gravity g and the initial angular amplitude θ_0 , which is dimensionless. We start from the hypothesis $T = m^\alpha g^\beta \ell^\gamma F(\theta_0)$ where F is a dimensionless function. We can then write the following dimensional equation, where $[x]$ designates the dimension of x :

$$\begin{aligned} [T] &= [m]^\alpha [g]^\beta [\ell]^\gamma \\ \mathsf{T} &= \mathsf{M}^\alpha (\mathsf{LT}^{-2})^\beta \mathsf{L}^\gamma \\ &= \mathsf{M}^\alpha \mathsf{L}^{\beta+\gamma} \mathsf{T}^{-2\beta} \\ \Rightarrow \alpha &= 0, \quad \beta = -\frac{1}{2}, \quad \gamma = \frac{1}{2} \end{aligned}$$

Dimensional analysis alone gives

$$T = \sqrt{\frac{\ell}{g}} F(\theta_0),$$

where F is a dimensionless function that this analysis cannot determine. The period is therefore independent of mass. In the small-angle approximation, $|\theta_0| \ll 1$, we have $F(\theta_0) \simeq 2\pi$, giving

$$T \simeq 2\pi \sqrt{\frac{\ell}{g}}.$$

Example. The Planck length is the natural length scale of a hypothetical theory unifying relativity, gravitation and quantum mechanics. It is constructed from the fundamental constants c (speed of light in vacuum, in m/s), G (universal gravitational constant, in $\text{m}^3\text{kg}^{-1}\text{s}^{-2}$) and $\hbar$ (reduced Planck constant $\hbar/2\pi$, in $\text{m}^2\text{kg}\cdot\text{s}^{-1}$). Determine an expression for the Planck length and calculate its value using Table 5.

Dimensional analysis gives the Planck length

$$\ell_{\text{Planck}} = \sqrt{\frac{\hbar G}{c^3}} \approx 1.6 \times 10^{-35} \text{ m},$$

which may be compared with a typical atomic size ($\sim 10^{-10} \text{ m}$) or proton size ($\sim 10^{-15} \text{ m}$).

Fact. The $\exp$, $\ln$, $\sin$, $\cos$ and $\tan$ functions accept as an argument only numbers without units¹.

Example. $\sin \theta$, where θ is expressed in radians;

$\cos(x/\ell)$, where x and ℓ have the dimension of length;

$e^{-t/T}$, where t and T have the dimension of time;

$\tan(ft)$, where f is a frequency and t is a time.

2.3 Fundamental Constants

The table below lists selected fundamental physical constants and their recommended CODATA 2022 values; see <https://physics.nist.gov/constants> for details.

¹These are also called dimensionless numbers.

QUANTITY	SYMBOL	VALUE	STATUS
Speed of light in the vacuum	c	$2.99792458 \times 10^8 \text{ m} \cdot \text{s}^{-1}$	exact
Universal gravitational constant	G	$6.67430(15) \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2}$	
Vacuum permeability	μ_0	$1.25663706127(20) \times 10^{-6} \text{ V} \cdot \text{s} \cdot \text{A}^{-1} \cdot \text{m}^{-1}$	
Vacuum permittivity	ϵ_0	$8.8541878188(14) \times 10^{-12} \text{ A} \cdot \text{s} \cdot \text{V}^{-1} \cdot \text{m}^{-1}$	$\frac{1}{c^2 \mu_0}$
Elementary charge	e	$1.602176634 \times 10^{-19} \text{ C}$	exact
Electron rest mass	m_e	$9.1093837139(28) \times 10^{-31} \text{ kg}$	
Avogadro constant	N_A	$6.02214076 \times 10^{23} \text{ mol}^{-1}$	exact
Boltzmann Constant	k_B	$1.380649 \times 10^{-23} \text{ J} \cdot \text{K}^{-1}$	exact
Planck Constant	h	$6.62607015 \times 10^{-34} \text{ J s}$	exact

Table 5: CODATA 2022 values of some basic constants.

CHAPTER 3

KINEMATICS

Kinematics *kinematics* is the branch of mechanics that describes the motion of bodies without considering its causes. It describes the trajectory of a body, usually treated as a point particle, through its position, velocity and acceleration as functions of time. Kinematics does not consider the masses of bodies or the forces acting on them.

Space and time In order to describe the movement of a body, it is necessary to make hypotheses about the nature of *space* and *time*. Classical mechanics, a theory developed by Isaac Newton, presupposes the existence of a space with 3 dimensions and a universal and absolute time, that is, independent of matter. Space and time are the framework within which physical processes take place and would exist even if there were nothing. They constitute the " theatre " of events.¹ In other words, this amounts to assuming that time is made of moments and flows in the same way for all observers² and that space, made of points, has the same structure regardless of the movement of observers. These hypotheses, as natural as they may appear, will be abandoned by Einstein with the theory of relativity. Finally, it must be noted that the notions of space and movement are already relative, even within the framework of classical mechanics³!

If motion takes place in three-dimensional Euclidean space, it can always be resolved into three *independent* one-dimensional motions along mutually perpendicular directions. This explains the fundamental importance of the one-dimensional motion studied first.

3.1 One-Dimensional Kinematics

Position, velocity and acceleration

One-dimensional motion takes place along a single direction in space. Since position is not absolute, we first choose a reference point, or origin O , on the line of motion. Choosing a positive direction then defines a one-dimensional coordinate system. Thus the system is specified by an axis, an origin and a positive direction (see Figure 1). We assume throughout that it is calibrated in metres.

Position In one dimension, the position of a body is its coordinate x (a scalar expressed in m): its signed distance from the origin, positive in the chosen positive direction and negative in the opposite direction.

Example. In Figure 1, the body (blue dot) has position $x_1 > 0$ at time t_1 ($x_1 = d_1$) and position $x_2 < 0$ at time t_2 ($x_2 = -d_2$). Its distance from the origin, denoted d_i ($i = 1, 2$), is non-negative.

Remark. As the position x of a body generally varies over time, it is convenient to consider it as a function of time: $x = x(t)$ It is customary to use the notation x_1 to designate $x(t_1)$ and the equivalence between these two notations is expressed by writing $x_1 \equiv x(t_1)$.

Displacement The displacement of a body between time t_1 and t_2 is defined by the relationship

$$\Delta x_{t_1, t_2} \stackrel{\text{def}}{=} x(t_2) - x(t_1) \quad (\text{m}) \quad (1)$$

¹According to Newton, " the absolute space, without relation to external things, remains always similar and immobile. True and mathematical absolute time, without relation to anything external, flows indifferently and is called duration ".

²In mechanics, an observer designates a calibrated coordinate system with a clock to measure time intervals.

³Think, for example, of the movements of planets seen from the Sun or the Earth.

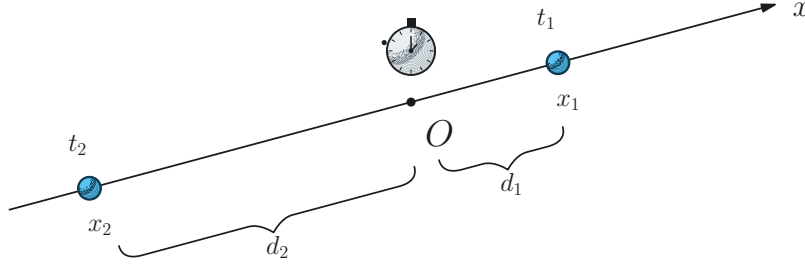


Figure 1: One-dimensional reference frame with a coordinate system (specified by its origin O , axis and positive direction, calibrated in metres) and a clock. The body, shown as a blue point, is at $x_1 = d_1 > 0$ when the clock reads t_1 and at $x_2 = -d_2 < 0$ when it reads t_2 . Here d_i ($i = 1, 2$) is the distance of the body from the origin O .

Average velocity The average velocity of a body between times t_1 and t_2 is defined by

$$\bar{v}_{t_1, t_2} \stackrel{\text{def}}{=} \frac{x(t_2) - x(t_1)}{t_2 - t_1} = \frac{\Delta x_{t_1, t_2}}{\Delta t} \quad (\text{m/s}) \quad (2)$$

Velocity The instantaneous velocity of a body at time t is the time derivative of its position at time t ,

$$v(t) \stackrel{\text{def}}{=} \lim_{\Delta t \rightarrow 0} \frac{\Delta x_{t, t+\Delta t}}{\Delta t} = \frac{dx(t)}{dt} \quad (\text{m/s}) \quad (3)$$

The mathematical definition of the derivative of $x(t)$ with respect to t is the middle expression in equation (3); the final expression is simply an alternative notation.

Average acceleration The mean acceleration of a body between time t_1 and time t_2 is defined by the relationship

$$\bar{a}_{t_1, t_2} \stackrel{\text{def}}{=} \frac{v(t_2) - v(t_1)}{t_2 - t_1} = \frac{\Delta v_{t_1, t_2}}{\Delta t} \quad (\text{m/s}^2) \quad (4)$$

Acceleration The instantaneous acceleration of a body at time t is the time derivative of its velocity at time t ,

$$a(t) \stackrel{\text{def}}{=} \lim_{\Delta t \rightarrow 0} \frac{\Delta v_{t, t+\Delta t}}{\Delta t} = \frac{dv(t)}{dt} \quad (\text{m/s}^2) \quad (5)$$

Displacement, velocity (average or instantaneous) and acceleration (average or instantaneous) do not depend on the choice of *origin*, but their signs depend on the *positive direction* of the coordinate system. In one dimension these physical quantities are represented by *scalars*.

The third time derivative of position is called *jerk*, and the fourth is called *snap* or *jounce*. Jerk measures the rate of change of acceleration.

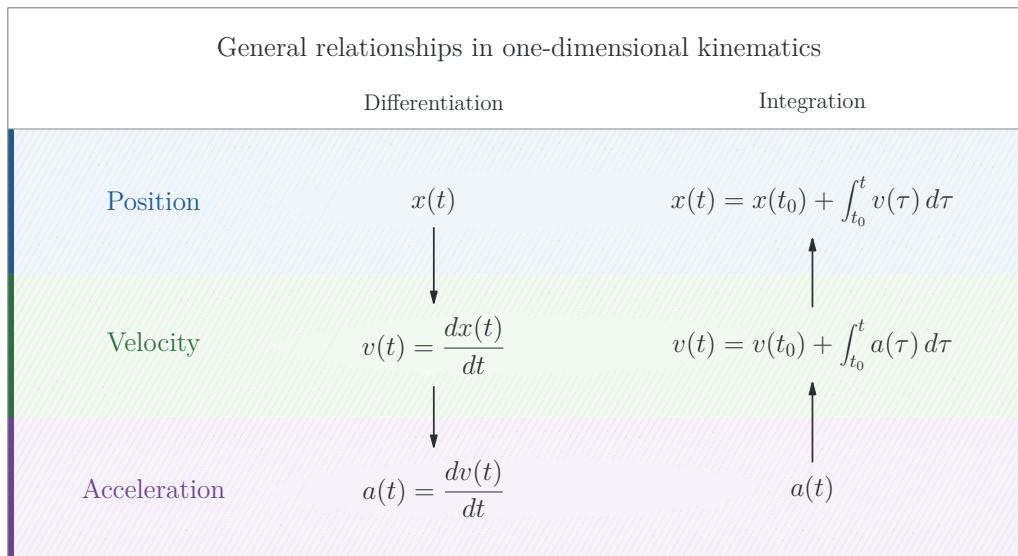


Figure 2: General relationships between position, velocity and acceleration at one dimension. To determine velocity from acceleration, and position from velocity, you must know the initial conditions $v(t_0)$ and $x(t_0)$.

Example. A body is subject to acceleration (expressed in m/s^2)

$$a(t) = \begin{cases} 0.1 & 0 \leq t \leq 5 \\ 0 & 5 < t \leq 10 \\ -0.5625 \sin[0.75(t - 10)] & 10 < t \leq 20 \end{cases}$$

shown in Figure 3. Find its velocity and position for all times $t \in [0, 20]$, given the initial conditions $x(0) = -3$ m and $v(0) = 0.25$ m/s. Also find its average velocity and average acceleration over the time interval $[0, 10]$.

The velocity and position are obtained by repeated integrations of the acceleration over the time intervals $[0, 5]$, $]5, 10]$ and $]10, 20]$. In order to obtain the velocity in the time interval $[0, 5]$, we integrate the acceleration for the first time and use the initial condition $v(0) = 0.25$ m/s ($t_0 = 0$):

$$v(t) = \int_0^t a(\tau) d\tau + v(0) = 0.1t + 0.25$$

A second integration and the initial condition $x(0) = -3$ m provide the position in the time interval $[0, 5]$:

$$x(t) = \int_0^t v(\tau) d\tau + x(0) = 0.05t^2 + 0.25t - 3$$

In particular, in $t = 5$ s, the velocity and position of the body are given respectively by $v(5) = 0.75$ m/s and $x(5) = -0.5$ m. To determine the velocity and position in the time intervals $]5, 10]$ and $]10, 20]$, we follow the same procedure by taking as initial time $t_0 = 5$ s and $t_0 = 10$ s respectively. For $t \in]5, 10]$, we obtain

$$v(t) = \int_5^t a(\tau) d\tau + v(5) = 0.75$$

and

$$x(t) = \int_5^t v(\tau) d\tau + x(5) = 0.75(t - 5) - 0.5$$

so that $v(10) = 0.75 \text{ m/s}$ and $x(10) = 3.25 \text{ m}$. For $t \in]10, 20]$ we then obtain

$$v(t) = \int_{10}^t a(\tau) d\tau + v(10) = 0.75 \cos[0.75(t - 10)]$$

and

$$x(t) = \int_{10}^t v(\tau) d\tau + x(10) = \sin[0.75(t - 10)] + 3.25$$

In the end, the velocity (expressed in m/s) is given by

$$v(t) = \begin{cases} 0.1t + 0.25 & 0 \leq t \leq 5 \\ 0.75 & 5 < t \leq 10 \\ 0.75 \cos[0.75(t - 10)] & 10 < t \leq 20 \end{cases}$$

and the position (expressed in m) by

$$x(t) = \begin{cases} 0.05t^2 + 0.25t - 3 & 0 \leq t \leq 5 \\ 0.75(t - 5) - 0.5 & 5 < t \leq 10 \\ \sin[0.75(t - 10)] + 3.25 & 10 < t \leq 20 \end{cases}$$

The graphs (x, t) , (v, t) and (a, t) associated with this movement are shown in Figure 3.

Average velocity and acceleration over the time interval $[0, 10]$ are obtained from (2) and (4), i.e.

$$\bar{v}_{0,10} = \frac{x(10) - x(0)}{10} = 0.625 \text{ m/s}$$

and

$$\bar{a}_{0,10} = \frac{v(10) - v(0)}{10} = 0.05 \text{ m/s}^2$$

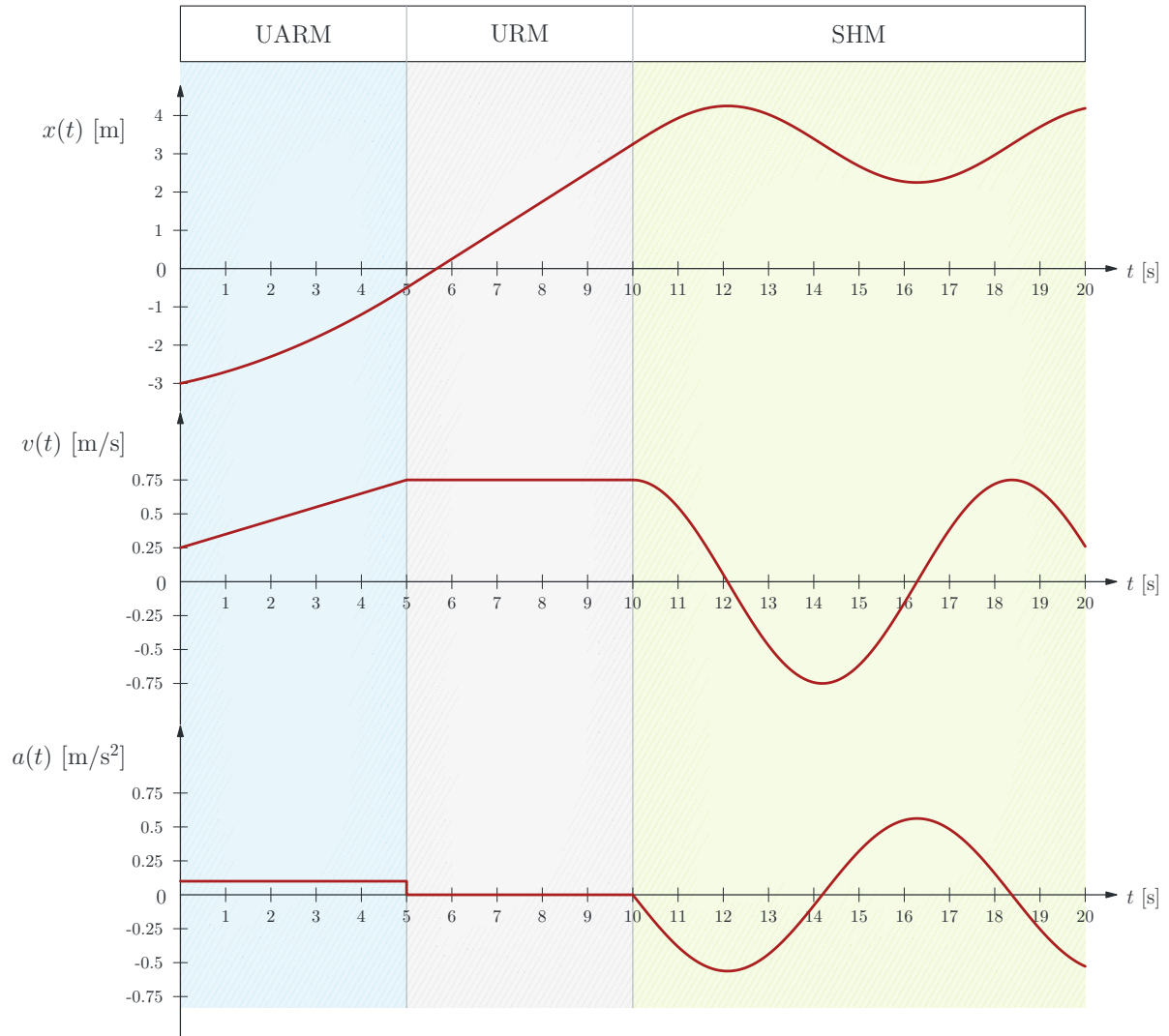


Figure 3: Example graphs of (x, t) , (v, t) and (a, t) for one-dimensional motion consisting of uniformly accelerated rectilinear motion (UARM) for $0 \leq t \leq 5$, uniform rectilinear motion (URM) for $5 < t \leq 10$, and simple harmonic motion (SHM) for $10 < t \leq 20$. The initial conditions are $x(0) = 3$ m and $v(0) = 0.25$ m/s. Here the acceleration is imposed externally, through external forces. The initial conditions for the URM and SHM stages are therefore the final positions and velocities of the UARM and URM stages, respectively.

The function $v(t)$, shown in the middle, is the derivative of $x(t)$ with respect to t . This derivative is the slope of the tangent to the graph: it increases during UARM, is constant during URM and oscillates during SHM. Similarly, $a(t)$ is the derivative of $v(t)$ with respect to t . Note that the acceleration varies continuously during SHM.

3.2 Three-Dimensional Kinematics⁴

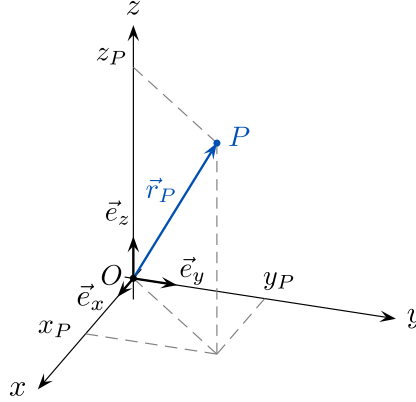


Figure 4: Orthonormal coordinate system $Oxyz$. The point P , with Cartesian coordinates x_P, y_P and z_P , is located by its position vector $\vec{r}_P$, directed from the origin to P . In Cartesian components, $\vec{r}_P = (x_P, y_P, z_P)$.

Position In three dimensions, the position of a body is defined as the vector that connects the origin of the coordinate system to the point in the space where the body is located (see Figure 4). In a system of Cartesian coordinates, the position vector can be written in terms of Cartesian components $x(t), y(t), z(t)$

$$\vec{r}(t) = \overrightarrow{OP} = x(t) \vec{e}_x + y(t) \vec{e}_y + z(t) \vec{e}_z = (x(t), y(t), z(t)) \quad (6)$$

Velocity The instantaneous velocity of a body at time t is the time derivative of its position at t , namely the vector quantity

$$\boxed{\vec{v}(t) \stackrel{\text{def}}{=} \frac{d\vec{r}(t)}{dt}} \quad (7)$$

or, in terms of Cartesian components

$$\vec{v}(t) = (v_x(t), v_y(t), v_z(t)) \stackrel{\text{def}}{=} \left(\frac{dx(t)}{dt}, \frac{dy(t)}{dt}, \frac{dz(t)}{dt} \right)$$

Remark. The time derivative of a function $f(t)$ is sometimes denoted by $\dot{f}(t)$ to simplify the notation. Velocity can thus be written as $\vec{v}(t) = \dot{\vec{r}}(t) = (\dot{x}(t), \dot{y}(t), \dot{z}(t))$.

Fact. *The velocity is always tangent to the trajectory.*

Proof. By definition of the derivative, we have $\vec{v}(t) = \lim_{\Delta t \rightarrow 0} \frac{\vec{r}(t+\Delta t) - \vec{r}(t)}{\Delta t}$. Therefore, $\vec{v}(t) \parallel \Delta \vec{r} \equiv \vec{r}(t + \Delta t) - \vec{r}(t)$ when $\Delta t \rightarrow 0$, where the vector $\Delta \vec{r}$ is tangent to the trajectory when $\Delta t \rightarrow 0$, hence the conclusion. $\square$

⁴One- or two-dimensional kinematics is only a special case of three-dimensional kinematics.

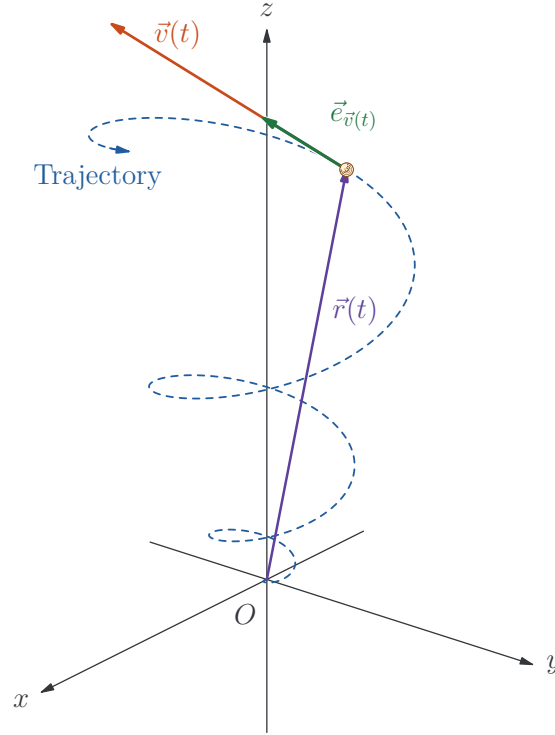


Figure 5: Trajectory of a body as well as its position and velocity at a given moment t .

Acceleration The instantaneous acceleration of a body at time t is the vector defined as the time derivative of its velocity at time t , i.e.

$$\boxed{\vec{a}(t) \stackrel{\text{def}}{=} \frac{d\vec{v}(t)}{dt}} = \frac{d^2\vec{r}(t)}{dt^2} \quad (8)$$

or, in terms of Cartesian components,

$$\vec{a}(t) = (a_x(t), a_y(t), a_z(t)) \stackrel{\text{def}}{=} \left(\frac{dv_x(t)}{dt}, \frac{dv_y(t)}{dt}, \frac{dv_z(t)}{dt} \right) = \left(\frac{d^2x(t)}{dt^2}, \frac{d^2y(t)}{dt^2}, \frac{d^2z(t)}{dt^2} \right)$$

Remark. In one dimension, position, velocity and acceleration are *scalar* quantities (real numbers). In two or three dimensions they are *vectors* (pairs or triples of real numbers when written in components). Do not confuse one-dimensional velocity, a scalar denoted by $v(t)$ that may be negative, positive or zero, with the magnitude of velocity in two or three dimensions, denoted by $|\vec{v}(t)|$ or $v(t)$, which is always non-negative and is called speed.

DECOMPOSITION OF ACCELERATION

Fact. For non-zero velocity, acceleration can be resolved into a component parallel to the velocity, hence tangent to the trajectory, called *tangential acceleration* ($\vec{a}_t$), and a component perpendicular to the velocity, called *centripetal acceleration* ($\vec{a}_c$). More precisely,

$$\vec{a}(t) = \underbrace{\frac{dv(t)}{dt} \vec{e}_{\vec{v}(t)}}_{\equiv \vec{a}_t(t)} + \underbrace{v(t) \frac{d\vec{e}_{\vec{v}(t)}}{dt}}_{\equiv \vec{a}_c(t)} = \vec{a}_t(t) + \vec{a}_c(t) \quad (9)$$

where $\vec{e}_{\vec{v}(t)} \equiv \frac{\vec{v}(t)}{|\vec{v}(t)|}$ is the unit vector pointing along the velocity at every time t (see Figure 5).

Proof. The product rule gives $\vec{a}(t) = \frac{d\vec{v}(t)}{dt} = \frac{d}{dt}(v(t)\vec{e}_{\vec{v}(t)}) = \frac{dv(t)}{dt} \vec{e}_{\vec{v}(t)} + v(t) \frac{d\vec{e}_{\vec{v}(t)}}{dt}$. It remains to show that these two components are perpendicular, or equivalently that $\vec{e}_{\vec{v}(t)} \perp \frac{d\vec{e}_{\vec{v}(t)}}{dt}$. Since $\vec{e}_{\vec{v}(t)}$ is a unit vector, we use

the identity $\vec{e}_{\vec{v}(t)} \cdot \vec{e}_{\vec{v}(t)} = 1$, valid $\forall t$. Differentiating with respect to time gives

$$\frac{d\vec{e}_{\vec{v}(t)}}{dt} \cdot \vec{e}_{\vec{v}(t)} = 0 \quad \forall t.$$

The dot product vanishes if and only if the two vectors are perpendicular, which proves the result. $\square$

Fact. *The centripetal acceleration is always directed towards the centre of curvature of the trajectory (see figure 6).*

Proof. Suppose zero tangential acceleration, so that $\vec{a}_c = \vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\vec{v}(t+\Delta t) - \vec{v}(t)}{\Delta t}$ and $|\vec{v}(t + \Delta t)| = |\vec{v}(t)|$. Therefore, $\vec{a}_c(t) \parallel \Delta \vec{v} \equiv \vec{v}(t + \Delta t) - \vec{v}(t)$ when $\Delta t \rightarrow 0$, where the vector $\Delta \vec{v}$ always points towards the centre of curvature of the trajectory, hence the conclusion. $\square$

ACCELERATION MAGNITUDES

Fact. *The magnitudes of the tangential and centripetal accelerations are*

$$a_t = \left| \frac{dv}{dt} \right| \quad \text{and} \quad a_c = \frac{v^2}{R_c}$$

with R_c the radius of curvature of the trajectory at the point where the body is located (see Figure 6).

Proof. The first relation follows by taking the magnitude of $\vec{a}_t(t) = \frac{dv(t)}{dt} \vec{e}_{\vec{v}(t)}$, since $\vec{e}_{\vec{v}(t)}$ is a unit vector by construction. The second relation is stated without proof. $\square$

Based on (9), we have

$$\begin{aligned} \vec{a}_t = 0 &\Leftrightarrow v(t) = \text{constant} \\ \vec{a}_c = 0 &\Leftrightarrow \vec{e}_{\vec{v}(t)} = \text{constant} \end{aligned}$$

TANGENTIAL AND CENTRIPETAL ACCELERATIONS

Fact. *Tangential acceleration $\vec{a}_t$ describes changes in speed, whereas centripetal acceleration $\vec{a}_c$ describes changes in the direction of velocity (a vector quantity).*

The magnitude of the total acceleration is (time dependence is implicit)

$$a = |\vec{a}| = \sqrt{\vec{a} \cdot \vec{a}} = \sqrt{a_t^2 + a_c^2} \quad (10)$$

The radius of curvature R_c at a point in the trajectory, which appears in the expression of centripetal acceleration, is defined as the radius of the circle tangential to the trajectory at that point, and which best fits the trajectory⁵. It is called an osculating circle or circle of curvature — see figure 6.

⁵The osculating circle best fits the trajectory in the sense that the first three terms of their Taylor expansions, through the second derivative, agree at that point. Equivalently, consider three nearby points along the trajectory and the unique circle passing through them. As all three approach a common point A , this circle approaches the osculating circle at A .

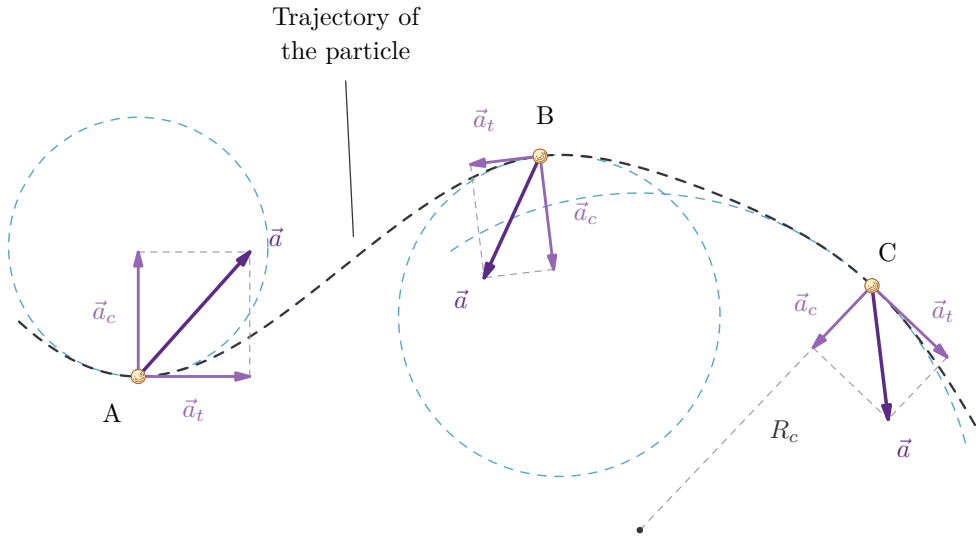


Figure 6: Total, tangential and centripetal acceleration at three points (A, B and C) along a trajectory. At any point, centripetal acceleration points towards the centre of the osculating circle and has magnitude v^2/R_c , where R_c is the radius of curvature (the radius of that circle), which varies along the trajectory.

Remark. The acceleration a person can withstand depends on its duration. An acceleration of $30\text{ g} \approx 300\text{ m/s}^2$ for a tenth of a second, such as that produced by an aircraft ejection seat, may be survivable. The reported maximum acceleration survived by a human is about $80\text{ g} \approx 800\text{ m/s}^2$, although accelerations of 15 g or more can be fatal. Crickets can jump with accelerations of about 18 g , and accelerations of about 135 g have been measured in fleas.

3.3 Changing Coordinate Systems

Consider two reference frames⁶ S and S' , whose axes are parallel and do not rotate relative to each other, with S' moving at velocity $\vec{V}(t)$ relative to S (see Figure 7). Positions transform from S to S' according to

$$\vec{R}(t) + \vec{r}'_p(t) = \vec{r}_p(t) \quad \Leftrightarrow \quad \vec{r}'_p(t) = \vec{r}_p(t) - \vec{R}(t),$$

where $\vec{R}(t) = \overrightarrow{OO'}$ joins the origins O of S and O' of S' . Differentiating with respect to time, and noting that $\frac{d\vec{R}(t)}{dt} = \vec{V}(t)$, gives the velocity transformation

$$\vec{v}'_p(t) = \vec{v}_p(t) - \vec{V}(t)$$

and the acceleration transformation

$$\vec{a}'_p(t) = \vec{a}_p(t) - \frac{d\vec{V}(t)}{dt}$$

⁶A *reference frame* consists of a set of axes in three-dimensional physical space attached to an observer (real or imaginary) with a clock. Events are identified by spatial coordinates and a time coordinate.

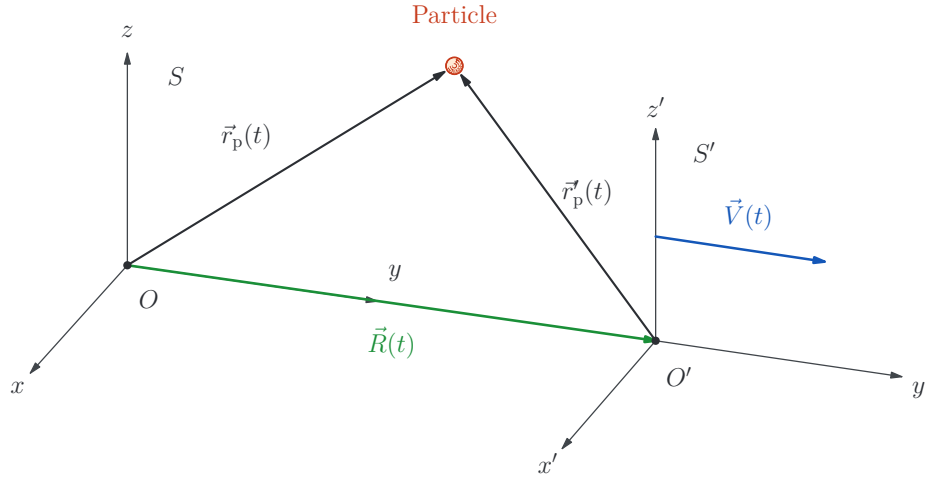


Figure 7: Two reference frames, one moving at velocity $\vec{V}$ in relation to the other, in which the movement of the same particle is analyzed.

3.4 Special Types of Motion

3.4.1 Uniform rectilinear motion (URM)

Uniform rectilinear motion follows a straight path (Figure 8) at constant velocity. An appropriate choice of coordinates always reduces it to one-dimensional motion, as illustrated in Figure 8. Position is then a linear scalar function of time: $x(t) = x_0 + v(t - t_0)$ where $x_0 \equiv x(t_0)$ is the initial position and $v(t) = v$ the constant velocity ($\Rightarrow a(t) = 0$). In two or three dimensions, the position vector $\vec{r}(t)$ is a linear vector function of time: $\vec{r}(t) = \vec{r}_0 + \vec{v}(t - t_0)$ where $\vec{r}_0 \equiv \vec{r}(t_0)$ is the initial position.

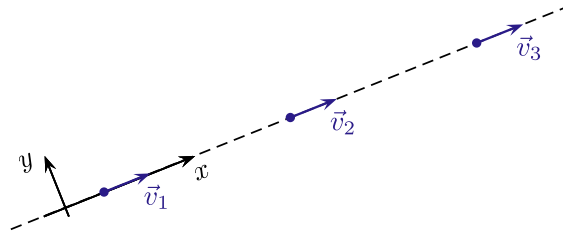


Figure 8: Trajectory (dashed) and velocity of a particle in URM at three instants. In URM, $\vec{v}_1 = \vec{v}_2 = \vec{v}_3$ and $\vec{a}_1 = \vec{a}_2 = \vec{a}_3 = 0$. An inclined coordinate system reduces the description to one spatial dimension.

3.4.2 Uniformly accelerated rectilinear motion (UARM)

Uniformly accelerated rectilinear motion follows a straight path (see Figure 9) with constant acceleration. The speed may increase or decrease. For the path to remain straight, the initial velocity and acceleration must be collinear, possibly in opposite directions. Otherwise, the motion combines UARM with URM (see projectile motion). Choosing suitable coordinates always reduces UARM to one dimension, as in Figure 9. Position is then a quadratic scalar function of time: $x(t) = x_0 + v_0(t - t_0) + \frac{1}{2}a(t - t_0)^2$, where $x_0 \equiv x(t_0)$ and $v_0 \equiv v(t_0)$ are the initial position and velocity. In summary,

$$\begin{cases} x(t) = x_0 + v_0(t - t_0) + \frac{1}{2}a(t - t_0)^2 \\ v(t) = v_0 + a(t - t_0) \\ a(t) = a \end{cases}$$

At two or three dimensions, the position of the body $\vec{r}(t)$ is, at any moment, a quadratic vector function of time: $\vec{r}(t) = \vec{r}_0 + \vec{v}_0(t - t_0) + \frac{1}{2}\vec{a}(t - t_0)^2$ with $\vec{r}_0 \equiv \vec{r}(t_0)$ its initial position and $\vec{v}_0 \equiv \vec{v}(t_0)$ its initial velocity. In summary, we have

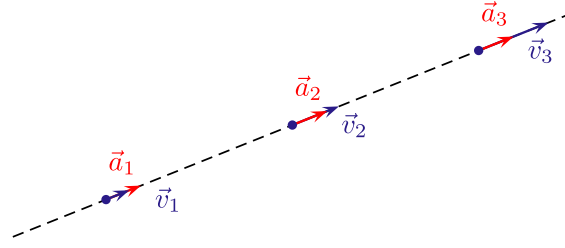


Figure 9: Velocity, acceleration and trajectory of a particle in one-dimensional UARM at three instants. We have $\vec{a}_1 = \vec{a}_2 = \vec{a}_3$, but $\vec{v}_1 \neq \vec{v}_2 \neq \vec{v}_3$.

$$\begin{cases} \vec{r}(t) = \vec{r}_0 + \vec{v}_0(t - t_0) + \frac{1}{2} \vec{a}(t - t_0)^2 \\ \vec{v}(t) = \vec{v}_0 + \vec{a}(t - t_0) \\ \vec{a}(t) = \vec{a} \end{cases}$$

3.4.3 Superposition of URM and UARM

In two or three dimensions, when the initial velocity is not parallel to a constant acceleration, motion can be described as the superposition of UARM and URM. Choose one basis vector ($\vec{e}_y$, for example) parallel to the acceleration and the other two ($\vec{e}_x$ and $\vec{e}_z$) perpendicular to it. Projecting the equations onto these axes then gives one-dimensional UARM along $\vec{e}_y$ and URM along the other axes.

Projectile motion

Projectile motion with air resistance neglected is an example, with gravitational acceleration $\vec{a} = \vec{g}$ of magnitude $g = 9.81 \text{ m/s}^2$ at our geographical location (see Figure 10).

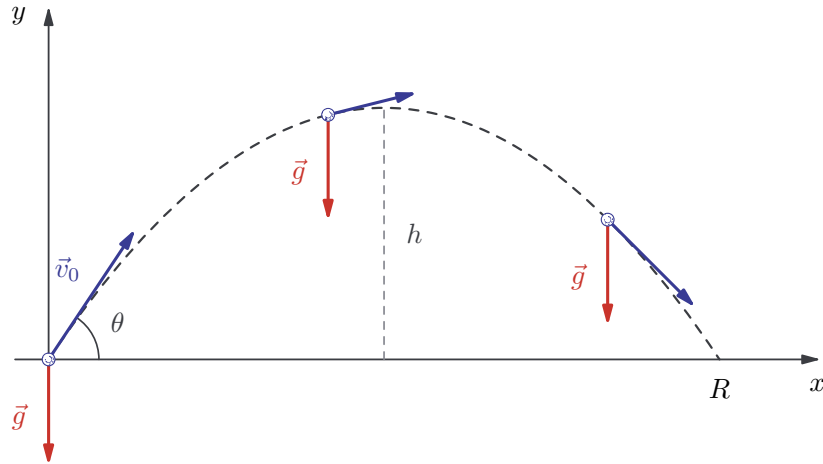


Figure 10: Projectile motion under gravitational acceleration $\vec{g}$, constant and identical at every point of the trajectory. The projectile undergoes UARM along y and URM along x . Velocity and acceleration are shown at three different times.

The position of the projectile subject to gravity acceleration changes over time according to

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + \begin{pmatrix} v_{x,0} \\ v_{y,0} \end{pmatrix} t + \frac{1}{2} \begin{pmatrix} 0 \\ -g \end{pmatrix} t^2$$

where $\vec{r}_0 = (x_0, y_0) = (0, 0)$ is its initial position and $\vec{v}_0 = (v_{x,0}, v_{y,0}) = (v_0 \cos \theta, v_0 \sin \theta)$ is its initial velocity. The Cartesian equation of the trajectory can be obtained by expressing the ordinate as a function of the abscissa

(by eliminating the time parameter),

$$y = (\tan \theta)x - \left(\frac{g}{2v_0^2 \cos^2 \theta} \right) x^2 \quad (11)$$

The equation (11) is that of a parabola since y appears as a quadratic function of x (see trajectory in Figure 10). The maximum height h and the range of the shot R correspond to the maximum of the parabola and its strictly positive zero, i.e.

$$h = \frac{v_0^2 \sin^2 \theta}{2g}, \quad R = \frac{v_0^2 \sin(2\theta)}{g}$$

3.4.4 Uniform circular motion (UCM)

A particle in circular motion follows a circle of radius r and centre C (see Figure 11). The motion is *uniform* if its speed is constant. The angular velocity ω is the angle, in radians, swept per unit time by the position vector measured from the centre C . If T is the time needed for a complete revolution, we have

$$\omega \stackrel{\text{def}}{=} \frac{2\pi}{T} \quad (\text{rad/s}) \quad (12)$$

and

$$v = \frac{2\pi r}{T} = \omega r \quad (13)$$

UCM has constant angular velocity or, equivalently, velocity of constant magnitude, by equation (13).

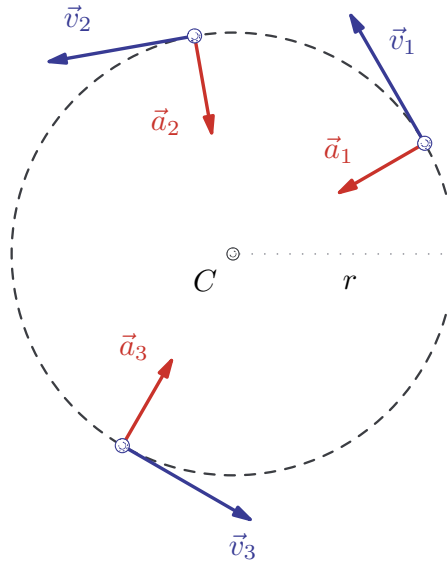


Figure 11: Velocity and acceleration of a particle in UCM at three different times. We have $|\vec{v}_1| = |\vec{v}_2| = |\vec{v}_3| = \omega r$ and $|\vec{a}_1| = |\vec{a}_2| = |\vec{a}_3| = \omega^2 r$, but $\vec{v}_1 \neq \vec{v}_2 \neq \vec{v}_3$ and $\vec{a}_1 \neq \vec{a}_2 \neq \vec{a}_3$. Neither vector is constant, since both continuously change direction; only their magnitudes remain constant.

Although the speed is constant, the velocity continuously changes direction. Tangential acceleration therefore vanishes, while centripetal acceleration is

$$a_c = \frac{v^2}{r} = \omega^2 r \quad (14)$$

since the radius of curvature R_c is at any moment equal to the radius r of the trajectory.

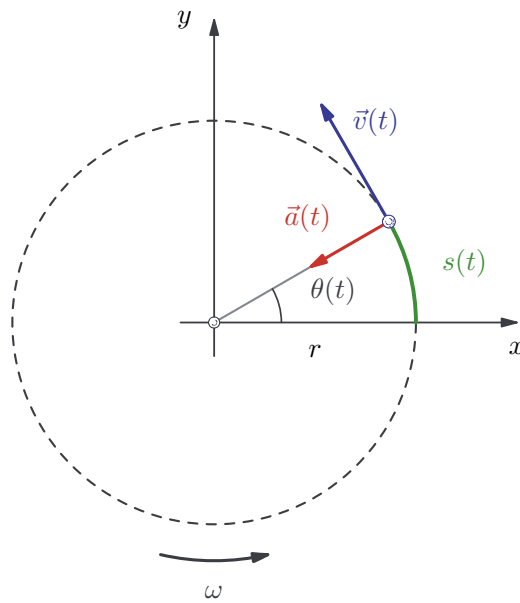


Figure 12: Angular position $\theta(t)$ of a circularly moving particle. The length of the circle arc swept by the position vector is given by $s(t) = r\theta(t)$. The purely centripetal acceleration is illustrated when $\theta(t)$ increases linearly over time.

This result can be obtained using polar coordinates (r, θ) . In UCM, the radius r is constant and θ increases linearly with time: $\theta(t) = \theta_0 + \omega t$, where θ_0 is the initial angular position (taken as zero below) and ω is the angular velocity. The distance travelled in time t is $s(t) = r\theta(t)$, so the speed is

$$v(t) = \frac{ds(t)}{dt} = \omega r$$

The particle's velocity and acceleration are readily obtained using two-dimensional Cartesian coordinates:

$$\begin{aligned}\vec{r}(t) &= r(\cos(\theta(t)), \sin(\theta(t))) \\ \vec{v}(t) &= \frac{d\vec{r}(t)}{dt} = \omega r(-\sin(\theta(t)), \cos(\theta(t))) \Rightarrow |\vec{v}(t)| = \omega r \\ \vec{a}(t) &= \frac{d\vec{v}(t)}{dt} = -\omega^2 r(\cos(\theta(t)), \sin(\theta(t))) \Rightarrow |\vec{a}(t)| = \omega^2 r\end{aligned}$$

Example. The Earth orbits the Sun at approximately constant speed along an approximately circular path. Determine its orbital speed.

The UCM period is one year and the orbital radius (the Earth-Sun distance) is about 150 million km. The Earth's orbital speed is therefore about $v = \omega r = 2\pi r/T \approx 30000$ m/s, or more than 100000 km/h.

[As Henri Poincaré said, if we are in a given place today and return to the same place tomorrow at the same time, we will actually be 31 million kilometers further.]

Example. In a classical planetary model of the hydrogen atom, inspired by Rutherford's nuclear model, suppose an electron moves in a uniform circular orbit around a proton. Take an orbital frequency of order 10^{15} Hz and an orbital radius of 0.6×10^{-10} m. This model is of historical interest only: quantum mechanics does not describe an electron as a particle following a definite orbit. What electron speed does this classical model predict?

For UCM, the speed is $v = \omega r = 2\pi r f$, giving $v \approx 377000$ m/s (more than 1 million km/h). Although very large on everyday scales, it remains small compared with the speed of light in vacuum.

Example. In the LHC, a circular particle accelerator with circumference about 27 km, a proton in uniform circular motion may have centripetal acceleration $a = 90 \text{ Gm s}^{-2} = 9.0 \times 10^{10} \text{ m s}^{-2}$. What is its speed?

For an UCM, the acceleration is purely centripetal and we have $a = v^2/r$ or $v = \sqrt{ra}$ with $r = 27 \times 10^3/(2\pi)$, i.e. $v \simeq 1.97 \times 10^7 \text{ m s}^{-1} \simeq 0.066 c < c$ (approximately $7.1 \times 10^7 \text{ km h}^{-1}$).

Example. Consider a device for measuring the speed of a bullet. The bullet passes through two thin polystyrene discs rotating uniformly at angular velocity $\omega = 2500$ revolutions/min, separated by 1 m. The measured angular separation between the two bullet holes is 30° . What is the bullet's speed?

The velocity is calculated by $v = \Delta x/\Delta t$ with $\Delta t = \Delta\theta/\omega = 30^\circ/(2500 \times 360^\circ/60 \text{ s})$, which gives $v = 500$ m/s (or 1800 km/h).

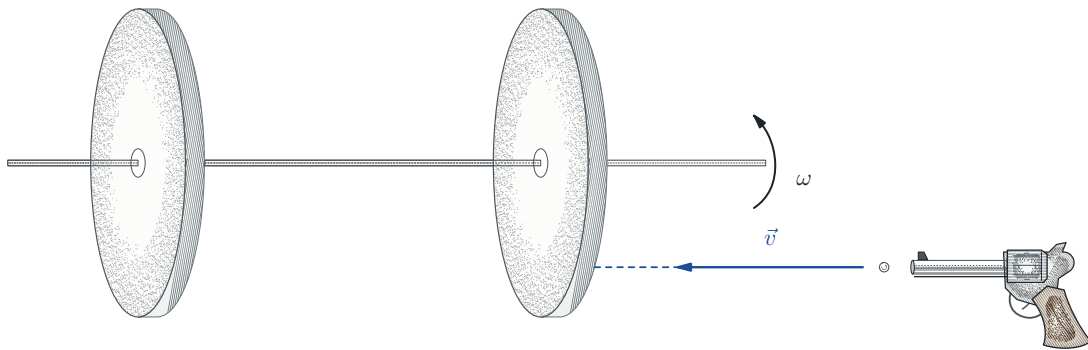


Figure 13: Device for measuring the speed of a bullet.

3.4.5 Simple harmonic motion (SHM)

In simple harmonic motion (SHM), a particle's position oscillates periodically at a single frequency. In one dimension, its position has the general form

$$x(t) = A \sin(\omega t + \varphi),$$

where A is the amplitude (m), ω the angular frequency (rad/s), and φ the phase (rad). The frequency and period are respectively

$$f = \frac{\omega}{2\pi} \quad \text{and} \quad T = \frac{1}{f} = \frac{2\pi}{\omega}.$$

Differentiating position gives the velocity and acceleration as functions of time:

$$v(t) = A\omega \cos(\omega t + \varphi)$$

$$a(t) = -A\omega^2 \sin(\omega t + \varphi)$$

It should be noted that acceleration is not constant in an SHM, but periodically oscillates at the same frequency as the position. The maximum velocity and acceleration of an SHM are given by

$$v_{\max} = |A| \omega = x_{\max} \omega$$

$$a_{\max} = |A| \omega^2 = x_{\max} \omega^2$$

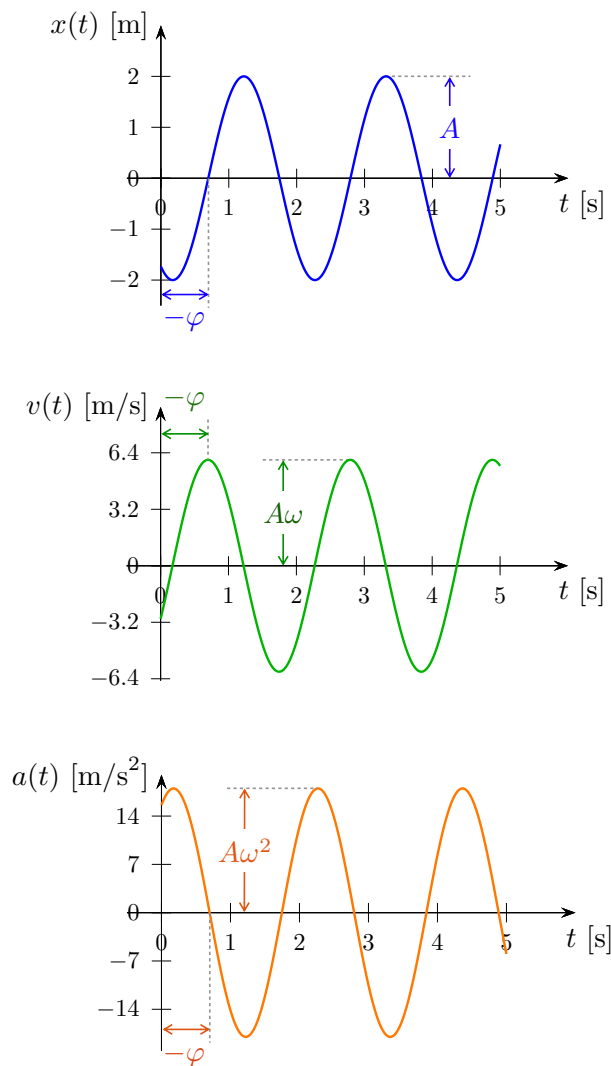


Figure 14: Position, velocity and acceleration in harmonic motion with amplitude $A = 2$ m, angular frequency $\omega = 3$ rad/s and phase $\varphi = -2\pi/3$ rad.

CHAPTER 4

DYNAMICS

Dynamics Dynamics is the branch of mechanics that studies the relationship between forces acting on bodies and the resulting motion. Forces and kinematic quantities must be described in a specified reference frame.

Remark. In physics, *body* or *object* usually denotes a rigid collection of particles: a rigid body. A *point particle* or *point particle* represents a body whose dimensions are small compared with the characteristic distances of its motion. It may then be modelled as a geometric point with mass m equal to the body's mass.

4.1 Forces

4.1.1 Introduction

Force Force is a quantitative description of an interaction between objects. Newton's laws relate forces to changes in motion and allow us to analyse deformation and static equilibrium.

Forces are represented by bound vectors and are therefore characterised by a *magnitude*, a *direction*, a *positive direction* and a *point of application* (Figure 1). The vector sum of all forces acting on a body is the *net force* (Figure 2). A force's point of application specifies where it acts on the body. This information is essential when studying rotation about a fixed axis. For pure translation, however, the points of application need not be specified, and the forces can be treated as free vectors.

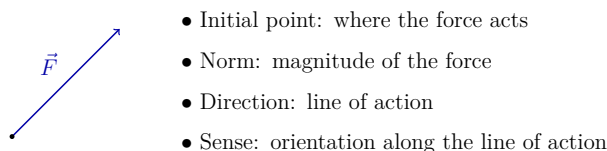


Figure 1: Characteristics of a force.

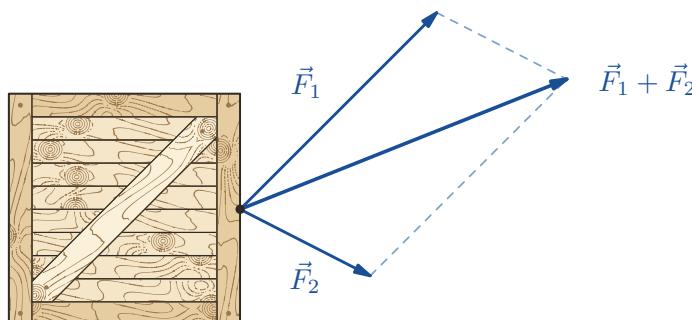


Figure 2: Resultant of two forces acting at the same point.

Force magnitude is expressed in newtons, with $\text{N} = \text{kg m s}^{-2}$ in the SI. One newton accelerates a body of mass 1 kg at 1 metre per second squared.

4.1.2 Types of forces

The physical phenomena observed to date are explained by four *fundamental* types of force:

- the *gravitational* force
- the *electromagnetic* force
- the *strong nuclear* force
- the *weak nuclear* force

In classical mechanics, gravity is the attractive interaction between bodies with mass, such as an apple and the Earth or the Earth and the Sun. It is by far the weakest fundamental interaction, yet it keeps us on the Earth's surface and holds the atmosphere around our planet. Gravity plays the dominant role in organising matter on astronomical scales.

The electromagnetic force, or *Lorentz force*, acts on a charged particle, such as an electron or proton, in an electromagnetic field. The electric component of this force is attractive between opposite charges and repulsive between like charges. Strong attraction between opposite charges favours the formation of electrically neutral structures such as atoms, molecules and crystals.

The strong nuclear force binds atomic nuclei together. The weak nuclear force is responsible for certain radioactive processes and plays an important role in nuclear fusion in stars. The interactions also differ in range: nuclear forces are *short range*, whereas gravitational and electromagnetic forces are *long range*, which explains their importance in macroscopic phenomena.

Fundamental force	Relative strength	Range (m)	Type of interaction
gravitational	1	∞	attraction
electromagnetic	$\sim 10^{36}$	∞	attraction or repulsion ¹
strong nuclear	$\sim 10^{38}$	$\sim 10^{-15}$	attraction or repulsion ²
weak nuclear	$\sim 10^{33}$	$\sim 10^{-18}$	transmutation

Table 1: The fundamental forces of nature and the properties of the interactions they describe.

Fundamental forces describe interactions *at a distance*. For example, the Sun attracts the Earth without a material medium between them; two magnets also interact across empty space. By contrast, *contact* interactions, such as friction, normal forces and spring forces, are macroscopic manifestations of microscopic fundamental interactions, mainly electromagnetism. These forces are therefore called *phenomenological*³, as opposed to fundamental forces.

Depending on the problem, a force may be constant or may depend on time, position, velocity or several of these quantities simultaneously⁴. In general it is a vector-valued function of several variables:

$$\vec{F} = \vec{F}(\vec{r}, \vec{v}, \dots, t)$$

or, in Cartesian components,

$$\begin{aligned} \vec{F} &= (F_x(\vec{r}, \vec{v}, \dots, t), F_y(\vec{r}, \vec{v}, \dots, t), F_z(\vec{r}, \vec{v}, \dots, t)) \\ &= (F_x(x, y, z, v_x, v_y, v_z, \dots, t), F_y(x, y, z, v_x, v_y, v_z, \dots, t), F_z(x, y, z, v_x, v_y, v_z, \dots, t)) \end{aligned} \quad (1)$$

¹according to the sign of electrical charges

²depending on the distance between hadrons (protons, neutrons, pions, ...). At very short distances the strong nuclear force is strongly repulsive, whereas at larger distances, of order one femtometre, it is attractive.

³The mathematical expression of a phenomenological force is generally as simple as possible in order to account for experimental observations in a given parameter regime. It is not directly deduced from fundamental principles.

⁴The force may also depend on the acceleration of the particle or even its higher-order temporal derivatives, but these cases are relatively uncommon (radiation reaction force, ...).

Example. A particle of electric charge q in electric and magnetic fields $\vec{E}$ and $\vec{B}$ that vary in space $\rightarrow$ taking different values at different points $\vec{r}$ and in time $\rightarrow$ changing as time passes t experiences the electromagnetic, or Lorentz, force

$$\vec{F}_{\text{Lorentz}}(\vec{r}, \vec{v}, t) = q(\vec{E}(\vec{r}, t) + \vec{v} \times \vec{B}(\vec{r}, t)).$$

This force depends simultaneously on the particle's position $\vec{r}$, velocity $\vec{v}$ and time t .

Example. Near the Earth's surface, a body experiences a force called its weight, given by

$$\vec{F} = \vec{P} = m\vec{g}$$

where $\vec{g}$ is the gravitational acceleration (in m/s^2). Over a small region this force is approximately constant, independent of the body's position $\vec{r}$, velocity $\vec{v}$ and time t , and points towards the Earth's centre.

Among all the forces that one encounters in nature, whether fundamental or phenomenological, some depend only on their point of application, i.e. $\vec{F} = \vec{F}(\vec{r}) = (F_x(\vec{r}), F_y(\vec{r}), F_z(\vec{r}))$. Moreover, among these, some may be written in the form⁵

$$\vec{F} = - \left(\frac{\partial U(\vec{r})}{\partial x}, \frac{\partial U(\vec{r})}{\partial y}, \frac{\partial U(\vec{r})}{\partial z} \right), \quad (2)$$

Here $U(\vec{r})$ is a real-valued function of position $\vec{r}$, called *potential energy*. Forces of this kind are said to be *conservative*⁶ and that they derive from the potential energy $U(\vec{r})$.

Example. Universal gravitation is conservative, as is the uniform gravitational force $\vec{F} = -mg\vec{e}_z$, since it is constant and can be written as $\vec{F} = - \left(\frac{\partial U(\vec{r})}{\partial x}, \frac{\partial U(\vec{r})}{\partial y}, \frac{\partial U(\vec{r})}{\partial z} \right)$ with $U(\vec{r}) = mgz + C$. The magnetic component of the Lorentz force, $q\vec{v} \times \vec{B}$, does not derive from a scalar potential energy $U(\vec{r})$, but does no work because it is perpendicular to the velocity. A non-conservative force is therefore not necessarily dissipative. The electric component $q\vec{E}$ is conservative when the field is electrostatic.

⁵notation $\frac{\partial U(\vec{r})}{\partial x}$ means the partial derivative of $U(\vec{r})$ over x , defined by $\frac{\partial U(\vec{r})}{\partial x} \stackrel{\text{def}}{=} \lim_{\Delta x \rightarrow 0} \frac{U(x+\Delta x, y, z) - U(x, y, z)}{\Delta x}$.

⁶and not *conserving* as we hear too often.

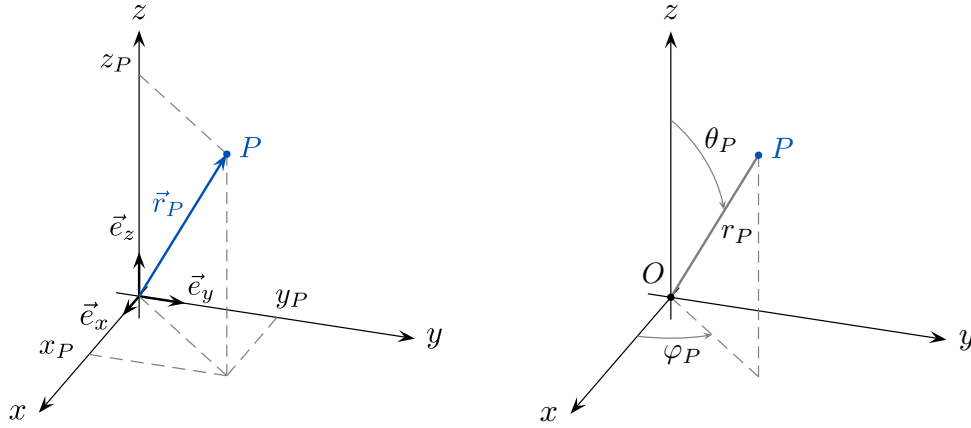


Figure 3: Orthonormal coordinate system $Oxyz$. Point P has Cartesian coordinates (x_P, y_P, z_P) (left) and spherical coordinates $(r_P, \theta_P, \varphi_P)$ (right). The conversion from spherical to Cartesian coordinates is given by $x_P = r_P \sin \theta_P \cos \varphi_P$, $y_P = r_P \sin \theta_P \sin \varphi_P$ and $z_P = r_P \cos \theta_P$. The position vector from the origin to point P is denoted $\vec{r}_P$.

4.2 Inertial frames

Let us begin by recalling the concept of reference frame introduced in the course of kinematics.

reference frame A *reference frame* consists of a set of spatial coordinate axes associated with an observer, together with a clock. An event is specified by three spatial coordinates and one time coordinate. Two reference frames may move relative to one another.

Example. In a system of Cartesian axes, an event taking place in a given place and at a given moment will be detected by the Cartesian coordinates x, y, z of its place and the temporal coordinates t of its moment.

inertial frame An *reference frame inertial* is a reference frame in which a particle subject to no external force remains at rest or moves with constant velocity. Such a frame is also called Galilean. Any frame whose axes remain parallel to those of an inertial frame and which moves at constant velocity relative to it is also inertial. Thus, knowing one inertial frame gives an infinite family of them.

Example. For experiments of short duration and limited spatial extent, a terrestrial reference frame can be treated in good approximation as inertial if the effects of Earth's rotation and orbital motion are negligible.

Galilean transformation The coordinate transformation between inertial frames in classical mechanics is called a Galilean transformation. Consider frames S and S' with parallel, non-rotating axes, moving at constant relative velocity $\vec{v}_0 = (v_{0,x}, v_{0,y}, v_{0,z})$ (Figure 4). The coordinates in S' are related to those in S as follows, provided S' and S coincide at time $t = 0$:

$$\text{Galilean transformation: } \begin{cases} t' = t \\ x' = x - v_{0,x}t \\ y' = y - v_{0,y}t \\ z' = z - v_{0,z}t \end{cases} \Rightarrow \begin{cases} v'_x = v_x - v_{0,x} \\ v'_y = v_y - v_{0,y} \\ v'_z = v_z - v_{0,z} \end{cases} \Rightarrow \boxed{\vec{a}' = \vec{a}}$$

In particular, acceleration is invariant under a Galilean transformation.

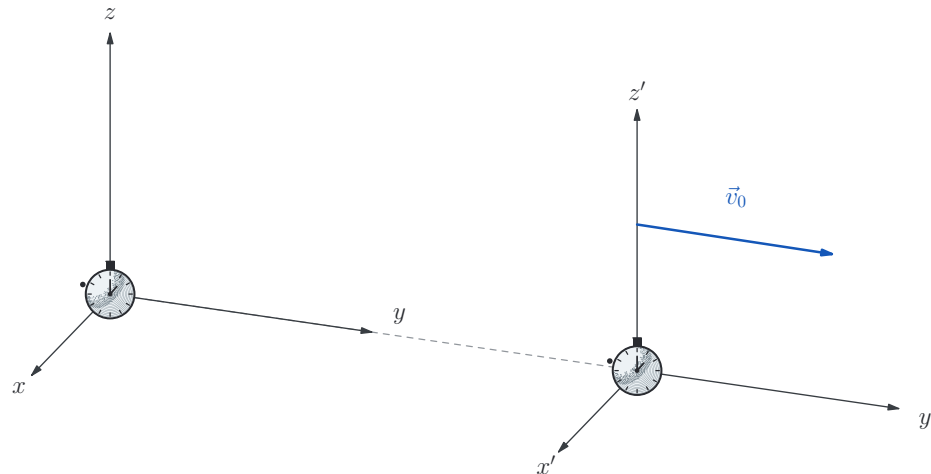


Figure 4: Two frames moving at constant relative velocity $\vec{v}_0$. If either frame is inertial, so is the other. In both frames, $\vec{a} = 0 \Leftrightarrow \vec{F} = 0$.

4.3 Newton's 3 Laws

THE PRINCIPLE OF INERTIA

NEWTON'S FIRST LAW

Inertial frames exist: reference frames in which a point particle that is free (subject to no external influence) remains at rest or in uniform rectilinear motion (URM). In such frames, any change in motion must result from one or more forces.

Newton's 1st law is also known as Galileo's principle. In Galileo's words, "Motion is motion and acts as motion only in relation to things that do not share it. For all things that participate in it, [...] motion is as though it did not exist. [...] Motion is like nothing." Thus two observers in uniform rectilinear relative motion cannot determine which of them is 'really' moving. Neither feels that motion, whatever their relative velocity. Uniform motion is indistinguishable from rest through experiments confined to the moving frame; changes in motion have observable effects.

Example. A person sitting in a train moving slowly at constant velocity cannot tell whether their own train or the train beside it is moving without looking at the platform.

Example. A juggler below deck on a large ship can juggle in the same way when the ship is stationary or moving at constant velocity.

FUNDAMENTAL RELATIONSHIP OF DYNAMICS

NEWTON'S SECOND LAW

In an inertial frame, the net force on a point particle equals its mass times its acceleration:

$$\vec{F} = \sum_i \vec{F}_i = m \vec{a} \quad (3)$$

Some remarks about the fundamental relationship of dynamics (3):

- ▷ In a given inertial frame, Newton's second law relates the net force to acceleration. Acceleration is invariant under a Galilean transformation; if a force law depends on velocity, its transformation must be examined explicitly.

- ▷ Mass is a positive scalar⁷ that, through equation (3), quantifies a particle's *resistance to acceleration*. For a fixed applied force, a greater mass gives a smaller acceleration. This resistance is called the particle's *inertia*.
- ▷ The force in relation (3) describes the particle's interaction with surrounding matter and fields. Its mathematical expression must be postulated from experimental evidence, as for gravitation between massive bodies or the Coulomb force between charged particles, so as to account for observed macroscopic phenomena. Only once a force law has been specified does equation (3) acquire its full meaning.⁸
- ▷ Newton's second law is a vector equation, so it is equivalent to 3 scalar equations:

$$\vec{F} = m \vec{a} \quad \Leftrightarrow \quad \begin{cases} F_x = m a_x \\ F_y = m a_y \\ F_z = m a_z \end{cases} \quad \text{in any system of Cartesian coordinates}$$

- ▷ Velocity is the first time derivative of position and acceleration is the second. Newton's second law can therefore be expressed entirely in terms of position and its time derivatives:

$$\vec{F} \left(\vec{r}(t), \frac{d\vec{r}(t)}{dt}, t \right) = m \frac{d^2 \vec{r}(t)}{dt^2} \quad (4)$$

In Cartesian components, Newton's second law gives three coupled second-order differential equations for $x(t), y(t), z(t)$. For a specified force law, the initial position $\vec{r}(0)$ and initial velocity $\vec{v}(0)$ determine the subsequent motion. These are the *initial conditions* required to obtain a unique solution of equation (4). In general, this system has no analytical solution, except in a few simple but important cases (planetary motion, the harmonic oscillator, free fall, ...). Many systems of interest are *chaotic*, and numerical methods are then required.

- ▷ Emphasise that law (3) holds only in *inertial* reference frames.
- ▷ Equation (3) in the form $\vec{F} = m\vec{a}$ assumes constant mass; it may then be rewritten as $\vec{F} = \frac{d}{dt}(m\vec{v}) = \frac{d\vec{p}}{dt}$. For an open system with variable mass, simply writing $\vec{F}_{\text{ext}} = d(m\vec{v})/dt$ is generally incorrect: the momentum balance must also account for momentum carried into or out of the system by matter.
- ▷ The classical form of equation (3) is a non-relativistic approximation. It fails when particle speed becomes comparable to the speed of light in vacuum, $c = 2.99792458 \times 10^8 \text{ m s}^{-1}$. The relativistic generalisation is

$$\vec{F} = \frac{d}{dt} \left(\frac{m_0 \vec{v}}{\sqrt{1 - (v/c)^2}} \right) \quad (\text{relativistic form})$$

where m_0 is the rest mass, independent of speed. Defining $\gamma = \frac{1}{\sqrt{1 - (v/c)^2}}$, with $v = |\vec{v}| < c$, gives relativistic momentum $\vec{p} = \gamma m_0 \vec{v}$ and force $\vec{F} = \frac{d\vec{p}}{dt}$. As v increases, momentum and energy increase while the mass m_0 remains constant. The required energy diverges as $v \rightarrow c$, so a massive particle cannot reach the speed of light c .

- ▷ Newton's first law is only a special case of (3): $\vec{F} = \vec{0} \Leftrightarrow \vec{a} = \vec{0}$

ACTION AND REACTION

⁷In Newtonian mechanics, the mass of a closed system is conserved and additive. The mass contained in an open system can vary by material exchange.

⁸Ernst Mach (1838-1916), the influential Austrian physicist, remarked: "It is readily recognised that laws I and II are contained in the definitions of force given above. According to these definitions, in the absence of any force only rest or uniform rectilinear motion is possible. After defining acceleration as the measure of force, repeating that the change in motion is proportional to force is a wholly unnecessary tautology. It would have sufficed to say that the definitions were not arbitrary mathematical definitions, but reflected experimentally observed properties of bodies."

NEWTON'S THIRD LAW

*When a particle 1 exerts a force $\vec{F}_{1/2}$ on a particle 2,
particle 2 exerts an equal and opposite force on particle 1 $\vec{F}_{2/1} = -\vec{F}_{1/2}$.*

Some remarks about Newton's 3rd law:

- ▷ In the context of Newtonian mechanics, this relationship concerns the two real forces of the same interaction.
- ▷ It is unchanged under a Galilean transformation. It does not generally apply to fictitious forces introduced in a non-inertial frame; in electromagnetism, the momentum carried by the field must also be included
- ▷ If $\vec{F}_{1/2}$ is called *action*, $\vec{F}_{2/1}$ will be called *reaction* and vice versa. Both forces being present *simultaneously*, one must not be seen as the cause of the other.
- ▷ Action and reaction are forces that are exerted on different particles and therefore have different application points
- ▷ $\vec{F}_{1/2}$ and $\vec{F}_{2/1}$ are opposed. Their collinearity with the line joining the points of application constitutes an additional hypothesis, called the strong form of Newton's 3rd law, satisfied in particular by the central forces
- ▷ This law is a postulate, not a result that can be proved

Remark. Newton's first two laws apply only in *inertial* frames. In accelerating frames, including rotating or freely falling frames, the appropriate inertial forces must be included to obtain predictions consistent with observations.

Newton's laws can initially seem counter-intuitive. Gravity often makes trajectories curved, while friction causes many moving bodies to slow down and stop. Both effects are fully consistent with Newton's laws once all the forces acting on a body, including gravity and friction, are taken into account.

4.4 Mass and Weight

The concept of mass covers several independent physical concepts *a priori*.

Inertial mass The *inertial mass* of a body appears in Newton's second law and measures its resistance to acceleration. It is an intrinsic property and must not be confused with amount of substance, which is measured in moles.⁹ It is a non-negative scalar, expressed in kilograms in the SI.

Gravitational mass The *gravitational mass* of a body is the scalar quantity appearing in the gravitational force law:

$$\vec{F}_{2/1} = G \frac{m_1 m_2}{r_{12}^2} \vec{e}_{\vec{r}_{12}} = -\vec{F}_{1/2}$$

where G is the universal gravitational constant, $\vec{r}_{12} = \vec{r}_2 - \vec{r}_1$, $r_{12} = |\vec{r}_{12}|$, and $\vec{e}_{\vec{r}_{12}} = \vec{r}_{12}/r_{12}$ points from particle 1 towards particle 2. The expression gives the force exerted by a body of mass m_2 on a body of mass m_1 , separated by distance r_{12} .

⁹In relativistic mechanics, mass is an intrinsic, invariant property of a body: it does not depend on velocity.

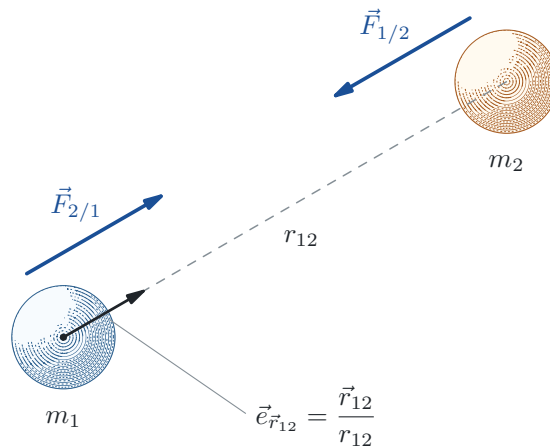


Figure 5: Universal gravity force.

The concepts of inertial and gravitational mass are brought together in general relativity. Its equivalence principle postulates that these masses are equal. Experiments support this equality, so it is customary to speak simply of *the mass* of a body, without specifying which kind.

Weight The *weight* of a body, denoted $\vec{P}$, is the force pulling it towards the ground in a terrestrial reference frame. Weight is a vector quantity measured in newtons. For a body of mass m , it is related to gravitational acceleration $\vec{g}$ by $\vec{P} = m\vec{g}$.

Remark. Weight is defined similarly on any celestial body (the Moon, Mars, ...). For example, an object weighs about six times less on the Moon than on Earth (see Table 2).

Place	g [m/s ²]
Earth (equator)	9.78
Earth (poles)	9.83
Earth (sea level)	9.81
Earth (Mount Everest)	9.78
Moon	1.62
Mars	3.80
Jupiter	25.90

Table 2: Acceleration of gravity in different places.

It should also be noted that the acceleration of gravity and weight depend on where you are on Earth. They vary slightly with latitude and altitude (see Table 2).

Apparent weight The *apparent weight* of a body is the reading of a weighing device. It equals the magnitude of the force exerted by its support on the body, and, by reaction, by the body on its support. In the figure below, this is the tension T measured by the spring balance. For the downward acceleration shown, we assume $0 \leq a < g$.

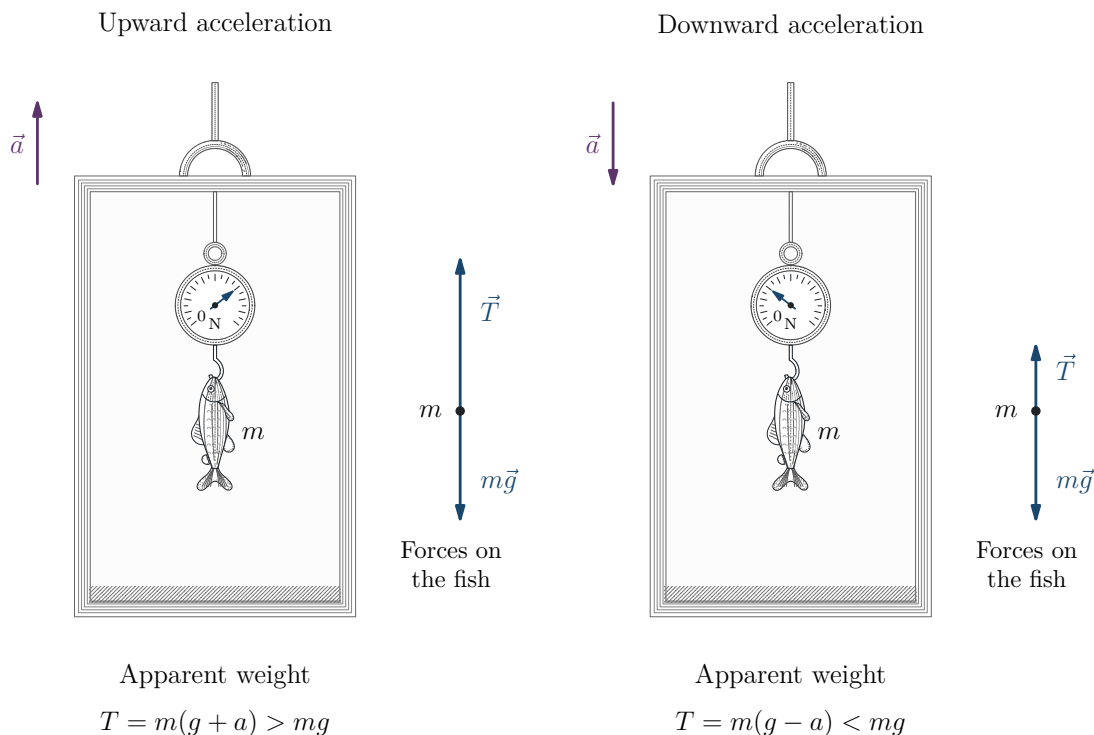


Figure 6: Mass, weight and apparent weight of a fish suspended in an accelerating lift.

4.5 Friction forces

Dry friction Dry friction is a contact force between solid surfaces that opposes their relative sliding or tendency to slide.

A dry-friction force never occurs alone (Newton's third law). Two bodies in contact exert equal and opposite frictional forces on each other. Figure 7 summarises their essential properties. The dimensionless coefficients of friction depend on the surfaces in contact. Without relative slipping, static friction adjusts to the applied forces according to $0 \leq |\vec{F}_{f,s}| \leq \mu_s N$. During sliding, kinetic friction approximately satisfies $|\vec{F}_{f,c}| = \mu_c N$ and opposes the relative velocity at the contact. Sticking and slipping refer to relative motion at the contact, not to rest or motion in an arbitrary frame. Table 2 gives some friction coefficients.

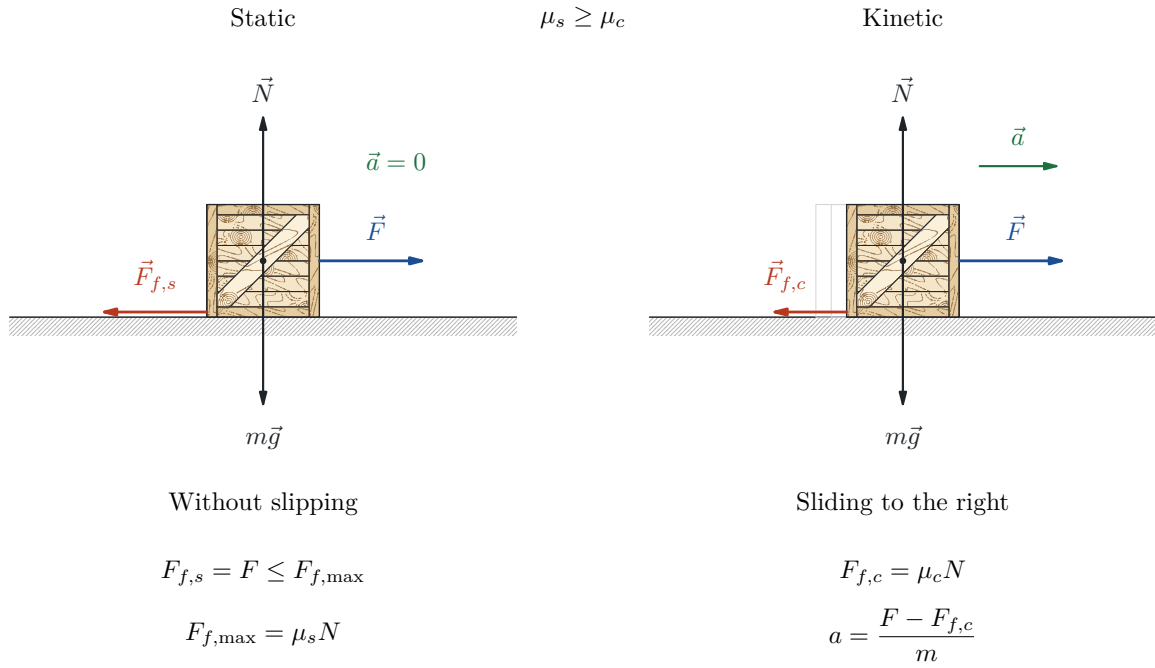


Figure 7: Static friction (left) and kinetic friction (right). Note that $\vec{F}_f$ and $\vec{N}$ point in different directions: the relations above concern force magnitudes only.

Contact surfaces	μ_s	μ_c
rubber on concrete	1.0	0.8
aluminium on steel	0.61	0.47
steel on steel	0.74	0.57
glass on glass	0.94	0.4
ice on ice	0.1	0.03

Table 3: Static and kinetic friction coefficients (numbers without dimension) for a few surfaces.

Fluid drag Fluid drag is the resistance experienced by a body moving relative to a fluid, whether a liquid or gas. Here $\vec{v} = \vec{v}_{\text{body}} - \vec{v}_{\text{fluid}}$ denotes the body's velocity relative to the fluid. In general, drag depends in a complex way on speed, fluid properties, body shape and other parameters.

However, when the body moves at low velocity relative to the fluid (regime *laminar*), the friction force follows in good approximation the following law

$$\vec{F}_f = -\alpha \vec{v}$$

where $\alpha > 0$ is a coefficient of friction (in kg/s) which depends on the nature of the fluid and the geometry of the body. In this case the intensity of the force is proportional to the velocity v between the body and the fluid. For example, in the case of a sphere, we have $\alpha = 6\pi\mu R$ with μ the dynamic viscosity of the fluid (in $\text{kg m}^{-1} \text{s}^{-1}$) and R the radius of the sphere.

At higher speeds relative to the fluid (*turbulent* regime), drag is better described by

$$\vec{F}_f = -\frac{1}{2} \rho v^2 S C_x \vec{e}_{\vec{v}},$$

where ρ is the fluid density (kg/m^3), S the frontal area (m^2), and $C_x > 0$ the dimensionless drag coefficient. The force magnitude is then proportional to the square of the speed. Table 3 gives some drag coefficients. The

dimensionless Reynolds number¹⁰,

$$\text{Re} = \frac{vd\rho}{\eta},$$

distinguishes the low- and high-speed regimes. Here v is the speed relative to the fluid, d a characteristic body size, ρ the fluid density and η its dynamic viscosity (in $\text{kg m}^{-1} \text{s}^{-1}$). The low-speed regime corresponds to $\text{Re} \ll 1$, and the high-speed regime to $\text{Re} \gg 1$.

Body 3D	Shape	Drag coefficient
sphere		0.47
hemisphere		0.42
cone		0.50
cube		1.05
streamlined		0.04

Table 4: Drag coefficient C_x (dimensionless) for some three-dimensional bodies.

Fact. *Friction is a non-conservative contact force that opposes relative sliding or the tendency to slide.*

Fact. *In the inertial frame where the ground is motionless, the work of the kinetic friction exerted by the ground on an object that slides is negative. If the support is mobile, or in another reference frame, the work of this force on a single object can be positive, zero or negative. Static friction can also provide positive, zero or negative work on a solid; it is not automatically dissipative. When there is slipping, the pair of friction forces usually converts the mechanical energy into internal energy of the bodies in contact and the medium.*

¹⁰The Reynolds number can be seen as the ratio between the relative velocity of the body relative to the fluid and a characteristic velocity that separates the laminar and turbulent regimes. This characteristic velocity depends on the properties of the fluid and the body. Specifically, we can write $\text{Re} = \frac{vd\rho}{\eta} = \frac{v}{v_c}$ with $v_c = \frac{\eta}{d\rho}$. For a ball of diameter $d = 1 \text{ mm}$ moving in the water at ambient temperature under atmospheric pressure (for which $\eta \approx 10^{-3} \text{ kg m}^{-1} \text{s}^{-1}$ and $\rho \approx 10^3 \text{ kg/m}^3$), the characteristic velocity is $v_c = 10^{-3} \text{ m/s}$.

CHAPTER 5

POWER, WORK, KINETIC AND POTENTIAL ENERGIES

5.1 Review of Forces

5.1.1 3d vs 1d

For one-dimensional motion, vectors need not be used explicitly. The net force $\vec{F}^1$ has a fixed direction along the line of motion. Choose an axis, say Ox , parallel to the net force $\vec{F}$; then

$$\vec{F} = F_x \vec{e}_x \quad \text{and it is usually written} \quad \vec{F} = F \vec{e}_x.$$

In one dimension, F usually denotes the component of the net force $\vec{F}$ along the line of motion. This component may be positive, zero or negative, whereas the magnitude $|\vec{F}|$ is non-negative.

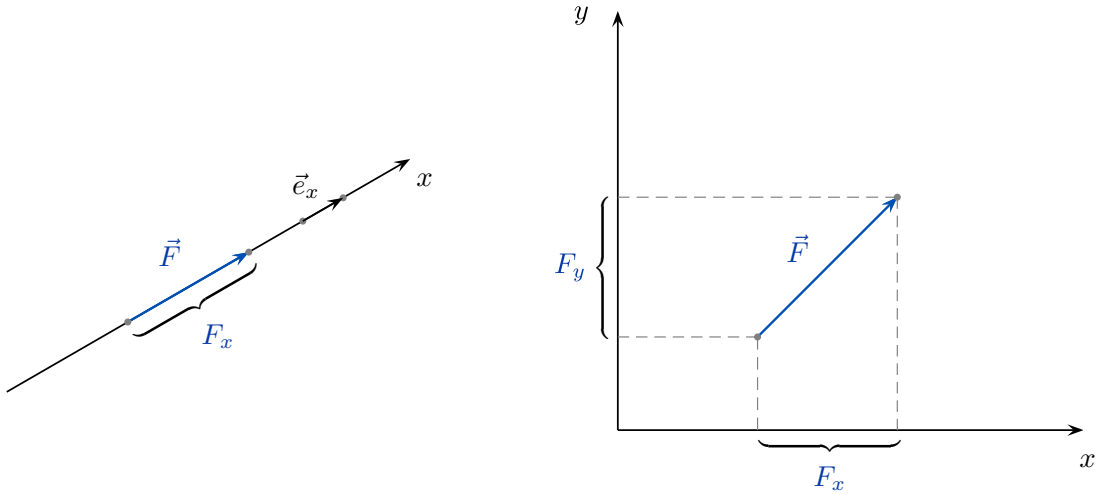


Figure 1: Left: at one dimension, a force is fully specified by its projection on an axis (Ox) which is parallel to it, $F_x = \vec{F} \cdot \vec{e}_x$. If $\vec{F}$ and $\vec{e}_x$ are of the same direction, $F_x = |\vec{F}| > 0$, and if $\vec{F}$ and $\vec{e}_x$ are of the opposite direction, $F_x = -|\vec{F}| < 0$. Right: at more than one dimension, a force is specified by its projections on orthogonal axes.

5.1.2 Constant forces and variable forces

Generally speaking, a force that is exerted on a particle may depend on its position $\vec{r}(t)$, its velocity $\vec{v}(t) = \frac{d\vec{r}(t)}{dt}$, and, in very rare cases, on the derivatives of higher orders of its position. A force may also depend on time, as when a charged particle is plunged into a variable electric field $\vec{E}(t)$. The force can therefore be seen as a function of several variables: $\vec{F} = \vec{F}(\vec{r}, \vec{v}, \dots, t)$.² Since for a given movement, it is possible to see the position and velocity of a particle as physical quantities that depend only on time, we can write the force as a function of the only time variable t , that is $\vec{F}(t) = \vec{F}(\vec{r}(t), \vec{v}(t), \dots, t)$.

¹a vector quantity

²The notation $\vec{F} = \vec{F}(\vec{r}, \vec{v}, \dots, t)$ means that the force depends on vector quantities such as the position $\vec{r}$ and velocity $\vec{v}$ of the particle on which it is exercised, as well as time t . Equivalently, $\vec{F} = \vec{F}(x, y, z, v_x, v_y, v_z, \dots, t)$ will be written in terms of cartesian components of $\vec{r}$ and $\vec{v}$.

Constant force A force is said to be *constant* if $\vec{F}(\vec{r}, \vec{v}, \dots, t) = \vec{F}_0$ with $\vec{F}_0$ a constant vector, or equivalently if $\vec{F}(t) = \vec{F}_0$ at any time t and for any movement.

A force that is not constant is said *variable*.

Example. In the approximation of a uniform gravity field near the Earth's surface, the gravity force $\vec{F} = m\vec{g}$ is constant. In this approximation, it does not depend on the time, position, or velocity of a particle.

Example. The electrostatic (Coulomb) force $\vec{F}_{\text{Coulomb}} = q\vec{E}_0$ on a particle of charge q in a uniform electrostatic field $\vec{E}_0$ is constant.

Example. The gravitational force between the Earth and a particle of mass m varies with their relative position.

5.2 Power, work and kinetic energy

When several forces act on a particle, only the component of their resultant along its velocity $\vec{v}$ produces tangential acceleration and changes its speed $v = |\vec{v}|$. This motivates the definition of power: the rate at which a force transfers energy to the particle.

Power The *power* of a force, the point of application of which moves at a velocity $\vec{v}$, is defined by

$$P \stackrel{\text{def}}{=} \vec{F} \cdot \vec{v} \quad (\text{power})$$

The power of the net force on a particle moving at velocity $\vec{v}$ is proportional to its tangential component, and therefore to the particle's tangential acceleration. Indeed, the dot product $\vec{F}_{\text{tot}} \cdot \vec{v}$ can be written as

$$\vec{F}_{\text{tot}} \cdot \vec{v} = F_{\text{tot},t} v = m a_t v$$

Applying Newton's second law to the net force ($\vec{F}_{\text{tot}} = m \frac{d\vec{v}}{dt}$ in an inertial frame) gives the following expression for power³

$$\vec{F}_{\text{tot}} \cdot \vec{v} = m \frac{d\vec{v}}{dt} \cdot \vec{v} = \frac{d}{dt} \left(\frac{1}{2} m \vec{v} \cdot \vec{v} \right) \quad (1)$$

The right-hand side of (1) suggests defining a new physical quantity: kinetic energy.

kinetic energy The kinetic energy of a particle of mass m and velocity $\vec{v}$ is defined by

$$K \stackrel{\text{def}}{=} \frac{1}{2} m \vec{v} \cdot \vec{v} = \frac{1}{2} m |\vec{v}|^2 \quad (\text{kinetic energy})$$

Kinetic energy is the energy associated with a particle's motion. For a particle at rest, $K = 0$. The greater its speed, the greater its kinetic energy. Kinetic energy is not absolute: it depends on the chosen inertial frame. Power and work also depend on the frame. Potential energy, defined later in this chapter, is determined only up to an additive constant.

Given the previous definition, the relationship (1) can still be written

$$P_{\text{tot}} = \vec{F}_{\text{tot}} \cdot \vec{v} = \frac{dK}{dt}$$

³With more details, we have

$$\frac{d}{dt} \left[\frac{1}{2} m \vec{v} \cdot \vec{v} \right] = \frac{d}{dt} \left[\frac{1}{2} m (v_x^2 + v_y^2 + v_z^2) \right] = \frac{1}{2} m \left[\frac{d}{dt} (v_x^2) + \frac{d}{dt} (v_y^2) + \frac{d}{dt} (v_z^2) \right] = \frac{1}{2} m \left[2v_x \frac{dv_x}{dt} + 2v_y \frac{dv_y}{dt} + 2v_z \frac{dv_z}{dt} \right] = m \vec{v} \cdot \frac{d\vec{v}}{dt}$$

The power of the resulting forces exerted on a particle is therefore equal to the rate of variation of the kinetic energy of the particle. By integration over a time interval $[t_i, t_f]$, we finally obtain

$$\Delta K_{if} \equiv K(t_f) - K(t_i) = \int_{t_i}^{t_f} \vec{F}_{\text{tot}} \cdot \vec{v} dt \quad (2)$$

The right-hand side of (2) also suggests defining another physical quantity: the work done by a force.

Work done by a force (general case) The work done by a force whose power is $P(t)$ over a time interval $[t_i, t_f]$ is defined by

$$W_{if} \stackrel{\text{def}}{=} \int_{t_i}^{t_f} P(t) dt = \int_{t_i}^{t_f} \vec{F} \cdot \vec{v} dt \quad (\text{work})$$

Remark. In the above definition of work, force and velocity are seen as functions of time: $\vec{F} = \vec{F}(\vec{r}(t), \vec{v}(t), t)$ and $\vec{v} = \vec{v}(t)$.

Let's summarize our path (“ $\rightarrow$ ” indicating a logical link):

change in $|\vec{v}| \rightarrow$ tangential acceleration ($\vec{a}_t$) $\rightarrow$ power of a force (P) $\rightarrow$
kinetic energy of a particle (K) and work done by a force (W)

The following result, derived from previous developments, is a fundamental relationship in mechanics.

WORK-KINETIC ENERGY THEOREM

The work of total force on a particle over a time interval $[t_A, t_B]$ is equal to the variation of kinetic energy of that particle over the same time interval, i.e.

$$W_{AB} = \Delta K_{AB} = K_B - K_A$$

This relationship is valid regardless of the nature of the forces present (conservative or non-conservative), provided that W_{AB} refers to the sum of their work.

Remark. The work-kinetic energy theorem is valid for all forces (friction force, gravity force, constant force, electrical force, magnetic force, pressure force, ...).

Example. A particle of mass m oscillates along the Ox axis with position $x(t) = A \sin(\Omega t + \varphi)$. Calculate the net force, its power and its work over one complete oscillation.

The net force on the particle is $\vec{F} = m\vec{a} = -m A \Omega^2 \sin(\Omega t + \varphi) \vec{e}_x$, using $\vec{a} = d^2\vec{r}/dt^2$, and its power is $P(t) = \vec{F}(t) \cdot \vec{v}(t) = (-m A \Omega^2 \sin(\Omega t + \varphi) \vec{e}_x) \cdot (A \Omega \cos(\Omega t + \varphi) \vec{e}_x) = -m A^2 \Omega^3 \sin(\Omega t + \varphi) \cos(\Omega t + \varphi)$. Its work over the interval $[0, 2\pi/\Omega]$, corresponding to one complete oscillation, is

$$W = -m A^2 \Omega^3 \int_0^{2\pi/\Omega} \sin(\Omega t + \varphi) \cos(\Omega t + \varphi) dt = 0$$

This result is expected: after one period the particle has the same speed, so its change in kinetic energy is zero and $W = \Delta K = 0$ by the work-kinetic energy theorem.

5.3 Work of a force along a path

In this section, we consider only forces that depend only on the position of their point of application (i.e. the position of the particle on which they are exerted): $\vec{F} = \vec{F}(\vec{r})$. It is important to note that this does not mean that these forces are necessarily conservative (see Section 5.4).

5.3.1 Work done by a force $\vec{F} = \vec{F}(\vec{r})$

Consider a particle moving from position A at time t_A to position B at time t_B . By definition, the work done by force $\vec{F}$ over interval $[t_A, t_B]$ is

$$W_{AB} = \int_{t_A}^{t_B} \vec{F} \cdot \vec{v} dt$$

Direct evaluation requires the velocity $\vec{v}$ of the force's point of application, here the particle itself. Often the detailed time dependence of the motion is unknown. When the force depends only on position, a change of variable $\vec{r} = \vec{r}(t)$ replaces the time integral of power by an integral along the path:

$$W_{AB} = \int_{t_A}^{t_B} \vec{F} \cdot \vec{v}(t) dt = \int_{t_A}^{t_B} \vec{F} \cdot \frac{d\vec{r}(t)}{dt} dt = \int_A^B \vec{F} \cdot d\vec{r}$$

that is

$$\boxed{W_{AB} = \int_A^B \vec{F} \cdot d\vec{r}} \quad (\text{work along a path})$$

We now see that work W_{AB} can be calculated solely on the basis of knowledge of the dependence of force on position ($\vec{F}(\vec{r})$) and of the path that connects point A to point B (the trajectory of the point of application of force).

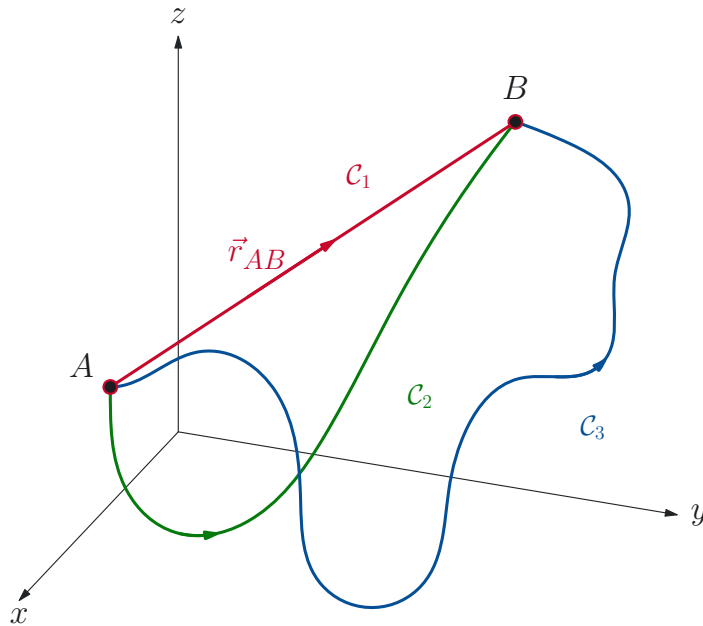


Figure 2: Three different paths C_1, C_2 and C_3 linking two points A and B , along which the work of a force can be calculated. The straight line path C_1 , represented in red, is along the displacement vector $\vec{r}_{AB} = \vec{r}_B - \vec{r}_A$. For conservative forces, the work along the three paths (red, blue and green) has the same value.

5.3.2 Case of constant force

For a constant force, $\vec{F} = \vec{F}_0$ the work can be calculated explicitly. We have

$$W_{AB} = \int_A^B \vec{F}_0 \cdot d\vec{r} = \vec{F}_0 \cdot \int_A^B d\vec{r} = \vec{F}_0 \cdot (\vec{r}_B - \vec{r}_A)$$

where $\vec{r}_{AB} \equiv \vec{r}_B - \vec{r}_A$ is the displacement vector linking the point A to the point B . The work of a constant force can still be written

$$\boxed{W_{AB} = \vec{F}_0 \cdot \vec{r}_{AB} = F_0 r_{AB} \cos \theta}$$

with θ the angle between the force vector and the displacement vector.

Example. Calculate the work done by gravity $\vec{F} = -mg\vec{e}_z$ on a particle of mass m moving from A to B .

The force of gravity $\vec{F} = -mg\vec{e}_z$ is a constant force. Therefore, its work at the end of the displacement of the particle from a point A to a point B is equal to $W_{AB} = -mg\vec{e}_z \cdot \vec{r}_{AB} = mg(z_A - z_B)$, regardless of the path followed.

5.3.3 Case of a 1d movement

In one dimension, the work done by force $\vec{F} = F_x \vec{e}_x$ from point A to point B along the Ox axis reduces to

$$W_{AB} = \int_{x_A}^{x_B} F_x dx.$$

The work–kinetic energy theorem keeps the form ($W_{AB} = \Delta K_{AB}$), with $K = \frac{1}{2}mv_x^2$ the kinetic energy for motion along the Ox axis.

Example. Calculate the work done by a constant force $\vec{F}_0 = F_0 \vec{e}_x$ on a particle of mass m , initially at rest, over time interval $[0, t]$.

The constant force $\vec{F}_0 = F_0 \vec{e}_x$ acting on mass m produces acceleration $\vec{a} = \vec{F}/m = (F_0/m) \vec{e}_x$. The particle therefore undergoes uniformly accelerated rectilinear motion (UARM), for which

$$x(t) = x_0 + \frac{1}{2} \frac{F_0}{m} t^2, \quad v_x(t) = \frac{F_0}{m} t$$

After time t , the particle has travelled distance $d = \frac{1}{2} \frac{F_0}{m} t^2$ and acquired kinetic energy $K = \frac{1}{2}mv^2 = \frac{1}{2}m(\frac{F_0}{m}t)^2 = \frac{1}{2} \frac{F_0^2}{m} t^2 = F_0 d$. The work done over interval $[0, t]$ is $W = \int_0^d F_x dx = \int_0^d F_0 dx = F_0 d$, confirming $\Delta K = W$.

Example. A body of mass m and initial speed v_0 slides on a rough horizontal surface. The coefficient of kinetic friction is μ_c . Find the speed v after it has travelled a distance d .

We determine v using an energy balance. The normal force does no work because $\vec{N}$ and $\vec{v}$ are perpendicular, and gravity also does no work. The work done by friction is $W = \int_0^d F_x dx = \int_0^d -\mu_c N dx = -\mu_c N d$, with $N = mg$ from vertical force balance. The work–kinetic energy theorem gives $\Delta K = W$, or $\frac{1}{2}mv^2 - \frac{1}{2}mv_0^2 = -\mu_c mgd$, hence

$$v = \sqrt{v_0^2 - 2\mu_c g d}$$

This expression applies for $0 \leq d \leq d_{\text{stop}} = v_0^2/(2\mu_c g)$. Beyond that distance, the body has stopped and $v = 0$.

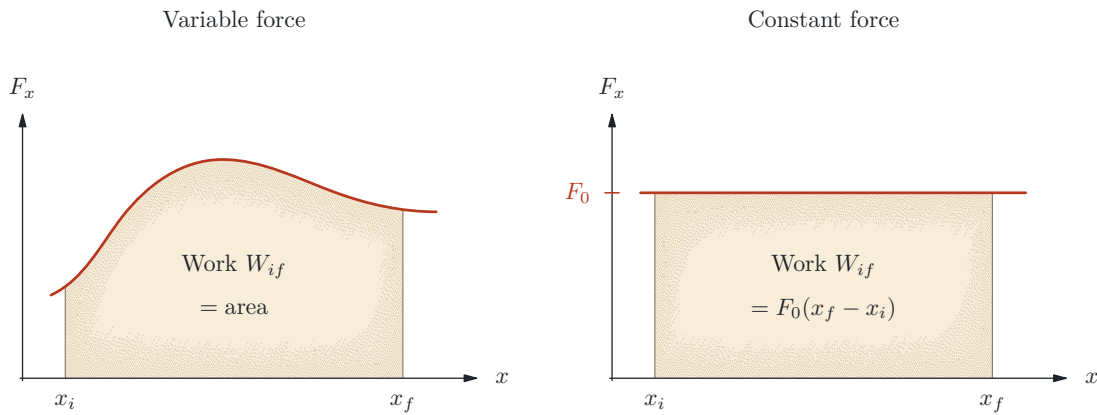


Figure 3: Work of a force (case 1d) as integral $W_{if} = \int_{x_i}^{x_f} F_x dx$ representing the algebraic area under the curve $F_x(x)$. For a constant force, the work is simply equal to $W_{if} = F_x(x_f - x_i) = F_x \Delta x$ with $\Delta x = x_f - x_i$ the algebraic displacement of the point of application of the force.

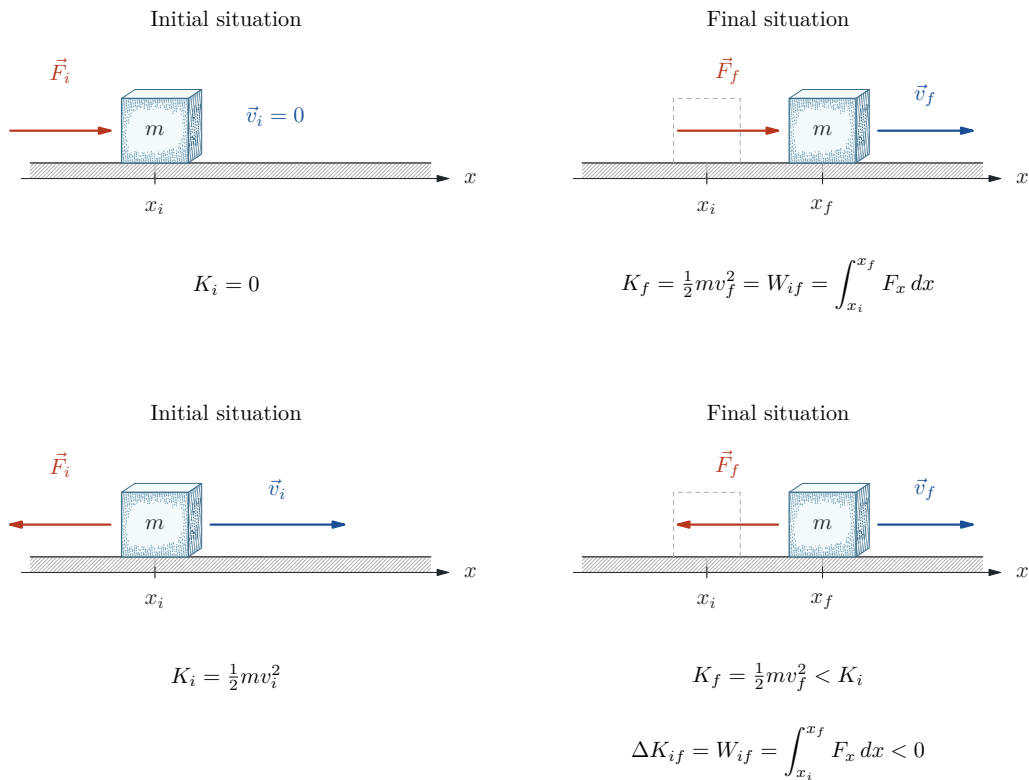


Figure 4: Work done along a one-dimensional path from x_i to x_f : positive work above and negative work below.

5.4 Conservative forces and potential energy

conservative force A force is *conservative*⁴ if it depends only on the position of the particle, $\vec{F} = \vec{F}(\vec{r})$, and its work between an initial point A and a final point B depends only on these endpoints, not on the path taken.

⁴In French, the correct technical adjective is “conservative”, not “conservatrice”: the latter suggests political conservatism rather than the physical property discussed here.

For conservative forces⁵, it is always possible to write

$$W_{AB} \stackrel{\text{def}}{=} U_A - U_B = -\Delta U_{AB} \quad (\text{work-potential energy relation})$$

with $U_A = U(\vec{r}_A)$ and $U_B = U(\vec{r}_B)$ where

$U(\vec{r})$ is the *potential energy* associated with conservative force $\vec{F}(\vec{r})$

The relationship between force and potential energy is given, in three dimensions, by

$$\vec{F}(\vec{r}) = \left(-\frac{\partial U(\vec{r})}{\partial x}, -\frac{\partial U(\vec{r})}{\partial y}, -\frac{\partial U(\vec{r})}{\partial z} \right) \quad (3)$$

and, at one dimension, by

$$F_x = -\frac{dU(x)}{dx}$$

For a non-conservative force, the work W_{AB} between A and B may depend on the path. If the force depends on time or velocity, it may also depend on how that path is traversed. There is therefore no single potential energy function satisfying $W_{AB} = U_A - U_B$. A non-conservative force need not be dissipative: its work can be negative, zero or positive.

CONSERVATIVE FORCE	POTENTIAL ENERGY
uniform gravity	gravitational
universal gravitation	gravitational
restoring	elastic
electrostatic	electric

Table 1: Examples of conservative forces and associated potential energies.

CONSERVATIVE FORCES	NON-CONSERVATIVE FORCES
gravity force, universal gravity force, restoring force, coulomb force	solid friction force, fluid drag force, external driving force explicitly dependent on time, electrical force due to induced electric field

Table 2: Examples of conservative forces and non-conservative forces.

Example. Let's calculate the work of a constant force, let's say the force of gravity $\vec{F}_0 = -mg\vec{e}_z$ which is exerted on a mass body m moving along the vertical axis Oz from a point A of height z_A to a point B of height z_B . Let's consider two different paths linking A to B : a path C_1 direct from A to B and a path C_2 from A to C (higher than B), and then from C to B . The work along these two paths is given respectively by

$$\begin{aligned} \text{Path } C_1 : W_{AB} &= \int_{z_A}^{z_B} (-mg) dz = mg(z_A - z_B) \\ \text{Path } C_2 : W_{AB} &= \int_{z_A}^{z_C} (-mg) dz + \int_{z_C}^{z_B} (-mg) dz = mg(z_A - z_C) + mg(z_C - z_B) = mg(z_A - z_B) \end{aligned}$$

We see that, in both cases, work has the same value, which depends only on the initial and final heights. The work is therefore independent of the path, because the force of gravity is conservative. The potential energy

⁵and only for conservative forces

associated with the force of gravity is such that $W_{AB} = U_A - U_B$, that is $mg(z_A - z_B) = U_A - U_B$ and we can choose

$$U(\vec{r}) = mgz + \text{constant}$$

We have moreover

$$\vec{F}(\vec{r}) = \left(-\frac{\partial U(\vec{r})}{\partial x}, -\frac{\partial U(\vec{r})}{\partial y}, -\frac{\partial U(\vec{r})}{\partial z} \right) = (0, 0, -mg) = -mg \vec{e}_z$$

Example. Raising a body of mass m through height h using a rope, an inclined plane and pulleys.

If the body leaves and arrives at rest, and the ropes and pulleys are ideal and the frictions neglected, the work of the external agent is the same in all cases.

$$W_{\text{ext}} = mgh$$

Coulomb Force Consider the Coulomb force that a charged particle q_1 exerts on another charged particle, of charge q_2 . The Coulomb force of q_1 on q_2 has for expression

$$\vec{F}_{q_1/q_2}(\vec{r}_{q_2}) = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{q_1 q_2}^2} \vec{e}_{q_1 q_2}$$

where ϵ_0 is the permittivity of the vacuum, $r_{q_1 q_2} = |\vec{r}_{q_1 q_2}|$ is the distance that separates q_1 and q_2 , and $\vec{e}_{q_1 q_2}$ is the unit vector of q_1 towards q_2 . By noting $\vec{r}_{q_1}$ and $\vec{r}_{q_2}$ the positions of the charges, we have

$$r_{q_1 q_2} = |\vec{r}_{q_2} - \vec{r}_{q_1}| \quad \text{and} \quad \vec{e}_{q_1 q_2} = \frac{\vec{r}_{q_2} - \vec{r}_{q_1}}{|\vec{r}_{q_2} - \vec{r}_{q_1}|}$$

The Coulomb force is conservative because it can be written in the form (3) with the Coulomb potential energy

$$U(\vec{r}_{q_2}) = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{q_1 q_2}} + \text{constant}$$

FORCE	WORK
$\vec{F} = \vec{F}(\vec{r}, \vec{v}, \dots, t)$	$W_{AB} = \int_{t_A}^{t_B} \vec{F} \cdot \vec{v} dt$
$\vec{F} = \vec{F}(\vec{r})$	$W_{AB} = \int_A^B \vec{F} \cdot d\vec{r}$
conservative $\vec{F}(\vec{r}) = \left(-\frac{\partial U}{\partial x}, -\frac{\partial U}{\partial y}, -\frac{\partial U}{\partial z} \right)$	$W_{AB} = -\Delta U_{AB}$
non-conservative	W_{AB} depends in general on the path $A \rightarrow B$ and, if the force depends on time or velocity, on how it is travelled
constant ($\vec{F} = \vec{F}_0$)	$W_{AB} = \vec{F}_0 \cdot \vec{r}_{AB}$
centripetal ($\vec{F} \perp \vec{v}$)	$W_{AB} = 0$
unidirectional ($\vec{F} = F_x \vec{e}_x$)	$W_{AB} = \int_{x_A}^{x_B} F_x(x') dx'$

Table 3: Work done by a force: summary of different cases.

5.5 Conservation of mechanical energy

Définition. The mechanical energy of a system is defined by

$$E \stackrel{\text{def}}{=} K + U$$

where U is the sum of the potential energies associated with conservative forces. The work-kinetic energy theorem leads to the following results.

ENERGY CONSERVATION AND ENERGY BALANCE

The mechanical energy of a system remains constant when the total work of non-conservative forces is zero, especially when only conservative forces act on it, i.e.

$$E = K + U = \text{constant} \quad \Leftrightarrow \quad \Delta(K + U)_{AB} = 0 \quad (\text{conservation of energy})$$

More generally, the change in mechanical energy of a system subject to conservative forces and non-conservative forces equals the work done by the non-conservative forces

$$\Delta E_{AB} \equiv \Delta(K + U)_{AB} = W_{AB}^{\vec{F}_{\text{non-conservative}}} \quad (\text{energy balance})$$

Remark. The total work done by non-conservative forces equals the change in mechanical energy. Such work may be negative, zero or positive. Dissipative friction usually converts mechanical energy into internal energy of the bodies in contact and their surroundings. When all forms of energy are included, *total* energy is conserved in an isolated system. Heat is a mode of energy transfer, not a form of stored energy.

restoring force Consider the restoring force exerted by a spring on a body of mass m attached to its end. The force obeys *Hooke's law*

$$F_r = -kx,$$

where x is the change in spring length and k the spring constant. Its work from A (coordinate x_A) to B (coordinate x_B) is

$$W_{A \rightarrow B} = \int_{x_A}^{x_B} F_r dx = - \int_{x_A}^{x_B} kx dx = -\frac{1}{2}k(x_B^2 - x_A^2)$$

Similarly, the work along path $A \rightarrow C \rightarrow B$ is

$$W_{A \rightarrow C \rightarrow B} = \int_{x_A}^{x_C} F_r dx + \int_{x_C}^{x_B} F_r dx = -\frac{1}{2}k(x_C^2 - x_A^2) - \frac{1}{2}k(x_B^2 - x_C^2) = -\frac{1}{2}k(x_B^2 - x_A^2)$$

Since the result depends only on the endpoints, the force has an elastic potential energy (Figure 5)

$$U(x) = \frac{1}{2}kx^2 + \text{constant}$$

satisfying $W_{AB} = -\Delta U_{AB} = U(x_A) - U(x_B)$. Without friction, the «mass-spring» system conserves its mechanical energy:

$$E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \text{constant}$$

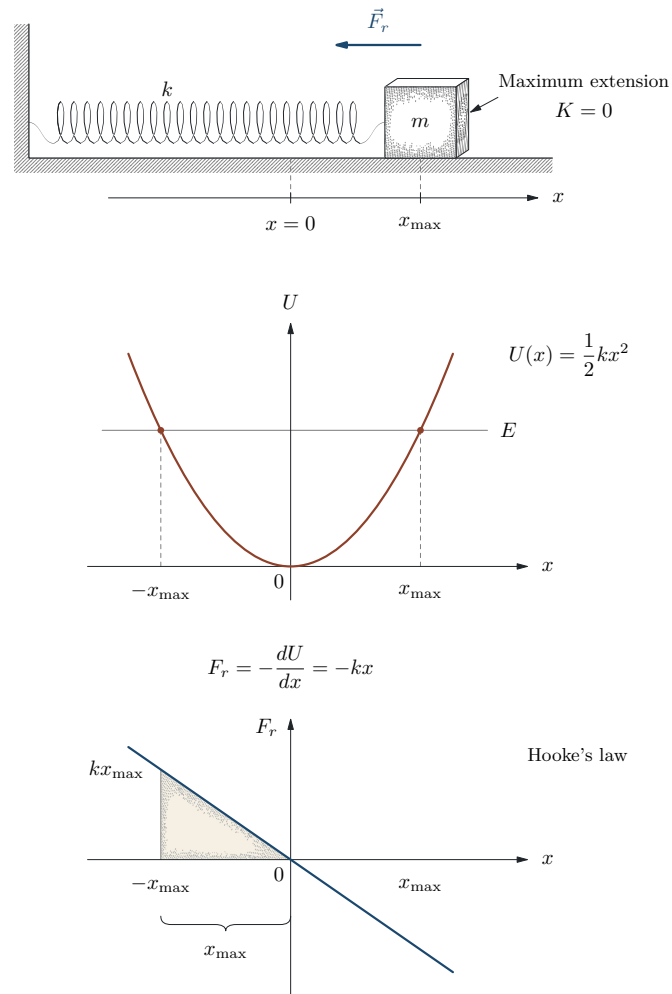


Figure 5: Mass-spring system (top), restoring force given by Hooke's law (bottom), and harmonic potential energy (middle).

Remark. Differentiating energy conservation, $\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \text{constant}$, with respect to time gives $v(ma + kx) = 0$. For a non-trivial trajectory, this yields $ma = -kx$ wherever $v \neq 0$, and at the turning points by continuity.

CHAPTER 6

MOMENTUM, CENTRE OF MASS AND COLLISIONS

6.1 Momentum

6.1.1 Particle

The momentum of a point particle of mass m and velocity $\vec{v}$ is defined by

$$\boxed{\vec{p} \stackrel{\text{def}}{=} m\vec{v}}$$

Let's start with a few remarks about this definition:

- ▷ $\vec{p}$ is a vector pointing along the velocity $\vec{v}$ (since $m > 0$ for matter or antimatter). Its magnitude is expressed in kg.m/s. At a given speed, a more massive particle has greater momentum
- ▷ Newton's 2nd law expressed in terms of momentum,

$$\boxed{\vec{F} = \frac{d\vec{p}}{dt}} \quad (\text{Newton's second law})$$

shows that a force applied to a particle over a time interval $[t_i, t_f]$ causes a momentum variation of the particle equal to

$$\Delta\vec{p}_{if} = \int_{t_i}^{t_f} \vec{F} dt \quad (\text{impulse})$$

6.1.2 Body (particle assembly)

The momentum of a body consisting of N particles of masses m_i and velocities $\vec{v}_i$ is defined by

$$\boxed{\vec{p}_{\text{tot}} \stackrel{\text{def}}{=} \sum_{i=1}^N \vec{p}_i = \sum_{i=1}^N m_i \vec{v}_i}$$

and we have

$$\boxed{\vec{F}_{\text{tot,ext}} = \frac{d\vec{p}_{\text{tot}}}{dt}} \quad (\text{Newton's second law})$$

CONSERVATION OF MOMENTUM

The total momentum of a material system is constant over time if and only if the total external force acting on that system is zero:

$$\vec{p}_{\text{tot}} = \text{constant} \quad \Leftrightarrow \quad \vec{F}_{\text{tot,ext}} = \vec{0}$$

In Newtonian mechanics, treating the universe as an isolated system with no external forces implies that its total momentum is conserved.

6.2 Movement of the centre of mass

centre of mass The position of the centre of mass of a system of N material particles of masses m_i and positions $\vec{r}_i$ ($i = 1, \dots, N$) is noted $\vec{r}_{\text{CM}}$ and is defined by the relation

$$\vec{r}_{\text{CM}} \stackrel{\text{def}}{=} \frac{1}{M} \sum_{i=1}^N m_i \vec{r}_i$$

with $M = \sum_{i=1}^N m_i$ the total mass of the system. The position of the centre of mass is therefore *mean position* of the particles of the system, *weighted* by their mass.

Remark. The centre of mass associated with the union of two sets of particles is simply given by the centre of mass of the two sets as if each of them were replaced by its own centre of mass, i.e.

$$\vec{r}_{\text{CM}} = \frac{M_1 \vec{r}_{\text{CM},1} + M_2 \vec{r}_{\text{CM},2}}{M_1 + M_2}$$

Example. Calculate the position of the centre of mass of the Earth system–Sun assuming that the Earth and the Sun are spherical bodies.

For the spherical, spherically symmetric bodies considered here, the centre of mass lies at the geometric centre. Place the Sun at the origin and the Earth on the Ox axis. Their centre of mass is then at $\vec{r}_{\text{CM}} = \frac{M_{\oplus} \vec{r}_{\text{CM},\oplus}}{M_{\oplus} + M_{\odot}}$, with $M_{\oplus} \approx 5.972 \times 10^{24}$ kg, $M_{\odot} \approx 1.989 \times 10^{30}$ kg and $\vec{r}_{\text{CM},\oplus} \approx (150 \times 10^9 \text{ m}, 0, 0)$, giving $\vec{r}_{\text{CM}} \approx (4.5 \times 10^5 \text{ m}, 0, 0)$. Thus it lies well inside the Sun, whose radius is $R_{\odot} \approx 7 \times 10^8 \text{ m}$.

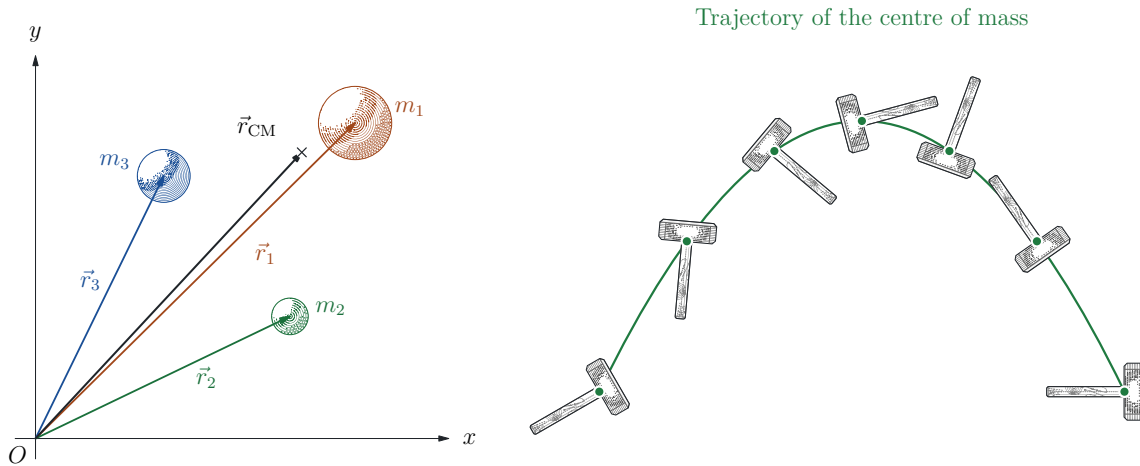


Figure 1: Left: position of the centre of mass of a three-body system. Right: parabolic trajectory of the centre of mass (green) of a hammer thrown into the air.

It follows from this definition that the velocity of the centre of mass is

$$\vec{v}_{\text{CM}} \stackrel{\text{def}}{=} \frac{d\vec{r}_{\text{CM}}}{dt} = \frac{1}{M} \sum_{i=1}^N m_i \vec{v}_i$$

The momentum (total) of the system can then be written

$$\vec{p}_{\text{tot}} \stackrel{\text{def}}{=} \sum_{i=1}^N m_i \vec{v}_i = M \vec{v}_{\text{CM}}$$

which, by taking the derivative from time, leads to

$$\vec{F}_{\text{tot,ext}} = M\vec{a}_{\text{CM}}$$

where $\vec{F}_{\text{tot,ext}}$ is the sum of the external forces on the system. We conclude that

MOVEMENT OF THE CENTRE OF MASS

The centre of mass of a system obeys an equation with the same form as Newton's second law. Its trajectory is that of a point particle with the total mass M of the system, subject to the net external force $\vec{F}_{\text{tot,ext}}$.

Example. The centre of mass of an object launched into the air follows a parabolic trajectory (see figure 1).

Gravitational potential energy and centre of mass In a uniform gravitational field, the potential energy of a system of particles is

$$U_{\text{grav}} = Mg z_{\text{CM}},$$

where z_{CM} is the vertical (Oz) coordinate of its centre of mass $\vec{r}_{\text{CM}}$ and $\vec{g} = -g\vec{e}_z$ is gravitational acceleration.

6.3 Collisions

Collision A collision is defined as the localized interaction in space and time between several bodies. Interaction can be done with or without contact.

Remark. Accurate modelling of a collision process is generally out of reach, justifying the use and interest of conservation laws in the study of such processes.

6.3.1 Classification of collision types

In a collision between two or more bodies, total kinetic energy need not be conserved. Some may be converted into internal motion, thermal energy or deformation, increasing internal energy.¹ A collision is *elastic* if total kinetic energy is the same before and after it. This can apply to fundamental particles, composite particles or macroscopic bodies; conversely, fundamental particles may undergo inelastic processes. Collisions are classified by the change in kinetic energy (Table 1).

COLLISION	Change in kinetic energy	Change in momentum
Superelastic	$\Delta K > 0$	$\Delta \vec{p} = 0$
Elastic	$\Delta K = 0$	$\Delta \vec{p} = 0$
Inelastic	$\Delta K < 0$	$\Delta \vec{p} = 0$
Perfectly inelastic	$\Delta K < 0, \quad \Delta \vec{p} = 0,$	$m_f = \sum_i m_i$

Table 1: Different types of collisions.

¹In physics, *heat* is energy exchanged through disordered microscopic interactions between the system and its surroundings. By contrast, *work* is energy transferred through ordered, macroscopic external forces.

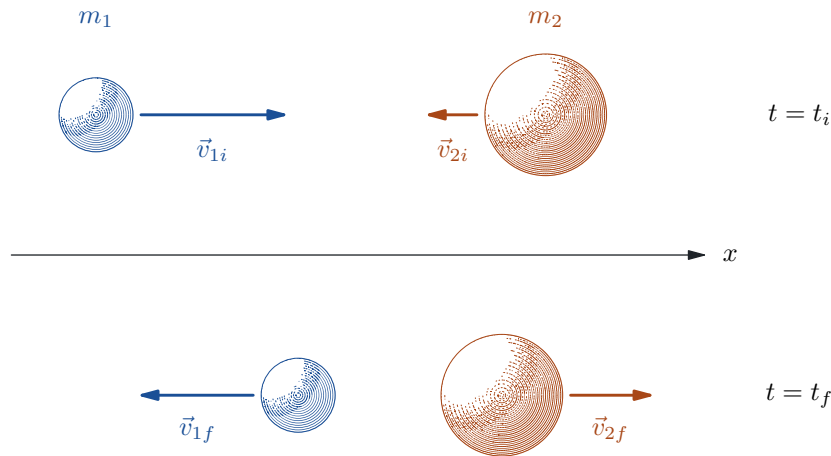


Figure 1: One-dimensional elastic collision between two rigid spheres.

Example. A collision between two electrons (charged point particles) is *elastic*.

Example. The collision between two billiard balls is in excellent approximation *elastic*.

Example. The explosion of a bomb can be seen as a collision *super-elastic* between its constituents.

Example. A collision between a bowling ball and a ball of modelling clay is generally *perfectly inelastic*.

Example. The collision between two cars is *inelastic*.

6.3.2 One-Dimensional Elastic Collisions

An elastic collision between two isolated bodies² conserves both total momentum and total kinetic energy. If motion is along a single spatial direction, as in Figure 1, the collision is one-dimensional, and the initial and final velocities of bodies 1 and 2 can be written as

$$\vec{v}_{1i} = (v_{1i}, 0, 0), \quad \vec{v}_{2i} = (v_{2i}, 0, 0)$$

and

$$\vec{v}_{1f} = (v_{1f}, 0, 0), \quad \vec{v}_{2f} = (v_{2f}, 0, 0)$$

In Figure 1, $v_{1i} > 0$ because body 1 initially moves towards increasing x , and $v_{1f} < 0$ because after the collision it moves towards decreasing x .

In terms of components, both conservation laws are written

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f} \quad (\text{conservation de } \vec{p}) \quad (1)$$

$$\frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2 \quad (\text{conservation de } K) \quad (2)$$

Knowing the masses m_1 and m_2 of the bodies and their initial velocities v_{1i} and v_{2i} , it is possible to determine the velocities after collision. Indeed, we have two equations for two unknowns (v_{1f} and v_{2f}). The equation (2) can be rewritten in the form

$$m_1 (v_{1i}^2 - v_{1f}^2) = m_2 (v_{2f}^2 - v_{2i}^2) \quad (3)$$

²that is, they are not subject to any external force.

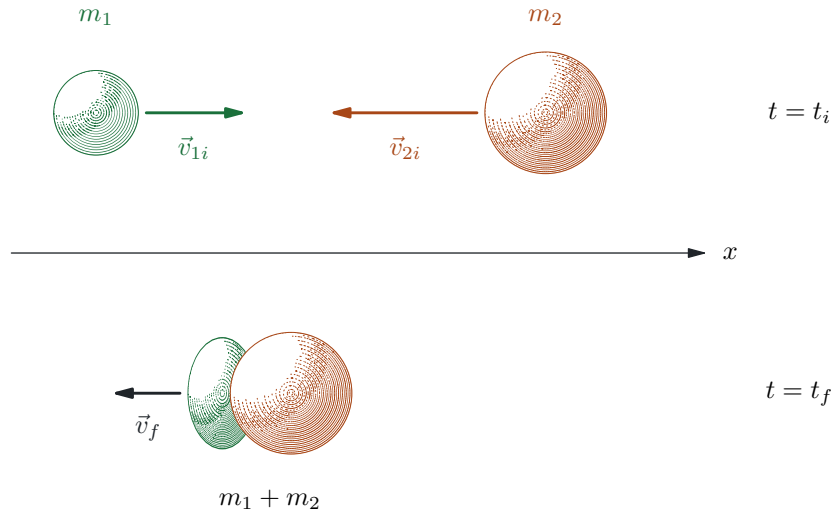


Figure 2: One-dimensional perfectly inelastic collision between a ball of modelling clay and a basketball.

The equation (3) can then be factored as

$$m_1 (v_{1i} + v_{1f})(v_{1i} - v_{1f}) = m_2 (v_{2f} + v_{2i})(v_{2f} - v_{2i}) \quad (4)$$

By separating the terms containing m_1 and m_2 from the equation (1), we obtain

$$m_1 (v_{1i} - v_{1f}) = m_2 (v_{2f} - v_{2i}) \quad (5)$$

By replacing (5) in (4), we arrive at

$$v_{1i} + v_{1f} = v_{2f} + v_{2i} \quad (6)$$

By multiplying (6) by m_2 , then subtracting (1) and resolving from v_{1f} , we result in

$$v_{1f} = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) v_{1i} + \left(\frac{2m_2}{m_1 + m_2} \right) v_{2i} \quad (7)$$

Similarly, we get for v_{2f}

$$v_{2f} = \left(\frac{2m_1}{m_1 + m_2} \right) v_{1i} + \left(\frac{m_2 - m_1}{m_1 + m_2} \right) v_{2i} \quad (8)$$

6.3.3 One-Dimensional Perfectly Inelastic Collisions

A process of inelastic collision between two isolated bodies obeys the law of conservation of the momentum. In the case of a perfectly inelastic collision, it is written after projection according to x

$$m_1 v_{1i} + m_2 v_{2i} = (m_1 + m_2) v_f$$

giving

$$v_f = \frac{m_1}{m_1 + m_2} v_{1i} + \frac{m_2}{m_1 + m_2} v_{2i}$$

The change in kinetic energy is

$$\Delta K = \frac{1}{2}(m_1 + m_2)v_f^2 - \left(\frac{1}{2}m_1 v_{1i}^2 + \frac{1}{2}m_2 v_{2i}^2 \right) < 0.$$

The energy converted mainly into internal energy is therefore

$$Q \approx -\Delta K > 0$$

Example. Consider the collision between an asteroid and the Earth, seen from the Earth's reference frame.

Let m_1 be the asteroid's mass and $m_2 \gg m_1$ the Earth's mass. In the Earth's reference frame, $v_{2i} = 0$, so

$$\Delta K = \frac{1}{2}(m_1 + m_2) \left(\frac{m_1}{m_1 + m_2} v_{1i} \right)^2 - \left(\frac{1}{2} m_1 v_{1i}^2 \right) \approx -\frac{1}{2} m_1 v_{1i}^2 = -K_{1,i}$$

Almost all the macroscopic kinetic energy is converted into internal energy, deformation and mechanical waves. For an asteroid 20 metres in diameter, of density 3000 kg/m^3 , approaching at 20 km/s , the energy released is approximately $\frac{1}{2}(\frac{4}{3}\pi 10^3) 3000 (20 \times 10^3)^2 \approx 2.5 \times 10^{15} \text{ J}$. This corresponds to about 0,60 megatonnes of TNT, or 40 bombs of 15 kilotonnes each, roughly the yield of the Hiroshima bomb.

6.3.4 Super-elastic one-dimensional collision

Example. In a non-relativistic fragmentation model, an unstable composite object of mass m breaks into two fragments of masses m_1 and m_2 . In the object's initial rest frame its velocity is zero. Projecting momentum conservation along the line of motion gives

$$0 = m_1 v_{1f} + m_2 v_{2f} \quad \Leftrightarrow \quad v_{2f} = -\frac{m_1}{m_2} v_{1f},$$

and the released kinetic energy is

$$\Delta K = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2 - 0 = \frac{1}{2} m_1 \left(1 + \frac{m_1}{m_2} \right) v_{1f}^2.$$

This classical calculation is only an approximation for slow fragmentation. Particle decay requires a relativistic treatment conserving the energy-momentum four-vector, with $E_i^2 = p_i^2 c^2 + m_i^2 c^4$ for each decay product.

Example. Big Bertha was a large siege gun used by the German army during the First World War. With a mass of 46.2 tonnes, it fired 400 kg shells at 333 m/s. Calculate the gun's recoil velocity.

The recoil velocity is $v_{2f} = -\frac{400}{46.2 \times 10^3} 333 \approx 2.88 \text{ m/s} \approx 10 \text{ km/h}$.

6.3.5 Two-dimensional collision

The law of conservation of the momentum for a two-dimensional collision between two bodies of identical mass m , one of which is initially at rest, is written

$$m \vec{v}_{1i} = m \vec{v}_{1f} + m \vec{v}_{2f} \quad (\text{conservation de } \vec{p})$$

$$\Leftrightarrow \vec{v}_{1i} = \vec{v}_{1f} + \vec{v}_{2f} \quad (9)$$

Taking the squared norm of equation (9) gives

$$v_{1i}^2 = v_{1f}^2 + v_{2f}^2 + 2 \vec{v}_{1f} \cdot \vec{v}_{2f} \quad (10)$$

The variation of kinetic energy can then be put into the form

$$\begin{aligned} \Delta K &= \frac{1}{2} m v_{1f}^2 + \frac{1}{2} m v_{2f}^2 - \frac{1}{2} m v_{1i}^2 \\ &= -m \vec{v}_{1f} \cdot \vec{v}_{2f} \end{aligned}$$

In the case of an elastic collision, $\Delta K = 0$, and we deduct

$$\vec{v}_{1f} \cdot \vec{v}_{2f} = 0 \quad \Leftrightarrow \quad \vec{v}_{1f} \perp \vec{v}_{2f} \quad \Leftrightarrow \quad \boxed{\theta + \varphi = 90^\circ} \quad (\text{collision elastic})$$

In the general case of an inelastic collision, $\Delta K < 0$, and we deduce from it

$$\begin{aligned} \vec{v}_{1f} \cdot \vec{v}_{2f} &= -\frac{\Delta K}{m} \\ \Leftrightarrow \quad v_{1f} v_{2f} \cos(\theta + \varphi) &= -\frac{\Delta K}{m} \\ \Leftrightarrow \quad \cos(\theta + \varphi) &= -\frac{\Delta K}{m v_{1f} v_{2f}} > 0 \\ \Rightarrow \quad \boxed{\theta + \varphi < 90^\circ} & \quad (\text{collision inelastic}) \end{aligned}$$

Remark. For a two-dimensional elastic collision, conservation of momentum and kinetic energy alone cannot determine all final velocity components: they provide three equations for four unknowns $(\theta, \varphi, v_{1f}, v_{2f})$.

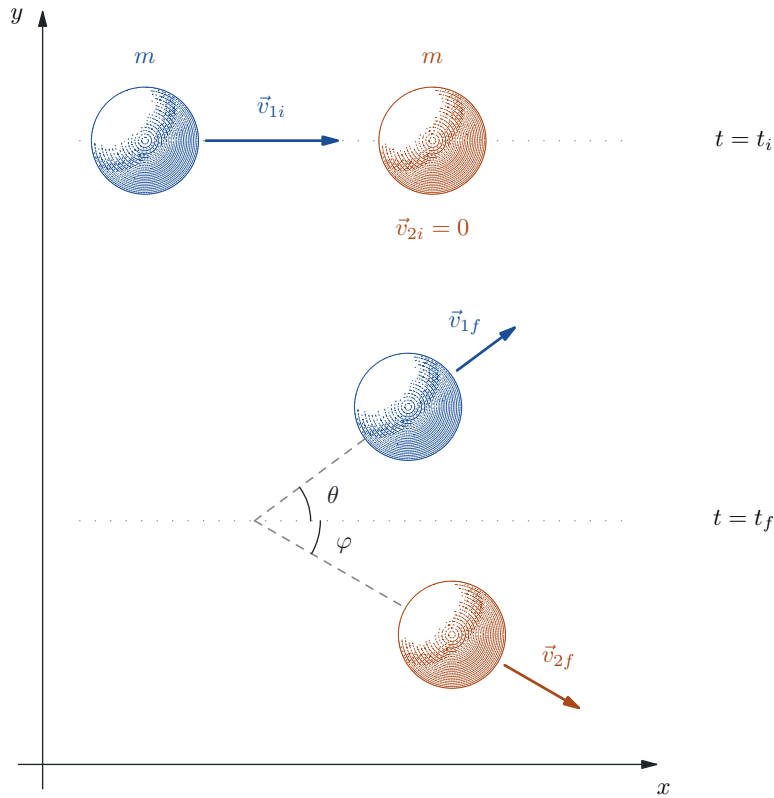


Figure 3: Two-dimensional collision between two balls of the same mass.

CHAPTER 7

TORQUE, ANGULAR MOMENTUM AND ROTATIONS

7.1 Torque of a force

Torque The torque¹ of a force $\vec{F}$ applied at P , about an origin O , is defined as

$$\vec{\tau} \stackrel{\text{def}}{=} \vec{r} \times \vec{F},$$

where $\vec{r}$ is the vector from O to P .

Torque is a vector perpendicular to the plane containing $\vec{r}$ and $\vec{F}$, with direction given by the right-hand rule. Its magnitude is $|\vec{\tau}| = |\vec{r} \times \vec{F}| = rF \sin \varphi = Fd$, expressed in $\text{N m} = \text{kg m}^2 \text{s}^{-2}$ in SI units. Here $d = r \sin \varphi$ is the *lever arm* of the force. Torque depends on the chosen origin because $\vec{r}$ does. For a point particle, $\vec{r}$ is the particle's position. For an extended body, $\vec{r}$ must run from the origin to the point where the force acts. This point may be replaced by any point on the line of action of $\vec{F}$, but not by an arbitrary point of the body. The distinction is essential in rotation and static equilibrium.

Example. Calculate the torque of gravity on a particle of mass m at position $\vec{r} = d(1, 2, -1)/\sqrt{6}$.

The gravity force is given in terms of Cartesian components by $\vec{F} = -mg \vec{e}_z = mg(0, 0, -1)$. Therefore, its torque is equal to $\vec{\tau} = \vec{r} \times \vec{F} = mgd(-2, 1, 0)/\sqrt{6}$.

¹The word «moment» here does not mean a «time interval». It comes from the Latin momentum, a contraction of *movimentum, derived from movere (to move), meaning impulse, motion or change.

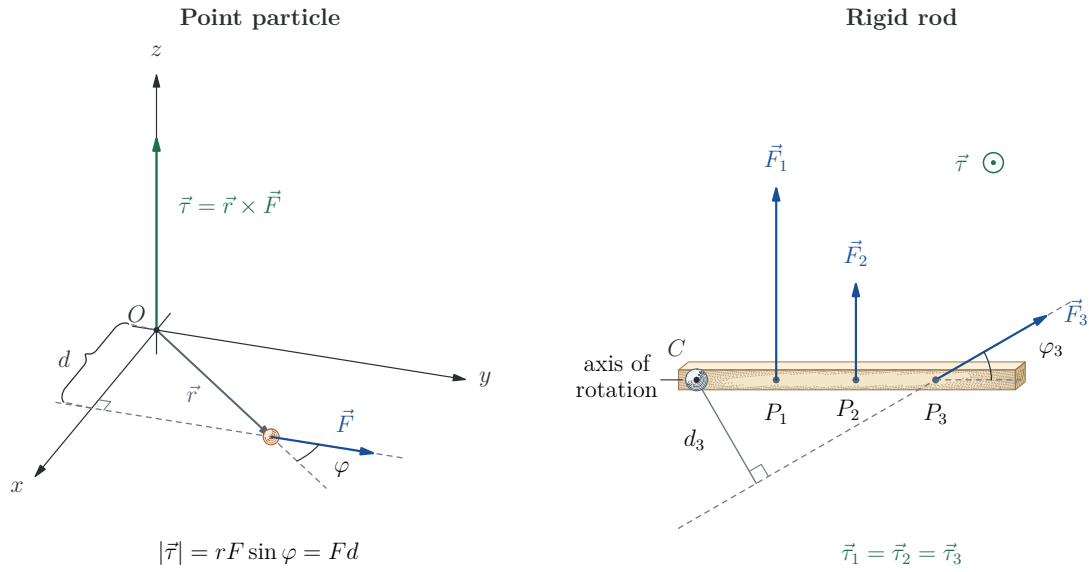


Figure 1: Torque on a point particle (left) and on a rigid rod attached to axis C (right). Its magnitude equals the force magnitude times the lever arm d , which depends on the point of application P . If the rod is initially at rest, a non-zero torque causes rotation about C and changes its angular momentum. Torque about rotation axis C (with origin $O \in C$) measures the effectiveness of the force in producing rotation about that axis.

7.2 Angular momentum

7.2.1 Particle

angular momentum of a particle The angular momentum² of a point particle of mass m about origin O , denoted $\vec{L}_O$ or simply $\vec{L}$, is defined by

$$\vec{L} \stackrel{\text{def}}{=} \vec{r} \times \vec{p} \quad (1)$$

Here $\vec{r}$ is the particle's position vector and $\vec{p} = m\vec{v}$ its momentum. Angular momentum is a vector quantity with SI units $\text{kg}\cdot\text{m}^2\cdot\text{s}^{-1}$, the rotational counterpart of linear momentum about O . Its magnitude quantifies «the particle's motion around O ». Its direction is perpendicular to the plane of the position and momentum vectors, along an axis through O . Even a particle moving along a straight line not passing through the origin effectively «rotates» about that origin, and therefore has non-zero angular momentum about O .

The definition (1) gives the following properties of a particle's angular momentum:

- ▷ If the position $\vec{r}(t)$ of the particle is known at any time t , then its angular momentum, $\vec{L}(t)$, can be determined at any time.
- ▷ $\vec{L}$ is perpendicular to the plane containing the particle's position and momentum vectors, and has magnitude $L = |\vec{r}| |\vec{p}| \sin \varphi = r_{\perp} p = r_{\perp} mv$ (see Figure 2).
- ▷ $\vec{L}$ depends on the origin O of the coordinate system chosen to calculate it (because the particle position depends on it), but not on the orientation of the axes.
- ▷ When the trajectory of a particle is a *straight line*, its angular momentum can be rendered *zero* at any time t by placing the origin of the coordinate system at a point in the trajectory.
- ▷ For a curved trajectory, no choice of origin makes the particle's angular momentum zero at all times t .

²The French terms *moment cinétique* and *moment angulaire* both denote angular momentum, emphasising its connection with rotation.

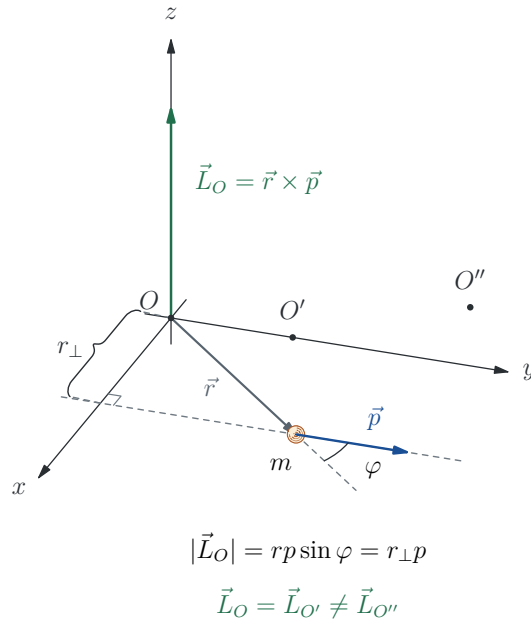


Figure 2: Angular momentum of a point particle ($\vec{r}$ and $\vec{p}$ in the Oxy plane). Angular momenta about O and O' are equal, but differ from that about O'' . The magnitude of $\vec{L}$ is $L = rp \sin \varphi = r_{\perp} p$, where $r_{\perp}$ is the distance from the line along $\vec{p}$ (dashed) to the origin O .

- ▷ For a particle subject to a central force, angular momentum is *constant* when the origin is placed at the centre of force. The force then has the form $\vec{F} = F(r) \vec{e}_r$, where $F(r)$ is a real-valued function of the distance $r = |\vec{r}|$ from the force centre and $\vec{e}_r = \vec{r}/r$ is the radial unit vector. Universal gravitation is an example of a central force.

In an inertial frame, for a fixed origin O , the angular momentum of a point particle evolves over time according to the equation

$$\frac{d\vec{L}_O}{dt} = \vec{r}_O \times \vec{F} \equiv \vec{\tau}_O$$

Proof. To determine the variations of the angular momentum (1) over time, it is sufficient to calculate its derivative relative to time,

$$\frac{d\vec{L}}{dt} = \frac{d}{dt}(\vec{r} \times \vec{p}) = \frac{d\vec{r}}{dt} \times \vec{p} + \vec{r} \times \frac{d\vec{p}}{dt} = \vec{r} \times \vec{F}$$

since $\vec{v} \parallel \vec{p}$ and $\vec{F} = d\vec{p}/dt$ according to Newton's 2nd law. □

CONSERVATION OF ANGULAR MOMENTUM : PARTICLE

A point particle's angular momentum is constant if and only if the torque acting on it is zero:

$$\vec{L} = \text{constant} \quad \Leftrightarrow \quad \vec{r} \times \vec{F} = 0 \quad \Leftrightarrow \quad \vec{F} = 0 \quad \text{or} \quad \vec{F} \parallel \vec{r} \quad (\text{single particle})$$

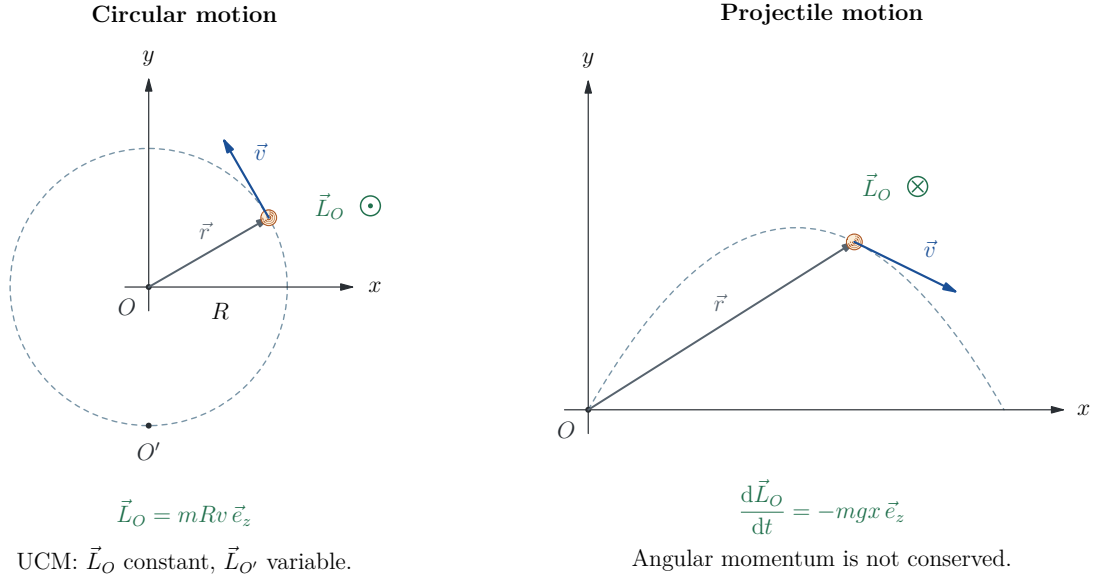


Figure 3: Angular momentum in circular and projectile motion. Left: with the origin at the centre of the circular path, angular momentum is $\vec{L} = mRv \vec{e}_z$ for anticlockwise motion. It is therefore constant for *uniform* circular motion, since R , $\underline{and}$ v are constant in UCM. With the origin O' on the circular path, angular momentum is no longer constant, even in UCM, because motion about O' is not uniform. Right: in projectile motion, angular momentum is not conserved for any choice of origin.

7.2.2 Body (particle assembly)

angular momentum of a body For a body consisting of N particles of masses m_i , positions $\vec{r}_i$ and velocities $\vec{v}_i$, where $i = 1, \dots, N$ is an integer number used to index particles, the angular momentum is defined as the vector sum of the angular momenta of all the constituent particles of the body, i.e.

$$\vec{L}_{\text{tot}} \stackrel{\text{def}}{=} \sum_{i=1}^N \vec{r}_i \times \vec{p}_i = \sum_{i=1}^N \vec{L}_i \quad (2)$$

In an inertial frame and relative to a fixed origin O , the angular momentum of a body evolves over time according to the equation

$$\frac{d\vec{L}_{\text{tot},O}}{dt} = \vec{\tau}_{\text{tot},\text{ext}}^O$$

where $\vec{\tau}_{\text{tot},\text{ext}}$ is the total torque of external forces (in number n)

$$\vec{\tau}_{\text{tot},\text{ext}} \stackrel{\text{def}}{=} \sum_{j=1}^n \left(\vec{r}_j \times \vec{F}_{j,\text{ext}} \right)$$

Internal forces do not contribute to changes in the total angular momentum.

Proof. To find the time evolution of angular momentum (2), differentiate $\vec{L}_{\text{tot}}$:

$$\frac{d\vec{L}_{\text{tot}}}{dt} = \frac{d}{dt} \left(\sum_{i=1}^N \vec{r}_i \times \vec{p}_i \right) = \sum_{i=1}^N \frac{d}{dt} (\vec{r}_i \times \vec{p}_i) = \sum_{i=1}^N \left(\frac{d\vec{r}_i}{dt} \times \vec{p}_i + \vec{r}_i \times \frac{d\vec{p}_i}{dt} \right) = \sum_{i=1}^N \left(\vec{v}_i \times \vec{p}_i + \vec{r}_i \times \frac{d\vec{p}_i}{dt} \right) = \sum_{i=1}^N \left(\vec{r}_i \times \frac{d\vec{p}_i}{dt} \right) = \sum_{i=1}^N \left(\vec{r}_i \times \vec{F}_i \right)$$

Here $\vec{F}_i$ is the net force on particle i . Distinguish internal from external forces by writing $\vec{F}_i = \vec{F}_{i,\text{int}} + \vec{F}_{i,\text{ext}}$ for each particle i , which gives

$$\frac{d\vec{L}_{\text{tot}}}{dt} = \underbrace{\sum_{i=1}^N \left(\vec{r}_i \times \vec{F}_{i,\text{int}} \right)}_{=0} + \sum_{i=1}^N \left(\vec{r}_i \times \vec{F}_{i,\text{ext}} \right) = \sum_{i=1}^N \vec{\tau}_{i,\text{ext}}$$

The torques of internal action-reaction pairs cancel under the stated assumptions about the forces between particles. $\square$

CONSERVATION OF ANGULAR MOMENTUM : BODY

A body's angular momentum is constant if and only if the net external torque on it is zero:

$$\vec{L} = \text{constant} \quad \Leftrightarrow \quad \sum_j \vec{r}_j \times \vec{F}_{j,\text{ext}} = 0 \quad (\text{for one body})$$

7.3 Dynamic and Static Equilibrium

Dynamic equilibrium A body is said to be in *dynamic equilibrium* when the momentum of its centre of mass and its angular momentum, calculated in relation to a fixed inertial origin O , are constant. The dynamic equilibrium conditions are therefore

$$\vec{p}_{\text{CM}} = \text{constant} \quad \text{and} \quad \vec{L}_O = \text{constant}$$

They are equivalent to the conditions

$$\vec{F}_{\text{tot,ext}} = 0 \quad \text{and} \quad \vec{\tau}_{\text{tot,ext}}^O = 0 \quad (3)$$

The first condition implies $\vec{a}_{\text{CM}} = 0$. The condition $\vec{\tau}_{\text{tot,ext}}^O = 0$ implies $d\vec{L}_O/dt = 0$, but does not in general imply zero angular acceleration about an arbitrary fixed axis.

DYNAMIC EQUILIBRIUM CONDITIONS

A body is at dynamic equilibrium if and only if

$$\vec{F}_{\text{tot,ext}} = 0 \quad \text{and} \quad \vec{\tau}_{\text{tot,ext}} = 0$$

When $\vec{F}_{\text{tot,ext}} = 0$, if the torque $\vec{\tau}_{\text{tot,ext}}$ is zero about one origin, it is zero about every origin.

Proof. Consider coordinate systems S and S' , with origins O and O' , respectively. The torque in S is $\vec{\tau}_{\text{tot,ext}}^O = \sum_i \vec{r}_i \times \vec{F}_i = \sum_i (\vec{r}'_i + \vec{r}_{OO'}) \times \vec{F}_i = (\sum_i \vec{r}'_i \times \vec{F}_i) + \vec{r}_{OO'} \times (\sum_i \vec{F}_i) = \vec{\tau}_{\text{tot,ext}}^{O'} + \vec{r}_{OO'} \times \vec{F}_{\text{tot,ext}}$, where $\vec{r}_{OO'}$ joins O to O' and $\vec{\tau}_{\text{tot,ext}}^{O'}$ is the torque in S' . If $\vec{F}_{\text{tot,ext}} = 0$, the torques $\vec{\tau}_{\text{tot,ext}}^O$ and $\vec{\tau}_{\text{tot,ext}}^{O'}$ are equal. $\square$

Static equilibrium A body is in *static equilibrium* when it is in dynamic equilibrium and both the initial velocity of its centre of mass and its initial angular velocity are zero. It then remains at rest, with kinematic conditions

$$\vec{v}_{\text{CM}}(t) = 0 \quad \text{and} \quad \vec{\omega}(t) = 0$$

They are totally equivalent to the conditions

$$\vec{F}_{\text{tot,ext}} = 0, \vec{p}_{\text{CM}}(0) = 0 \quad \text{and} \quad \vec{\tau}_{\text{tot,ext}} = 0, \vec{L}(0) = 0$$

Two force application points In this case, dynamic equilibrium conditions (3) are written

$$\vec{F}_1 + \vec{F}_2 = 0 \quad \text{and} \quad \vec{r}_1 \times \vec{F}_1 + \vec{r}_2 \times \vec{F}_2 = 0$$

Here $\vec{F}_j$ is the sum of forces applied at points $\vec{r}_j$ ($j = 1, 2$). Both conditions hold when

$$\vec{F}_1 = -\vec{F}_2 \quad \text{and} \quad (\vec{r}_1 - \vec{r}_2) \times \vec{F}_1 = 0$$

i.e. when the forces are opposite and have the same direction as the vector connecting their respective point of application (see Figure 4).

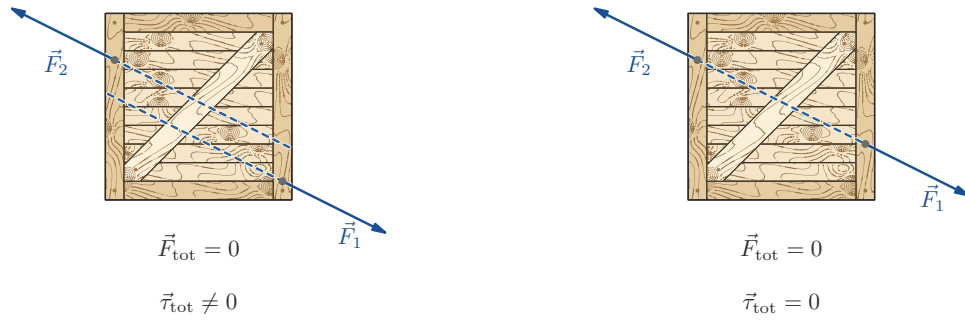


Figure 4: Total torque for two force application points.

Three force application points In this case, dynamic equilibrium conditions (3) are written

$$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = 0 \quad \text{and} \quad \vec{r}_1 \times \vec{F}_1 + \vec{r}_2 \times \vec{F}_2 + \vec{r}_3 \times \vec{F}_3 = 0$$

Here $\vec{F}_j$ is the sum of forces applied at $\vec{r}_j$ ($j = 1, 2, 3$). The second condition requires the three forces to be parallel or their lines of action to meet at a common point (see Figure 5).

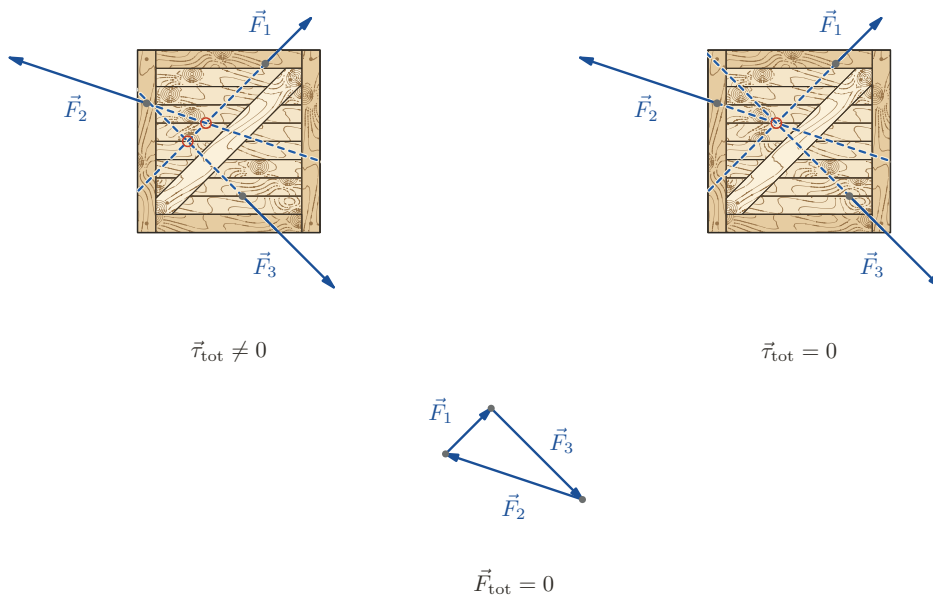


Figure 5: Total torque for three force application points.

7.4 Rotation around a fixed axis

Rotational kinetic energy The kinetic energy of a rigid body in rotation around a fixed axis C is the sum of the kinetic energies of the particles of this body. As each of the particles is in circular motion with the same angular velocity ω (not necessarily constant), we have

$$K_{\text{rot}} = \sum_I \frac{1}{2} m_i v_i^2 = \frac{1}{2} \sum_I m_i (\omega r_{i\perp})^2 = \frac{1}{2} I_C \omega^2$$

where $r_{i\perp}$ is the distance between the particle i and the axis C and where we have introduced at the last equality the moment of inertia I_C (see definition below).

moment of inertia The moment of inertia of a rigid body in relation to a fixed axis of rotation C is defined by

$$I_C \stackrel{\text{def}}{=} \sum_i m_i r_{i\perp}^2$$

where m_i is the mass of the particle i and $r_{i\perp}$ is its distance from the axis C . The moment of inertia depends on several factors: the axis of rotation, the shape of the body, its density, its inhomogeneities, ... A body usually has different moments of inertia around different axes of rotation. Some examples of moments of inertia are given in Figure 6. The moment of inertia of a body quantifies its resistance to a change of its rotational movement around an axis.

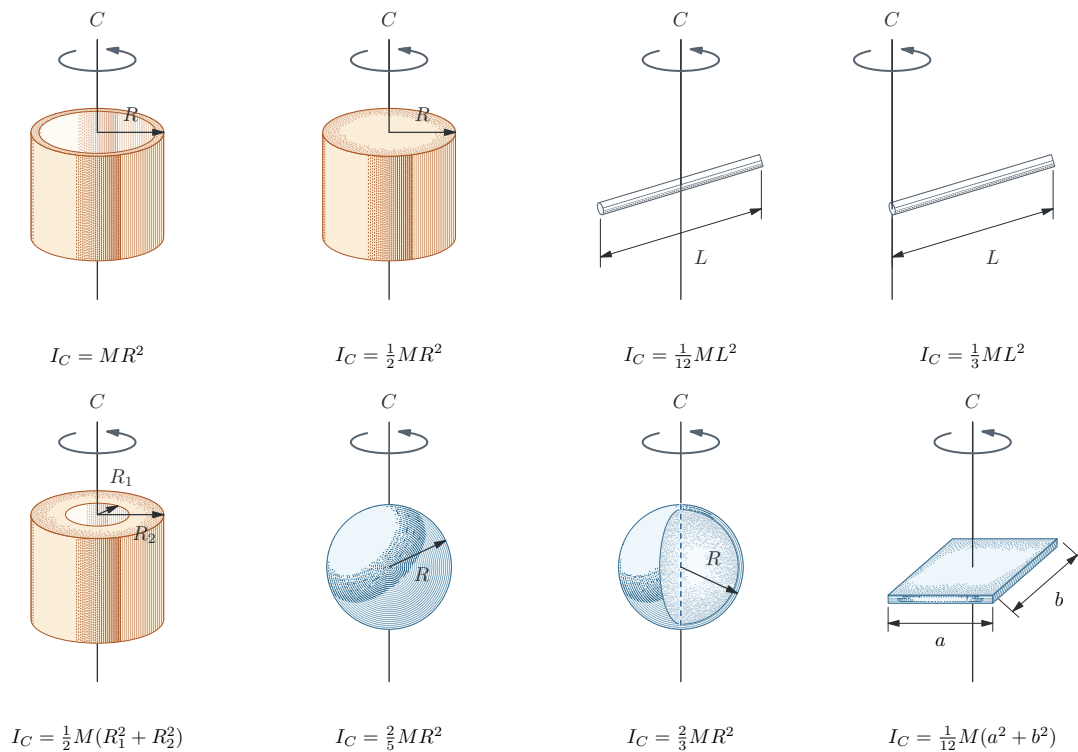


Figure 6: moments of inertia of a few rigid bodies around a fixed axis C .

PARALLEL-AXIS THEOREM (HUYGENS' THEOREM)

The moments of inertia of a body in relation to two parallel axes C and C' , of which C passes through the centre of mass of the body (see Figure 7), are connected by

$$I_{C'} = I_C + Md^2$$

where d is the distance between axes C and C' and M is the total mass of the body.

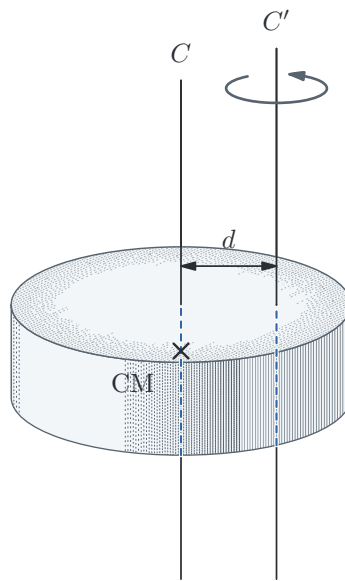


Figure 7: Axes C and C' and distance d such as in parallel axis theorem.

Example. Find the moment of inertia of a uniform rod about an axis C' perpendicular to the rod and passing through one end, given its moment of inertia $I_C = \frac{1}{12}ML^2$ about a parallel axis through the centre.

The distance between the parallel axes C and C' is $L/2$. A simple application of the parallel axes theorem then gives $I_{C'} = I_C + M \left(\frac{L}{2}\right)^2 = \frac{1}{3}ML^2$.

angular momentum The angular momentum of a rigid body rotating about a fixed axis is

$$\vec{L} = \vec{L}_\perp + \vec{L}_\parallel,$$

where $\vec{L}_\perp$ and $\vec{L}_\parallel$ are the components of $\vec{L}$ perpendicular and parallel to rotation axis C . The parallel component is

$$\vec{L}_\parallel = I_C \vec{\omega},$$

where $\vec{\omega}$ is the angular velocity vector: its magnitude is $|\vec{\omega}|$ and its direction specifies the axis and sense of rotation. In terms of angular momentum $\vec{L}_\parallel$, rotational kinetic energy is

$$K_{\text{rot}} = \frac{1}{2} I_C \omega^2 = \frac{L_\parallel^2}{2I_C}$$

angular acceleration Just as a force produces linear acceleration, a torque produces angular acceleration and changes a body's angular momentum $\vec{L}_\parallel$ (Figure 8). We have

$$\frac{d\vec{L}_\parallel}{dt} = I_C \vec{\alpha} = \vec{\tau}_\parallel,$$

where $\vec{\tau}_\parallel$ is the component of the net external torque along the rotation axis.

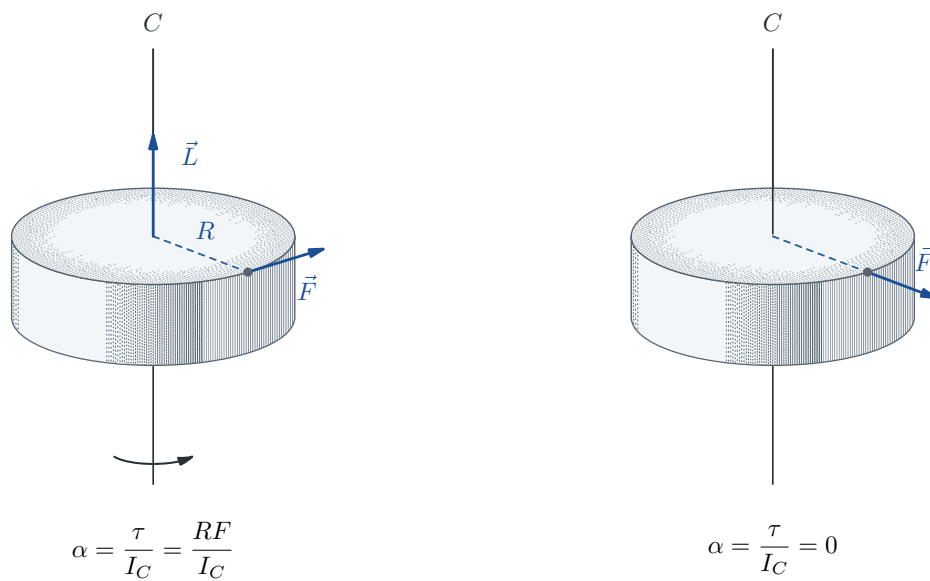


Figure 8: Angular momentum of a rigid body rotating about a fixed axis (C). If the rotation axis is a symmetry axis, as here, the angular momentum is parallel to it: $\vec{L} = \vec{L}_{\parallel} = I_C \vec{\omega}$, where I_C is the moment of inertia about C . The vector $\vec{\omega}$ has magnitude equal to the angular speed ω (rad/s), points along the rotation axis, and is oriented according to the right-hand rule. A non-zero torque component along that axis (left) produces angular acceleration α (rad/s²) and changes the corresponding angular-momentum component.

7.5 Analogy of rotation and translation

ROTATION ABOUT A FIXED AXIS	TRANSLATION (1D)
Angular velocity $\omega = \frac{d\theta}{dt}$	Linear velocity $v = \frac{dx}{dt}$
Angular acceleration $\alpha = \frac{d\omega}{dt}$	Linear acceleration $a = \frac{dv}{dt}$
Kinetic energy $K_R = \frac{1}{2}I\omega^2$	Kinetic energy $K = \frac{1}{2}mv^2$
Angular momentum $L = I\omega$	Momentum $p = mv$
Torque $\tau_{\text{tot,ext}} = \frac{dL}{dt} = I\alpha$	Force $F_{\text{tot,ext}} = \frac{dp}{dt} = ma$
Work $W = \int_{\theta_i}^{\theta_f} \tau d\theta$	Work $W = \int_{x_i}^{x_f} F_x dx$
Power $P = \tau\omega$	Power $P = Fv$

Figure 9: Comparison of rotation and translation. Symbols x, v, a and F refer to *components* position, velocity, acceleration and force along the direction of one-dimensional motion. Similarly, L and τ refer to *components* angular momentum and torque along the fixed rotation axis.

7.6 Overview

The figure below provides a visual summary of the three fundamental concepts of Newtonian mechanics and their conservation laws.

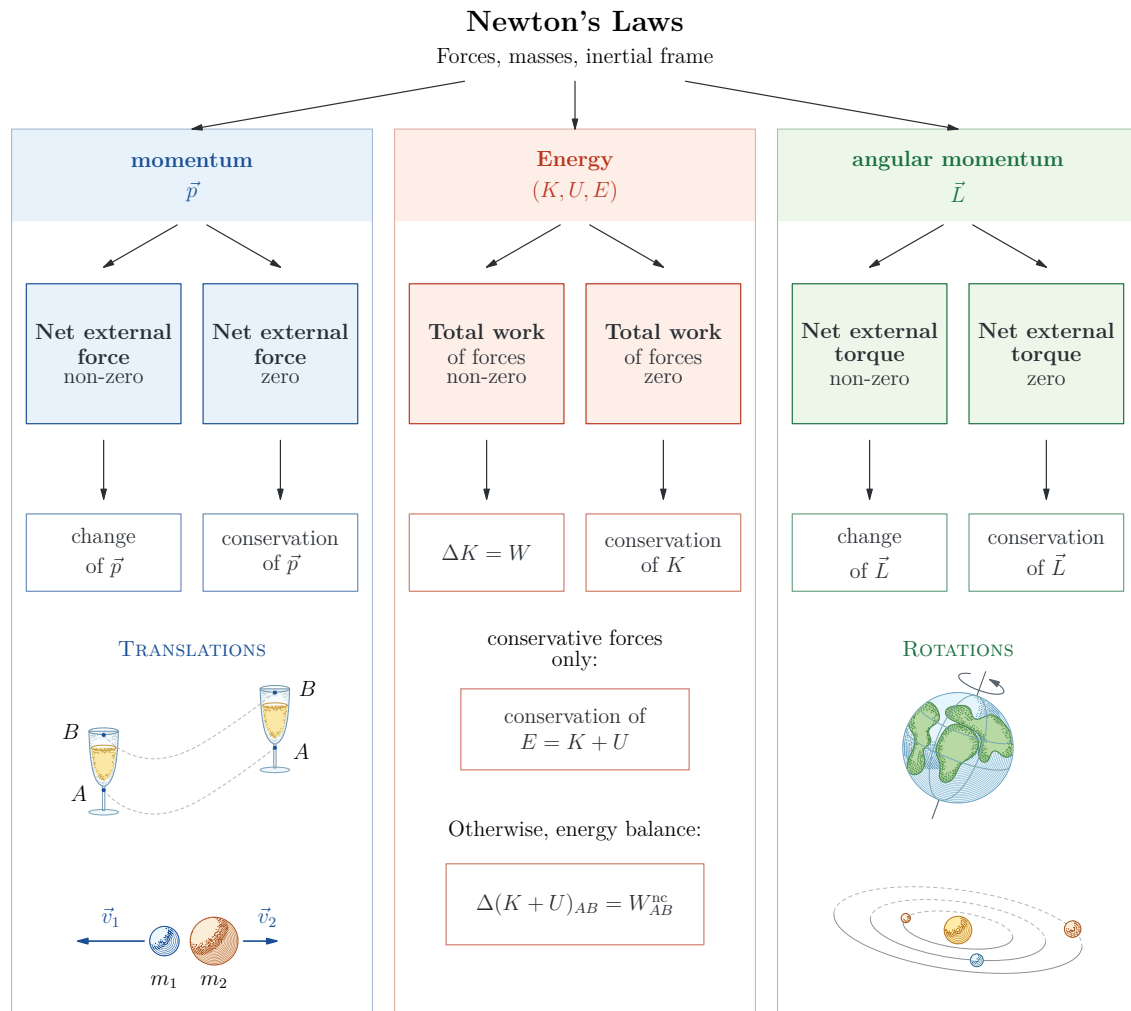


Figure 10: Situations where the basic physical quantities (momentum, energy and angular momentum) are conserved and where they vary.

CHAPTER 8

MECHANICAL WAVES

Wave A *wave* is a disturbance of a physical quantity that propagates through a medium, or through vacuum, as electromagnetic and gravitational waves do.

Mechanical wave A *mechanical wave* is a disturbance or deformation that propagates through a material medium.

A stadium wave (disturbance) in a crowd (material medium) is a mechanical wave. The disturbed quantity is the height of the crowd (in m).

The waves (disturbance) on the surface of the sea (material medium) are mechanical waves. The disturbed physical quantity is the water level (in m).

Sound waves are mechanical waves: compressions that propagate through a gas, a liquid or a solid. The disturbed quantity is the pressure in the medium (N/m^2).

Remark. There are two types of variation in a wave: (i) spatial variations in disturbance in the medium at a given moment and (ii) temporal variations in disturbance at a fixed position in the medium.

Remark. The disturbance travels through the medium at the wave's propagation speed. Particles of the medium may oscillate locally, including along the propagation direction for a longitudinal wave, without travelling along with the wave.

Remark. A wave does not necessarily correspond to a periodic physical phenomenon (think about the propagation of a pulse in a string or that of the waves produced by the fall of a stone in the water).

Transverse wave A wave is *transverse* when the disturbance in the medium is in a direction *orthogonal* to its propagation.

A stadium wave in a crowd is a *transverse* mechanical wave.

A pulse travelling along a string is a *transverse* mechanical wave.

Light is a *transverse* electromagnetic wave, not a mechanical wave.

Longitudinal wave A wave is *longitudinal* when the disturbance in the medium is *same* to the direction of propagation. Particles oscillate locally about equilibrium, allowing the wave to carry energy and information without necessarily transporting matter overall.

Sound in air is a *longitudinal* mechanical wave.

A seismic wave (produced during an Earthquake) may be *longitudinal*.

8.1 One-Dimensional Mechanical Waves

8.1.1 Travelling waves

In the absence of external forces and dissipation (friction), a mechanical disturbance in the model considered travels *at constant velocity without deforming*; this is a travelling wave. Let $y(x, t)$ denote the disturbance as a function of position x and time t . It may be positive, zero or negative. For a wave on a string, y is the transverse displacement from equilibrium (m); for a sound wave, y is the pressure difference from atmospheric pressure (N/m²). The amplitude is the maximum value of $|y|$, denoted by $A \geq 0$ for a sinusoidal wave. Let

$$y(x, 0) = f(x)$$

the spatial profile of the wave at the initial time. If the wave moves at a constant velocity v towards the increasing values of x (see Figure 1), then

$$y(x, t) = y(x - vt, 0) = f(x - vt)$$

and if the wave moves at constant velocity v towards the decreasing values of x , then

$$y(x, t) = f(x + vt)$$

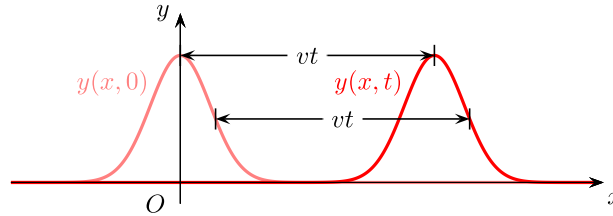


Figure 1: A disturbance travelling towards increasing x at constant speed v , without changing shape.

In a linear medium, the *superposition principle* states that when several waves overlap, *their displacements add algebraically*:

$$y_{\text{tot}}(x, t) = y_1(x, t) + y_2(x, t) + \dots$$

For the linear one-dimensional wave equation in a homogeneous, non-dispersive medium with constant wave speed v , the general solution is the superposition of a profile f travelling at speed v towards decreasing x and a profile g travelling at the same speed towards increasing x

$$\boxed{y(x, t) = \underbrace{f(x + vt)}_{\leftarrow} + \underbrace{g(x - vt)}_{\rightarrow}} \quad (1)$$

8.1.2 Sinusoidal travelling waves

The general form of a sinusoidal travelling wave is

$$y(x, t) = A \sin \left[\frac{2\pi}{\lambda} (x \pm vt) + \varphi \right] \quad (2)$$

where A is the amplitude, v the propagation speed, φ the phase and λ the wavelength. This is a travelling wave because $y(x, t)$ can be written as $f(x \pm vt)$ with $f(\cdot) = A \sin \left[\frac{2\pi}{\lambda} \cdot + \varphi \right]$. The wavelength λ determines its spatial period, since

$$y(x + n\lambda, t) = y(x, t) \quad \forall x, t \in \mathbb{R}, n \in \mathbb{Z}$$

The displacement (2) can also be written as

$$y(x, t) = A \sin \left[\frac{2\pi}{\lambda} (x \pm vt) + \varphi \right] = A \sin \left[2\pi \left(\frac{x}{\lambda} \pm \frac{t}{T} \right) + \varphi \right] = A \sin [kx \pm \omega t + \varphi] \quad (3)$$

where T is the period of the wave that sets its temporal periodicity,

$$y(x, t + nT) = y(x, t) \quad \forall x, t \in \mathbb{R}, n \in \mathbb{Z},$$

k is the wavenumber (m^{-1}), and ω is the angular frequency (rad.s^{-1}). The frequency f , defined by $f = \omega/2\pi = 1/T$, has a single value: the wave is monochromatic. Since the forms in (3) describe the same displacement, we have:

SINUSOIDAL TRAVELLING WAVE

A sinusoidal travelling wave has displacement

$$y(x, t) = A \sin \left[\frac{2\pi}{\lambda} (x \pm vt) + \varphi \right] = A \sin \left[2\pi \left(\frac{x}{\lambda} \pm \frac{t}{T} \right) + \varphi \right] = A \sin [kx \pm \omega t + \varphi],$$

with

$$\circ k = \frac{2\pi}{\lambda} \quad (\text{wavenumber})$$

$$\circ \omega = \frac{2\pi}{T} = 2\pi f \quad (\text{angular frequency})$$

$$\circ v = \frac{\lambda}{T} = \lambda f = \frac{\omega}{k} \quad (\text{speed})$$

It oscillates at a single frequency f and propagates at constant speed v without changing shape. In this non-dispersive model, propagation speed depends only on the physical properties of the medium, not on the wavelength λ or frequency f , which may vary while their product remains fixed.

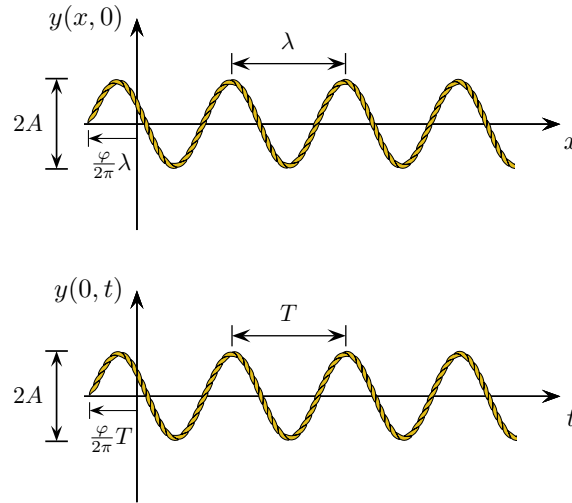


Figure 2: A one-dimensional sinusoidal wave shown at a fixed time ($t = 0$, top) as a function of position along the string, and at a fixed position ($x = 0$, bottom) as a function of time.

8.1.3 Energy transport

Consider a sinusoidal wave travelling horizontally along a string. As the wave passes, the string moves vertically and acquires kinetic energy, while its elastic deformation stores potential energy. Both forms of energy are transported along the string. A short segment centred at position x , of length Δx and mass $\Delta m = \mu \Delta x$, where μ is the linear mass density in kg/m , oscillates vertically with displacement, velocity and acceleration given by

$$\begin{aligned} y(x, t) &= A \sin [kx - \omega t + \varphi] \\ v_y(x, t) &= \frac{\partial y(x, t)}{\partial t} = -A\omega \cos [kx - \omega t + \varphi] \\ a_y(x, t) &= \frac{\partial^2 y(x, t)}{\partial t^2} = -A\omega^2 \sin [kx - \omega t + \varphi] \end{aligned}$$

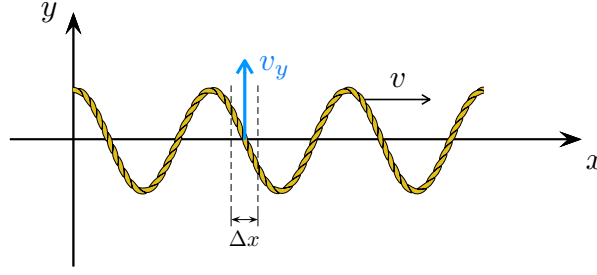


Figure 3: A wave travelling horizontally along a string at speed v . String elements vibrate vertically with velocity v_y as the wave passes.

The kinetic energy of the piece of string is given by

$$\Delta K = \frac{1}{2} \Delta m v_y^2 = \frac{1}{2} \mu \Delta x A^2 \omega^2 \cos^2 [kx - \omega t + \varphi]$$

and its potential elastic energy, which depends on the local slope of the string in the approximation of the small slopes, by

$$\Delta U = \frac{1}{2} T \left(\frac{\partial y}{\partial x} \right)^2 \Delta x = \frac{1}{2} \mu \Delta x A^2 \omega^2 \cos^2 [kx - \omega t + \varphi]$$

Here T is the string tension and we have used $T/\mu = (\omega/k)^2$. For this single sinusoidal travelling wave, the segment's instantaneous mechanical energy is

$$\Delta E(x, t) = \Delta K + \Delta U = \mu \Delta x A^2 \omega^2 \cos^2 [kx - \omega t + \varphi]$$

Its time average over a period is

$$\langle \Delta E \rangle = \frac{1}{2} \mu \Delta x A^2 \omega^2.$$

The average mechanical energy is directly proportional to the length of the piece of string, to the square of the maximum amplitude of the wave and to the square of its angular frequency.

8.1.4 Wave equation for a string

To derive the wave equation for a string, assume (i) small-amplitude disturbances ($|\partial y / \partial x| \ll 1$, with no restriction on wavelength λ) and (ii) uniform tension throughout the string, so that horizontal acceleration can be neglected.

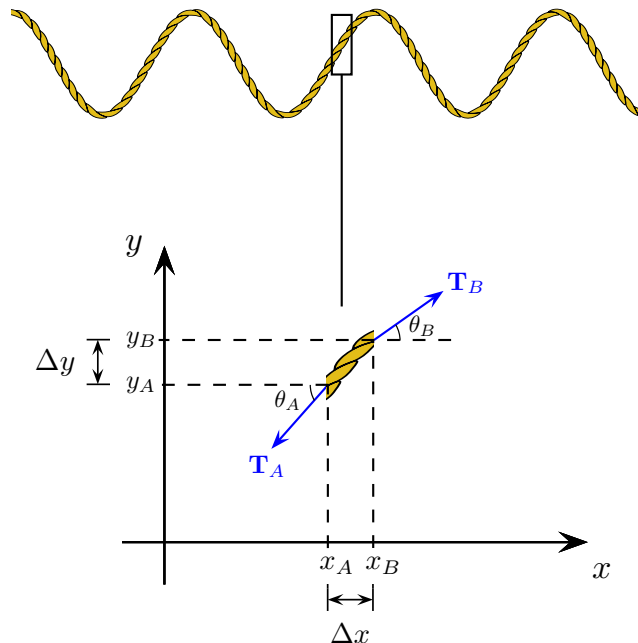


Figure 4: Forces on a string element.

Consider a piece of string whose ends have for abscissas x_A and $x_B = x_A + \Delta x$ and for ordinates y_A and $y_B = y_A + \Delta y$ (see Figure 4). The projection on the axis Oy of Newton's second law for the piece of string gives

$$\begin{aligned} \sum F_y &= \Delta m a_y \\ \Leftrightarrow T [\sin \theta_B - \sin \theta_A] &= \mu \Delta \ell \left(\frac{\partial^2 y}{\partial t^2} \right) \\ \Leftrightarrow T \left[\left(\frac{\partial y}{\partial x} \right)_{x_B} - \left(\frac{\partial y}{\partial x} \right)_{x_A} \right] &= \mu \Delta x \left(\frac{\partial^2 y}{\partial t^2} \right) \\ \Leftrightarrow \frac{\mu}{T} \left(\frac{\partial^2 y}{\partial t^2} \right) &= \lim_{\Delta x \rightarrow 0} \frac{(\partial y / \partial x)_{x_A + \Delta x} - (\partial y / \partial x)_{x_A}}{\Delta x} = \frac{\partial^2 y}{\partial x^2} \end{aligned}$$

To move from the first to the second line, we used the fact that $|\mathbf{T}_A| = |\mathbf{T}_B| = T$ and we noted $\Delta \ell$ the length of the piece of string. Then, to move from the second to the third line, we used the fact that for waves of low amplitude, $\sin \theta_i \approx \tan \theta_i = (\partial y / \partial x)_{x_i}$ with $i = A, B$ and $\Delta \ell = \sqrt{\Delta x^2 + \Delta y^2} \approx \Delta x$. Taking the limit of a piece of string of length tending towards zero, we finally end up with the wave equation

$$\boxed{\frac{\mu}{T} \left(\frac{\partial^2 y}{\partial t^2} \right) = \frac{\partial^2 y}{\partial x^2}} \quad (4)$$

This is a partial differential equation: the displacement y depends on both x and t , and the equation contains derivatives with respect to both variables. We seek a particular solution in the form of a monochromatic travelling wave

$$y(x, t) = A \cos(kx - \omega t)$$

This form is a solution of the wave equation (4) provided that

$$k^2 = \frac{\mu}{T} \omega^2 \quad \Leftrightarrow \quad \boxed{v = \sqrt{\frac{T}{\mu}}} \quad (5)$$

In this model, all waves on the string propagate at speed $v = \sqrt{T/\mu}$, independently of frequency and amplitude. The propagation speed (5) increases with string tension and decreases with linear mass density. For example, tension $T = 50$ N and linear density $\mu = 50$ g/m give $v \approx 114$ km/h. Using (5), the wave equation (4) can be written in the generic form

$$\frac{\partial^2 y(x, t)}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y(x, t)}{\partial t^2} \quad (6)$$

It is possible to show that the most general solution of (6) is of the form (1). The superposition principle now appears as a direct consequence of the linearity of the wave equation. If $y_1(x, t)$ and $y_2(x, t)$ are solutions of (6), then the sum of amplitudes $y_1(x, t) + y_2(x, t)$ will also be so (and more generally any linear combination $c_1 y_1(x, t) + c_2 y_2(x, t)$ with $c_1, c_2 \in \mathbb{R}$).

8.2 Doppler effect

Doppler effect The Doppler effect corresponds to a change in the perceived frequency¹ (measured) for a periodic wave due to the movement of the source and/or receiver.

8.2.1 Source and receiver at rest

In Figure 5, source S and receiver R are at rest, so there is no Doppler effect: the frequency measured by R , f_p , equals the source frequency $f = \omega/2\pi$. The wavelength is then

$$\lambda = v_{\text{sound}} T = \frac{v_{\text{sound}}}{f} \quad (7)$$

¹and possibly wavelength

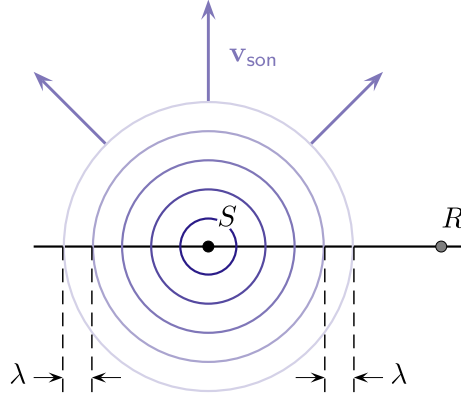


Figure 5: A stationary point source S emits a periodic sound wave of angular frequency ω in all directions. Pressure variations have the form $\Delta p(\mathbf{r}, t) \equiv p(\mathbf{r}, t) - p_0 = \frac{\delta p}{r} \sin[kr - \omega t]$, where p_0 is the equilibrium pressure and $r = |\mathbf{r}|$ the distance from the source. Pressure maxima, defined by $\sin[kr - \omega t] = 1$ and shown as blue circular wavefronts centred on the source, are emitted every $T = 2\pi/\omega$ seconds and travel at the speed of sound v_{sound} . More distant wavefronts were emitted earlier.

8.2.2 Moving source and stationary receiver

Here the source S moves a distance $v_S T$ between the emission of successive wavefronts (Figure 6, left), giving

$$\lambda' = \lambda - v_S T = \lambda - \frac{v_S}{f} \quad \text{and} \quad \lambda'' = \lambda + v_S T = \lambda + \frac{v_S}{f}$$

Once emitted, wavefronts propagate at the speed of sound in the medium (v_{sound}), independently of the motion of the source or receiver. The frequency measured by the receiver is therefore

$$f_p = \frac{v_{\text{sound}}}{\lambda'} = \frac{v_{\text{sound}}}{\lambda - v_S/f} \quad \text{and} \quad f_p = \frac{v_{\text{sound}}}{\lambda''} = \frac{v_{\text{sound}}}{\lambda + v_S/f}$$

according to whether the source approaches the receiver (−) or recedes from it (+). Using (7) gives

$$f_p = \left(\frac{v_{\text{sound}}}{v_{\text{sound}} \mp v_S} \right) f$$

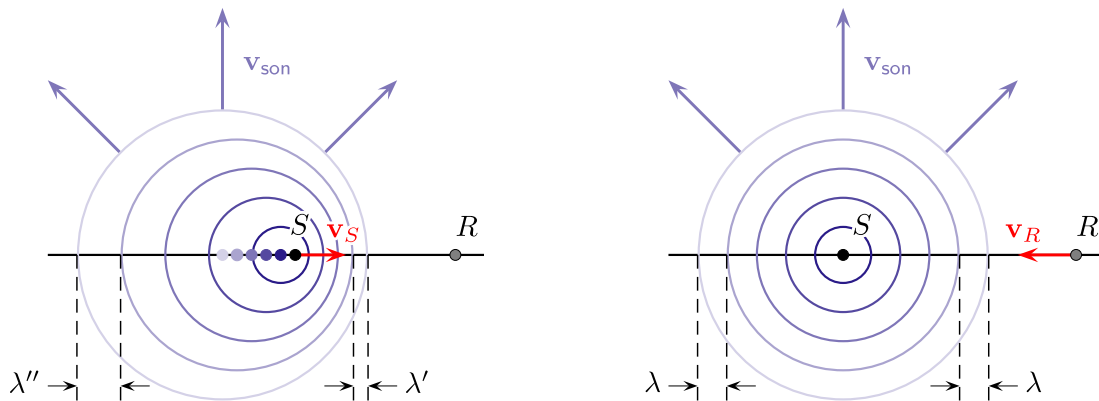


Figure 6: Illustration of the Doppler effect. Left: moving source and resting receiver. When the source moves to the receiver, the receiver receives wave fronts at a higher frequency ($f_p = \frac{v_{\text{sound}}}{\lambda'} > \frac{v_{\text{sound}}}{\lambda} = f$) than when it is at rest. On the contrary, if the source moved away from the receiver, the latter would receive wave fronts at a lower frequency ($f_p = \frac{v_{\text{sound}}}{\lambda''} < \frac{v_{\text{sound}}}{\lambda} = f$) than if it were at rest. Right: resting source and mobile receiver. When the receiver moves to the source, it perceives wave fronts at a higher frequency than if it were at rest. $f_p > f$. On the contrary, if the receiver moved away from the source, it would perceive wave fronts at a lower frequency than if it were at rest ($f_p < f$).

8.2.3 Stationary source and moving receiver

Consider that the receiver R moves towards the wave fronts emitted by the source (see Figure 6, right part) or moves away from it. It perceives two successive maxima separated by a time interval $T' = \lambda/(v_{R/\text{sound}})$ where $v_{R/\text{sound}}$ is the relative velocity between the wave fronts and the receiver. When the receiver moves towards the wave fronts, $v_{R/\text{sound}} = v_{\text{sound}} + v_R$, while if it moves away from it, $v_{R/\text{sound}} = v_{\text{sound}} - v_R$.

$$f_p = \frac{1}{T'} = \frac{v_{R/\text{sound}}}{\lambda} = \frac{v_{\text{sound}} \pm v_R}{\lambda}$$

and, using the relationship (7),

$$f_p = \left(\frac{v_{\text{sound}} \pm v_R}{v_{\text{sound}}} \right) f$$

8.2.4 Moving source and receiver

By combining the results of the previous sections, we obtain the general formula of the Doppler effect

$$f_p = \left(\frac{v_{\text{sound}} \pm v_R}{v_{\text{sound}} \mp v_S} \right) f$$

valid when the wave, source and receiver move along the same line, though not necessarily in the same sense.